Lecture Notes

Superconductivity

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Notes

This Lecture Notes is prepared for the students who are taking the Year 4 module of Superconductivity at the University of Birmingham[1]. The module consists of 22 sessions of 50 minutes of lectures in Autumn semester. The lecture note, which is the main text for this module, accordingly consists of 22 sections in one-to-one correspondence in order. Each section may contain a few `Exercises`, which should be considered as an integral part of the module. There are several mandatory readings from the books[2] which are available online via Canvas:

- *Introduction to Superconductivity* by AC Rose-Innes and EH Rhoderic. This is a very accessible text with intuitive explanations. Follow the whole chapters.
- *Superconductivity, Superfluids and Condensates* by JE Annett. Follow the chapters on general superconductivity and Ginzburg-Landau theory.
- *Superconductivity of Metals and Cuprates* by JR Waldram: Follow the first two chapters on the BCS theory.

The lecture proceeds in quasi-chronological order such that the presentation of the subject be *deductive* rather than inductive. Basically, I wish the students to learn the subject *through discovery* for themselves, led by a series of {`experimental data, plausible arguments, reasonable assumptions and logical development`}. In particular, I emphasize the experimentation and observations to come first, which is followed by theoretical development for explanation. This approach is generally more difficult from the student perspective, but I believe it would be rewarding. I hope the students to take this as an opportunity to get some flavour of what it would be like to make physics discoveries and progresses, especially with this lively subject of superconductivity where there has been fruitful interplay between experiments and theories for over a century till present.

[1] For those who are not taking this module at Birmingham, use this note at your own risk as I do not guarantee the correctness and accuracy of the contents.

[2] Most contents in the readings are covered in this Lecture Notes. However, it still remains useful to read the books of varying ideas of approach and rigor. Bear in mind that the books may use different notation and contain some typos.

Welcome address for 2020

Hello, everyone. Welcome to the Y4 module Superconductivity. My name is Mingee Chung (also known as Minki Jeong). I joined the University in 2017 like most of you, and since then I learned a lot, made new friends, enjoyed visiting city centre and so on, here at Birmingham, like you did. This year, as we all know, and all affected, we are facing great challenges and difficulties in many aspects amid pandemic. Indeed, I was very disappointed to learn that I should record lectures and deliver online as I miss close interactions in the classroom.

Nevertheless, if I may try to see any positive sides, besides that we enjoyed three fruitful years on the beautiful campus, let me recall my college days just twenty some years ago (I know, it already sounds like grandparents talking). Nothing much would have been possible if impacted by such pandemic, because, believe it or not, there was no WiFi, no smartphone not to mention tablet, and very limited internet connection spending tens of minutes to download a single image and never dreaming of a live streaming. On the other hand, with all the technologies and platforms (e.g., `Canvas, Panopto, Zoom` etc.) made available today, we can certainly mitigate the impact of pandemic on our learning physics.

If there is anything that should recover first, for the best of learning, I would say the *interaction*, between you and me, and also among yourselves. The lectures will be pre-recorded and released weekly alongside the lecture note. However, we will also have a weekly on-campus, face-to-face discussion session for limited number of students in turn each time; this will be live streamed for those not in their turn. You can post your comments, complaints or questions any time on the Discussion page in the Canvas (you are also more than welcomed to reply to your peer students), drop me a mail casually or arrange a Zoom meeting; doesn't matter how small or big things you want to talk, literally all welcomed!

I should also emphasise that interactions are key to the phenomena of superconductivity. You must have a good understanding of the behaviour of a single electron in a potential, or interaction between two electrons or ions via Coulomb force and so on. But, what if there are *many* electrons and ions that are interacting, and all the wavefunctions overlapping? Superconductivity is a canonical example of such *collective* manifestations that give rise to macroscopic quantum phenomena at human scale (or even neutron star scale!). To understand superconductivity, one therefore needs to synthesise their knowledge of quantum mechanics, statistical physics, electrodynamics and condensed matter physics. This will be challenging, but also intellectually stimulating. I hope that you will share my excitement I had when I first encountered this subject that I still holds.

Thank you for choosing this module as your option, and I look forward to talking to you soon.

All the best
Mingee

"Any sufficiently advanced technology is indistinguishable from magic."

Arthur C. Clarke
Profiles of the Future: An Inquiry into the Limits of the Possible

Part I
Phenomenology

We review basic electromagnetic and thermodynamic properties of materials in their superconducting states and develop phenomenological descriptions based on Maxwell's equations and thermodynamic functions.

"It is the mark of an instructed mind to rest assured with that degree of precision that the nature of the subject admits, and not to seek exactness when only an approximation of the truth is possible."

Aristotle
Nicomachean Ethics

1 Resistance Zero

You might have encountered the phenomena of superconductivity in some form, either from demonstration right in front of your eyes or from the YouTube clips or something. Then I presume that you are familiar with, or at least recall a characteristic image similar to that shown in Fig. 1. The photo shows a piece of magnet levitating above a superconductor that is cooled by liquid nitrogen. Cool. Isn't this amazing? Isn't this surprising?



Figure 1. A magnet levitating on a superconductor. Photo credit http://hebergement.u-psud.fr/supraconductivite

Hmm, well honestly, not so much! Levitation is cool, I agree. But, we know there are other examples of levitation of different kinds, which are far from being any magical (though still cool). See, for example, Fig. 2. From the left, the first image shows a frog being levitated by a magnetic field owning to the frog's diamagnetic property; magnetic levitation. The middle image shows small and light styrofoam particles being locked in space at the nodes of sound waves; acoustic levitation. In the last one, a person is lifted up in space by air pressure. So, the levitation itself, I would say, is not something extraordinary[3].

But, wait. Do you also recall the image something like shown in Fig. 3? There someone is holding a magnet with fingers and a superconductor is hanging below, without touching,

[3]Here we are only talking at the level of their *appearance*. Certainly, the reason behind the levitation with superconductor *is* extraordinary as we will see later.

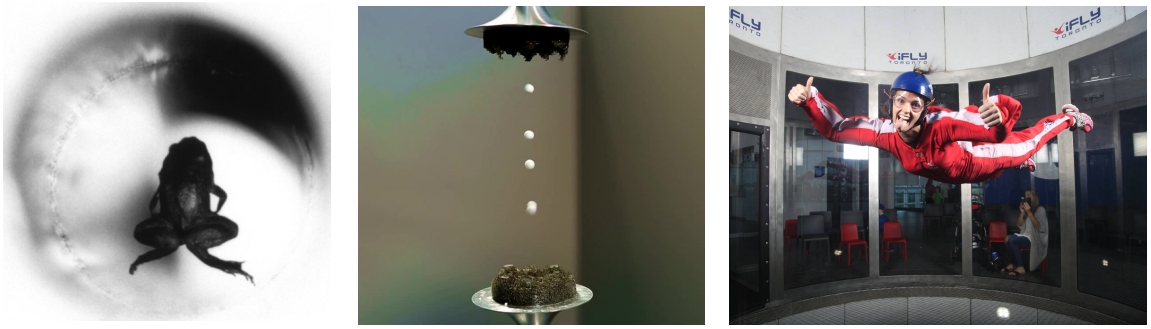


Figure 2. Left: a frog levitating in a magnetic field [1]. Middle: light styrofoam particles trapped in a standing wave of sounds (from "Acoustic Levitation", Wikipedia). Right: a person floating on air pressure (photo credit iFLY Toronto.

without connecting cords or anything! This is surprising, as it is not like anything we know of.

In this module, we aim to understand those phenomena shown in Figs 1 and 3, particularly how such *magnetic* phenomena are associated with the absence of *electrical* resistance that was the original observation of superconductivity. This is not as simple as it may sound, and indeed it took about fifty years for the whole physics community to work out the solutions. To follow closely their steps, let's start our journey from the very beginning of the whole story.



Figure 3. A superconductor hanging, without touching, below a magnet holded by a hand. Photo credit http://hebergement.u-psud.fr/supraconductivite

1.1 Discovery in total surprise

In 1911, the research group of Prof. H. Kamerlingh Onnes at Leiden, Netherlands, observed that the resistance of Hg (mercury) sample abruptly dropped to zero when cooled below 4.18 K, as shown in Fig. 4. The researchers couldn't believe what they observed, and this became sensational internationally. But, what was your impression when looking at Fig. 4. Can you tell what exactly could be surprising here? That they got utterly surprised

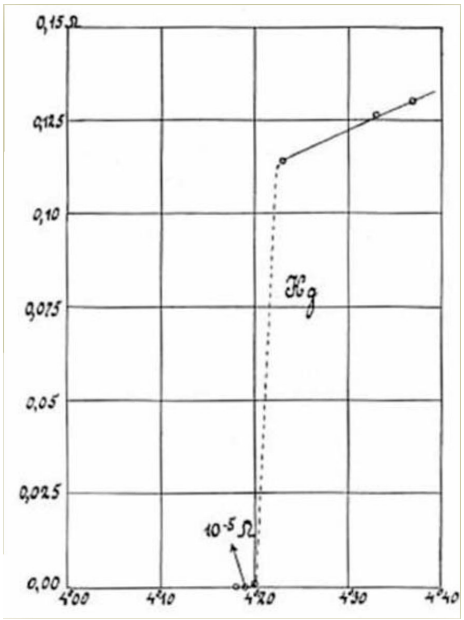


Figure 4. Electrical resistance (ohms) of mercury as a function of temperature (in kelvin) [2].

implies that they had expected something totally different. To this end, we should ask the followings first:

- What would you have expected for the temperature (T) dependence of the resistance of Hg or any metals?
- Why mercury rather than other metals?
- Has the resistance reached a true zero, or just a very tiny value? How do you know?

Let's bear these questions in mind (you are welcomed to guess) which we will find answers throughout this section.

1.2 Discovery from endeavor

Measuring electrical resistance sounds quite simple and so does obtaining just a few data points as shown in Fig. 4. Indeed, the idea of cooling a metal to very low temperatures, close to absolute zero, was shared by many scientists at the time. So, what did they have in mind?

In metals, there are itinerant electrons and localised ions which interact among themselves. If one considers just a single electron moving in the background of a perfectly periodic lattice of positively charged ions, as F. Bloch has proved, the electronic wavefunction is not scattered, that, is the resistivity $\rho = 0$, for $T = 0$. At finite temperatures, the electrons would scatter off phonons (vibrating lattice), for instance. Typically, $\rho \propto T$ for temperatures down to $\sim \theta_D/3$, where θ_D is the Debye temperature, and less rapidly at lower

temperatures, i.e., $\rho \propto T^5$ for $T \lesssim \theta_D/10$[4]. However, there must be other contributions to the resistivity such as impurities or other electrons.

Exercise 1

What is typical scale of Debye temperature and Fermi temperature? How do they compare with the superconducting transition temperatures?

A useful starting point to take into account different contributions to the resistivity is the Ohm's law,

$$\mathbf{j} = \sigma \mathbf{E}, \tag{1.1}$$

where $\mathbf{j}$ is the current density and $\mathbf{E}$ the electric field. The conductivity σ is, within the Drude model[5], expressed as

$$\sigma = \frac{ne^2\tau}{m_e}, \tag{1.2}$$

where n is the charge carrier (electron) density, e the charge of electron, τ the average life time between collisions, and m_e the mass of the electron. The life time would be determined by the collision of the electrons with impurities, other electrons as well as phonons.

The Ohm's law can be written equivalently as

$$\mathbf{E} = \rho \mathbf{j} \tag{1.3}$$

and the resistivity ρ expressed as

$$\rho = \frac{m_e}{ne^2}\tau^{-1}, \tag{1.4}$$

where τ^{-1} is the average scattering rate of the electrons. This is useful for our purposes as the rate from different contributions can be added up such that

$$\tau^{-1} = \tau_{\text{imp}}^{-1} + \tau_{\text{e}}^{-1} + \tau_{\text{ph}}^{-1} + ..., \tag{1.5}$$

where the scattering rate from the impurity $\tau_{\text{imp}}^{-1} = const.$, from the electrons $\tau_{\text{e}}^{-1} \propto T^2$ and from the phonons $\tau_{\text{ph}}^{-1} \propto T^5$ at low temperatures. Therefore, as one approaches the absolute zero, one would expect

$$\rho = \rho_0 + \alpha T^2 + ... \tag{1.6}$$

be dominant, as sketched on the left in Fig. 5. Depending on the level of purity and the impact of impurities, one would expect a systematic variation of the $\rho(T)$ curve. This is where the ideas of using Hg kicks in as it is the purest elemental metal one could get, and thus one might hope to observe clearly the $\rho(T) \propto T^2$ behaviour down to the lowest temperature.

[4]Derivation of these are beyond the scope of this module. For those who are interested, please refer to a textbook *Solid State Physics* by N. W. Ashcroft and N. D. Mermin

[5]The model considers a classical gas of electrons, and was first put forward even three years before the discovery of electrons. Please refer to any solid state textbook for the derivation of Eq. (1.1)

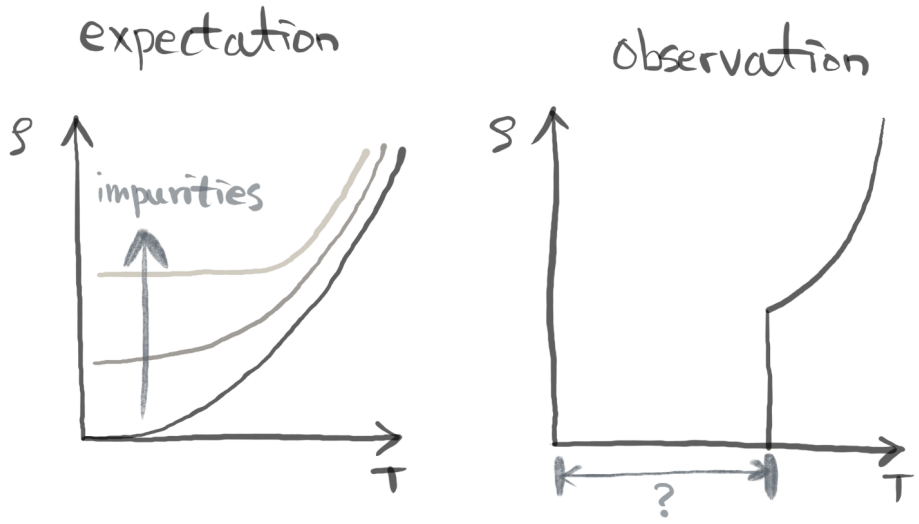


Figure 5. Schematic drawing of the resistance as a function of temperature as expected (left) and observed (right)

The actual observation, as sketched on the right in Fig. 5 (or as shown in Fig. 4) makes a sharp contrast to the expectation. Most of all, the resistance literally dropped abruptly to zero at some *finite* temperature, 4.18 K for Hg, below which our *metallic* picture ceased to apply. Does this imply a transition to a new phase? Moreover, it was observed later on that *impure* and *alloyed* samples also show the resistance dropping to zero at finite temperatures. So, what a surprise.

OK, then the Onnes group was just lucky enough to do the experiment first in the race to make this serendipitous discovery? It is actually more than that, much. There was a fundamental and practical challenge. That is to achieve ever lower temperatures. In fact, there was only a single research group in the world back then which could cool materials below 4.2 K. It was in 1908 when the Onnes group first succeeded in liquefying helium at 4.2 K, the only remaining element to *condense* by then. Soon after, they went further to be able cool down to 1 K by heavily pumping the liquid helium. This is never a small work; see the complexity of pumps, valves and pipes shown in Fig. 6 which captured a scene of the Onnes laboratory in the period of the discovery.

Exercise 2

Why helium, rather than a lighter element hydrogen, was the last element that has been liquefied?

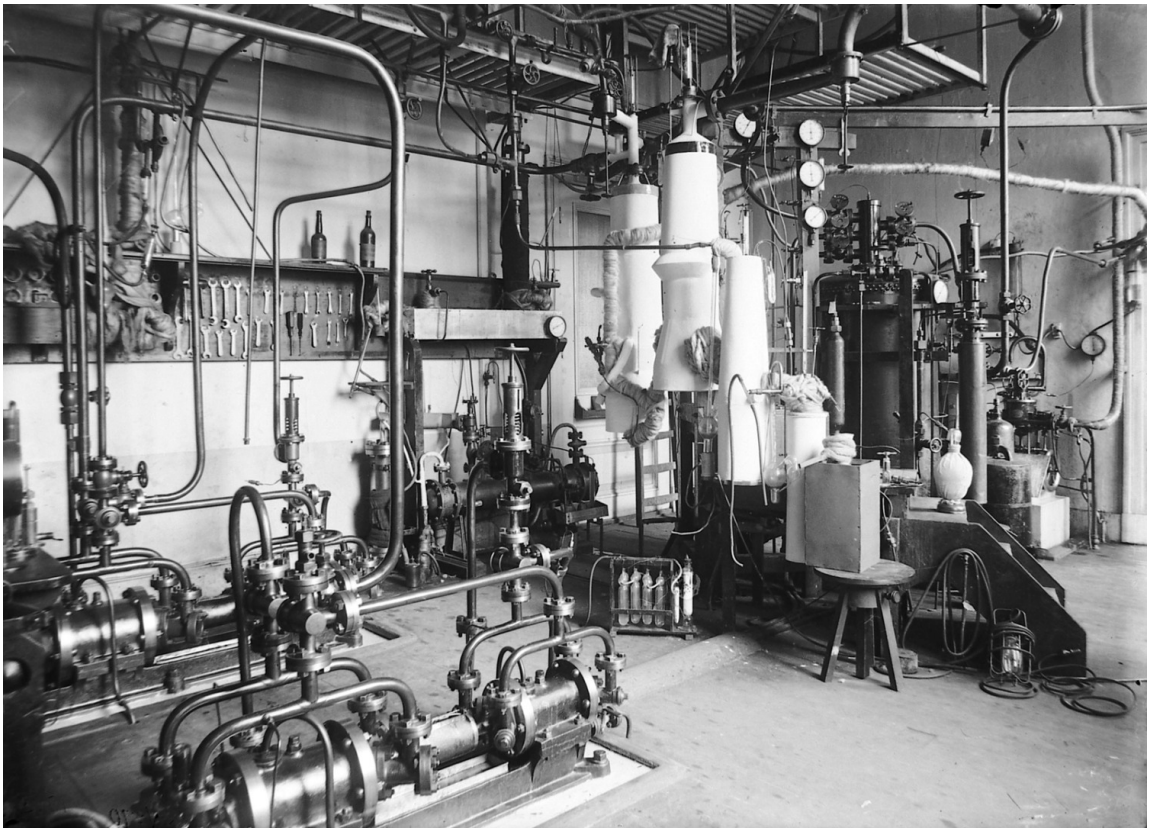


Figure 6. The low temperature laboratory led by K. Onnes at Leiden, Netherlands, around the times of the groundbreaking discoveries of liquid helium and superconductivity ©Rijksmuseum Boerhaave.

1.3 Total absence

How do we know for sure that the resistance of the Hg sample has dropped to zero, but not a very tiny value? One cannot measure zero, but can only push down an upper limit with increased precision, e.g., < 0.01 than < 0.1. The resistance (R) of a given sample is deduced by measuring a voltage drop (V) that occurs when the current (I) flows as $V = IR$. The current is provided by a current source, and the voltage is measured by a voltmeter. Figure 7 shows a schematic configuration for the resistance measurements, known as the four-probe method.

Exercise 3

Why do we need such configuration as shown in Fig. 7 for measuring resistance?

The residual resistance in the voltmeter would limit the accuracy. To measure ever smaller value of resistance with precision, with the given voltmeter, one may wish to increase the current to get the largest variation in voltage. However, it turned out that the current density (j) large than a certain value destroys superconductivity, i.e., a superconducting sample restores non-zero resistance as in metallic state for $j > j_c$.

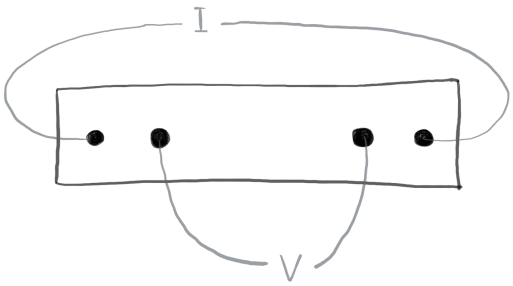


Figure 7. Schematic configuration for the resistance measurements.

Instead, one may resort to *dissipation* arising from R, if any. For measurable dissipation from tiny R, one would need a very long wire. Or better, prepare a ring. Then one can write

$$RI + L\frac{dI}{dt} = 0 \tag{1.7}$$

where L is the inductance of the ring. The solution is written as

$$I(t) = I(0)\exp\left[-\left(\frac{R}{L}\right)t\right]. \tag{1.8}$$

Thus, once a current is set up in a superconducting ring, by measuring a time constant L/R for a decaying current, one can detect a tiny R.

Now there arises two questions:

- How can one set up a current to start?
- how can one measure the changing current without interruption?

To answer the first question, consider the Lenz law. For instance, if one applies (e.g., increasing from zero to a certain value) a magnetic flux (Φ) through the area enclosed by the ring, electric current will be induced such that its effect opposes the flux change[6]. One can write

$$RI + L\frac{dI}{dt} = -\frac{d\Phi}{dt} \tag{1.9}$$

by adding an e.m.f. term due to the flux change on the right. For negligible R, noting that $\Phi = S \cdot B$, where S is the area enclosed by the ring and B the magnetic flux density, one obtains

$$\frac{d}{dt}\left(SB + LI\right) = 0. \tag{1.10}$$

Inside the parenthesis, the first term represents the applied flux and the second term the induced flux. The above equation tells us that if $R = 0$, i) a *persistent* current is set up and ii) the total magnetic flux remains constant. The former says nothing but that the current,

[6]Apparently, by removing the existing flux, one can equally induce a current but along the opposite direction.

once set up, does not change as long as the magnetic flux does not change. Note that the latter concerns total flux only, but not its vectorial density distribution; the induced flux density $\mathbf{B}_{\rm ind}(x)$ by the current would differ in general from the originally applied flux density $\mathbf{B}_{\rm a}(x)$, where x refers to some general coordinates. This may be more easily understood by looking into Fig. 8. On the left is shown magnetic flux density as applied at $T > T_c$ then cooled below T_c. When the applied flux density is removed, only the induced flux density and the resulting currents remain as shown on the right.

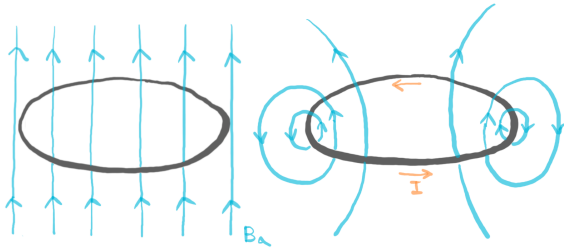


Figure 8. Left: magnetic flux density as applied at $T > T_c$ then cooled below T_c. Right: when the applied flux density removed and only the induced flux density and the resulting currents remaining.

To answer the second question, one may also observe the changing B rather than the changing I. There are different methods to measure magnetic flux or its density such as Nuclear Magnetic Resonance (NMR). Anyway, some researchers have actually monitored variation of B throughout an extended time span, and found no change over 2 years! This translates to an upper limit of the resistivity around $10^{-26}\Omega$m, which is 18 orders of magnitude smaller than the resistivity of Cu at 300 K. This is a huge difference. In fact, the resistivity of Cu is about 20 orders of magnitude smaller than that of insulators. In other words, this zero-resistance state is distinct from metals as much as metals differ from insulators. Therefore, one may presume that it is a new state of matter. Note that in the above discussion a superconductor and a perfect conductor are not really differentiated. But, should they?

Exercise 4

W may extend our discussion of a superconducting (or perfectly conducting) coil to a solenoid from which one can construct an electromagnet, or to a hollow cylinder which enables shielding interior from external magnetic field (how?). Indeed, if a solenoid carries persistent currents without resistance, one can generate a magnetic field without Joule heating or external power source. To learn how this dream can be realised, refer to textbooks or google *superconducting switch*. But why don't you try to devise one yourself first?

2 Perfect Expulsion

As electricity and magnetism are intertwined via Maxwell's equations, it is natural to ask how a superconductor would respond to a magnetic field. Indeed, we already learned that a superconducting ring preserves a total magnetic flux. We can apply this to extend the discussion to a bulk superconductor. One can draw in mind arbitrary rings inside the bulk where each ring would preserve its own magnetic flux. Put otherwise, anywhere inside a bulk superconductor, magnetic flux simply cannot change with time, i.e., $\partial \mathbf{B}/\partial t = 0$. This also implies, from the Faraday's law of induction, $\mathbf{E} = 0$ everywhere inside a bulk superconductor, which is consistent with the Ohm's law giving $\mathbf{E} = 0$ for $\rho = 0$. So far, we built our arguments essentially on a perfect conductor. Then one may wonder if a superconductor is a synonym of a perfect conductor. We will see in the following that a superconductor is more than a perfect conductor.

2.1 Magnetism of a perfect conductor

Let's examine more closely the meaning of Eq. (1.10), or its consequence $\partial \mathbf{B}/\partial t = 0$. Suppose we prepare a bulk metal which only becomes a perfect conductor below a certain temperature, T_c (see the first column in Fig. 9). When the sample is cooled in zero applied field below T_c, nothing happens. When a field is applied below $T < T_c$, no flux would penetrate the sample to respect the $\partial \mathbf{B}/\partial t = 0$ condition. Now consider a case where the field is applied when $T > T_c$ (the second column in Fig. 9). The magnetic flux would penetrate into the sample as a metal is paramagnetic. Now cool the sample in a field to $T < T_c$. One would observe the flux stays still inside the sample. What if now we remove the applied field? The flux inside the material would again stay still, only outside the sample the field would reorganise.

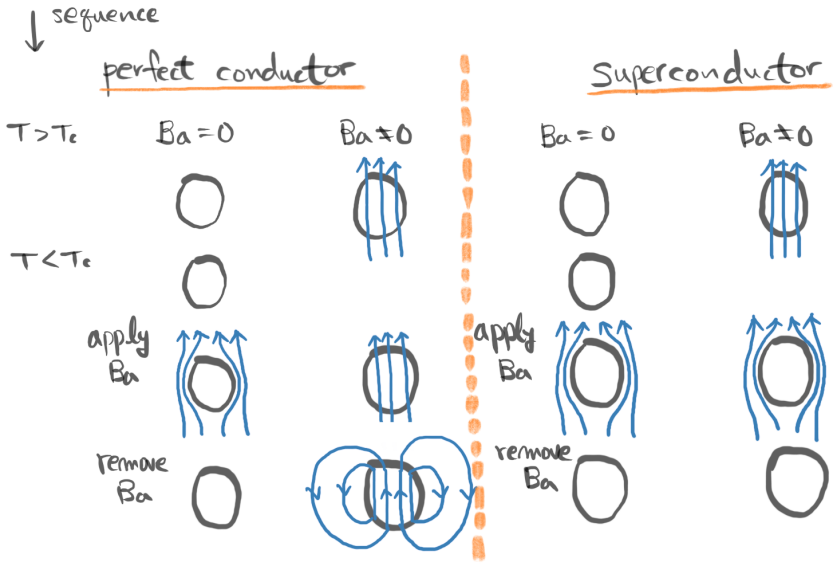


Figure 9. Comparison between the magnetic field responses of a perfect conductor (left) and a superconductor (right).

2.2 Magnetism of a superconductor

It turned out that a superconductor behaves differently from the expectation for a perfect conductor. The case where a field is applied when $T < T_c$ does not differ. However, for the case where a field is applied for $T > T_c$ and then cooled in field, the flux is expelled from the interior (the last column in Fig. 9). And there is a single way that this observation be compatible with $\partial \mathbf{B}/\partial t = 0$; that is, $B = 0$. In addition, the behaviour sketched on the right in Fig. 9 was observed to be history independent and reversible, implying that a superconductor is in a thermodynamic equilibrium state.

To recap, for a perfect conductor, $B = const.$ while for a superconductor $B = const. = 0$. The latter means a magnetic flux is excluded from a superconductor. To find out its implication, recall that

$$\mathbf{B} = \mu_0 \left(\mathbf{H}_a + \mathbf{M} \right), \tag{2.1}$$

where $\mathbf{H}_a$ is an applied magnetic field intensity and $\mathbf{M}$ the magnetisation induced in material[7]. Since $B = 0$ for a superconductor, one obtains

$$\mathbf{M} = -\mathbf{H}_a. \tag{2.2}$$

The minus sign indicates diamagnetism (positive sign for paramagnetism), meaning the material is magnetised in opposite direction to the applied field direction. And the value of unity indicates perfectness (otherwise less than one). This *perfect diamagnetism* is called the Meissner(-Ochsenfeld) effect, and is a distinctive signature of superconductivity. As one defines magnetic susceptibility with $\mathbf{M} = \chi \mathbf{H}_a$, $\chi = -1$ for a superconductor while $\chi \sim 10^{-4}$ for metals. One may understand the implication of Eqs. (2.1) and eq:diamagnetism by drawing magnetic flux density as shown in Fig. 10.

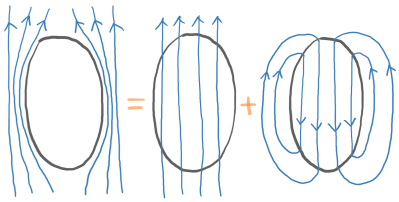


Figure 10. Net flux as a sum of applied field and flux from magnetisation.

2.3 Penetration

The sample magnetisation can be understood equivalently as the appearance of surface currents. However, the current cannot be confined to an infinitesimal thickness as it will induce $j \to \infty$. So there must be some very small yet finite thickness, where $B = 0$ may cease to apply. One can imagine, for instance, B field profile across the surface of the superconductor as shown in Fig. 11.

[7] Confusingly, both **B** and **H** are often called a magnetic field, though people often call a **B** field or **H** field. More properly, **B** is called a magnetic flux density or magnetic induction, and **H** is called a magnetic field intensity or magnetic field strength, though both are vector quantities.

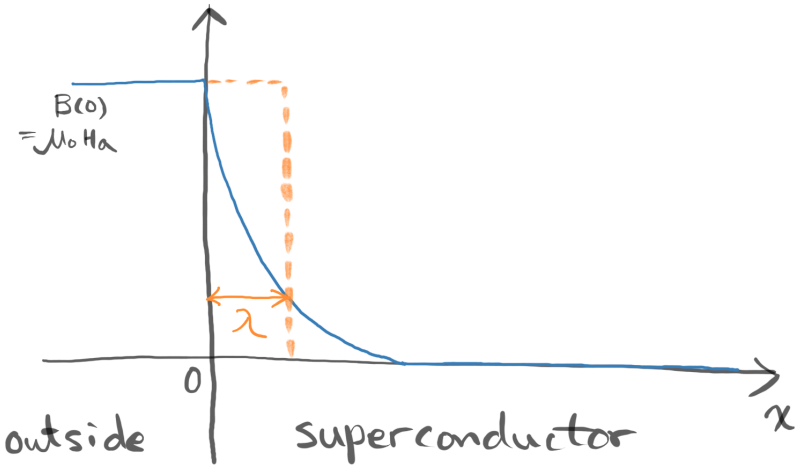


Figure 11. Exponential suppression of a magnetic flux when entering a superconducting bulk.

One may assume exponential suppression of the flux when entering into a superconductor,

$$B(x) = B(0) \exp\left[-x/\lambda\right], \tag{2.3}$$

such that

$$\int_0^\infty B(x)dx = \lambda B(0), \tag{2.4}$$

where the characteristic length λ is called the penetration depth. If the penetration depth is negligibly narrow, one may claim the bulk superconductor respect perfect diamagnetism. The penetration depth can be measured experimentally and turned out to be typically submicron size. This is extremely tiny value compared to the size of bulk sample, so the notion of perfect diamagnetism still holds.

The penetration depth can be measured as a function of temperature as shown in Fig. 12. The length was found to diverge rapidly as the sample is warmed toward T_c following the empirical form

$$\lambda = \frac{\lambda_0}{\sqrt{1-(T/T_c)^4}}. \tag{2.5}$$

Therefore, perfect diamagnetism holds for most temperature range of $T < T_c$ where λ remains very small.

Exercise 5

How can we measure the penetration depth experimentally?

2.4 Periodic table

We have so far considered superconductivity of Hg or some unspecified materials. By now you may wonder what would happen to other elemental metals. Would they all become

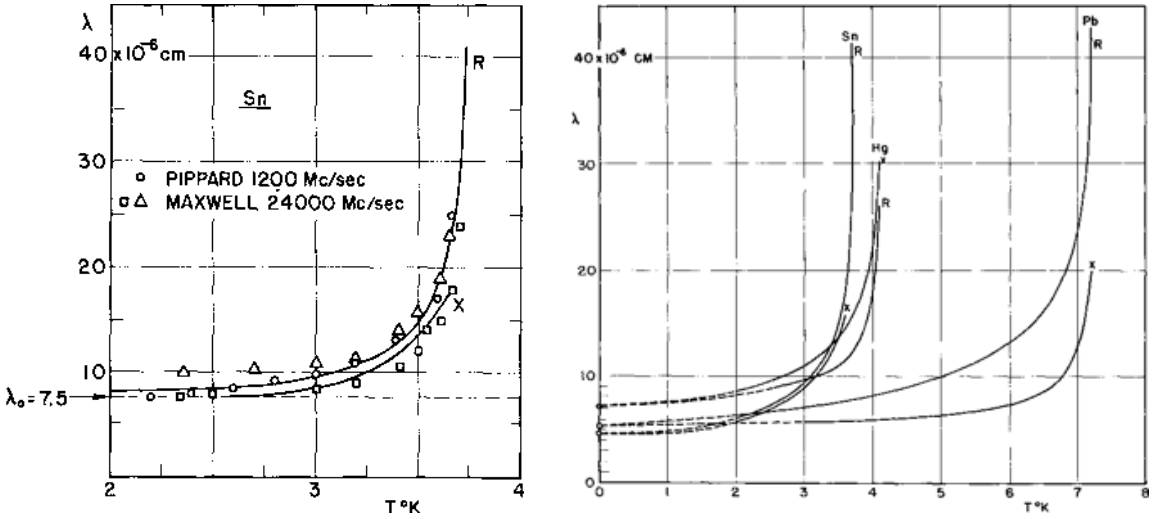


Figure 12. Penetration depth as a function of temperature for several elemental metals [3].

superconducting when cooled, or is Hg really unique? What about non-metallic elements, alloys or compounds? We will not answer everything at once, but let us have a quick look into the periodic table as shown in Fig 13. It turned out that about a half of elemental metals become superconductors at sufficiently low temperatures. The size of a bar in Fig. 13 represents the transition temperature.

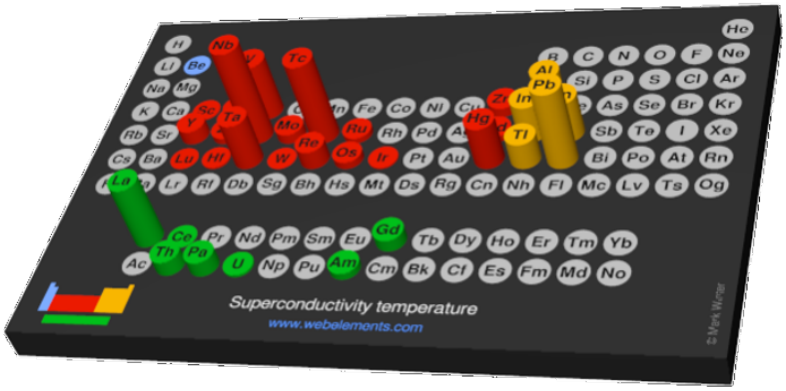


Figure 13. Periodic table highlighting superconducting elements.

More detailed information can be found from Fig. 14, where it is shown which element need assistance (e.g., applied pressure) to become a superconductor. Note that this table is not a constant of time, but is continuously revised. For instance, looking into ^{83}Bi, the table says it becomes superconducting at 8.7 K under an applied field of 9 Pa. However, it was found in 2017 that Bi becomes superconducting under ambient pressure when cooled below 0.5 mK [4].

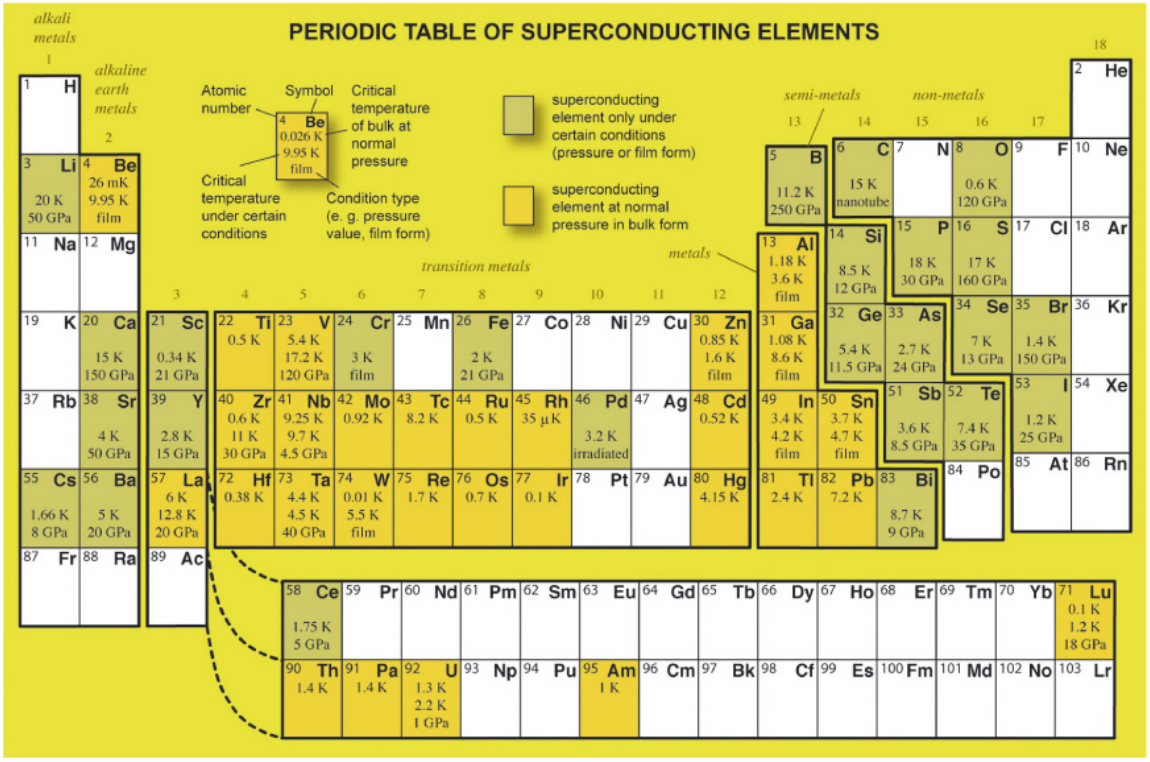


Figure 14. Periodic table of superconducting elements.

Exercise 6

What is your observation from Figs. 13 and 14? Do you find any systematic?

Exercise 7

From the periodic table (Figs. 13 and 14) many elements become superconducting (occasionally far) below 1 K. We learned that 1 K temperature can be obtained by pumping liquid helium which is known to be the last element liquefied. Then, how could you reach temperature below 1 K?

3 London Brothers

We have made two key observations for a superconductor:

- $\rho = 0$,
- $B = 0$,

that is, zero resistance and perfect diamagnetism, respectively. Now the question is how we can formulate or write down equations to describe the properties of superconductors which respect $\rho = 0$ and $B = 0$. For instance, we may check how the superconductors would follow the Maxwell's equations. And there is a constitutive equation, called the Ohm's law, which we had used before. But, what is the Ohm's law? It is the law that is valid when it works! For instance, it breaks down when the current is too large and leads to the so-called non-ohmic behaviour (i.e., V is not linear in I anymore). After all, the Ohm's law is not useful if $\rho = 0$. This motivates us to look for the replacement of the Ohm's law.

3.1 No scattering

The Drude model of electron transport can be written as

$$m_e \left(\frac{d\mathbf{v}}{dt} + \frac{\mathbf{v}}{\tau} \right) = e\mathbf{E}, \tag{3.1}$$

where the first term on the left represents acceleration and the second term drag due to various scattering. For a perfect conductor where there is no scattering, or $\tau \to \infty$, the drag term vanishes. If we substitute with the definition of a current density,

$$\mathbf{j} = ne\mathbf{v}, \tag{3.2}$$

we obtain

$$\frac{d\mathbf{j}}{dt} = \frac{ne^2}{m_e}\mathbf{E}, \tag{3.3}$$

one of the two London equations, which you may compare with the Ohm's law Eqs. (1.3) and (1.4). This equation tells us that to maintain a constant current, the electric field should remains zero, $E = 0$.

One can further find a magnetic field version of it using Maxwell's equations. First, take the curl on both sides of Eq. (3.3) find

$$\nabla \times \frac{d\mathbf{j}}{dt} = \frac{ne^2}{m_e} (\nabla \times \mathbf{E}) \tag{3.4}$$

Hoping to replace $\mathbf{j}$ and $\mathbf{E}$ with B, let's first substitute with the Faraday's law of induction, $\nabla \times \mathbf{E} = -d\mathbf{B}/dt$, which gives

$$\frac{d}{dt} (\nabla \times \mathbf{j}) = -\frac{ne^2}{m_e} \frac{d\mathbf{B}}{dt} \tag{3.5}$$

Now further substituting with the Ampere's law, $\nabla \times \mathbf{B} = \mu_0 \mathbf{j}$, gives

$$\frac{1}{\mu_0} \frac{d}{dt} [\nabla \times (\nabla \times \mathbf{B})] = -\frac{ne^2}{m_e} \frac{d\mathbf{B}}{dt} \tag{3.6}$$

One more substitution with the vector identity, $\nabla \times (\nabla \times \mathbf{B}) = \nabla (\nabla \cdot \mathbf{B}) - \nabla^2 \mathbf{B}$, along with the use of the Maxwell's no monopole, $\nabla \cdot \mathbf{B} = 0$, finally leads to

$$\frac{d}{dt}\left[\nabla^2 \mathbf{B} - \frac{\mu_0 n e^2}{m_e}\mathbf{B}\right] = 0 \tag{3.7}$$

OK, so everything is expressed in terms of $\mathbf{B}$. Now, what's the meaning of the equation? Seen from the first term in the bracket, one may find that it could be useful to define a length scale λ_L as

$$\frac{\mu_0 n e^2}{m_e} = \frac{1}{\lambda_L^2}. \tag{3.8}$$

Then the equation can be rewritten, for the simplest case with a single coordinate x, as

$$\frac{\partial^2}{\partial x^2}\left(\frac{\partial \mathbf{B}}{\partial t}\right) = \frac{1}{\lambda_L^2}\left(\frac{\partial \mathbf{B}}{\partial t}\right). \tag{3.9}$$

One then obtains the solution,

$$\frac{\partial \mathbf{B}}{\partial t} = \frac{\partial \mathbf{B}_a}{\partial t}\exp\left(-x/\lambda_L\right). \tag{3.10}$$

The solution tells us that however you change the applied field from outside the superconductor, $(\partial \mathbf{B}_a/\partial t)$, the field inside will not change up to the exponentially suppressing term near the surface. This agrees with our previous arguments for a perfect conductor. Indeed, so far we have considered the case of no scattering of electrons that is a definition of a perfect conductor.

3.2 Respecting Meissner & Ochsenfeld

Apparently, the simplest way of making the above equation be compatible with the experimental observation for superconductors is to replace $\partial \mathbf{B}/\partial t$ with $\mathbf{B}$: Replace

$$\left[\nabla^2 \mathbf{B} - \frac{\mu_0 n e^2}{m_e}\mathbf{B}\right] = const. \tag{3.11}$$

with

$$\left[\nabla^2 \mathbf{B} - \frac{\mu_0 n e^2}{m_e}\mathbf{B}\right] = 0, \tag{3.12}$$

which immediately leads us to the solution

$$B = B_a \exp\left(-x/\lambda_L\right). \tag{3.13}$$

The solution points at that $B = 0$ inside a superconductor up to a penetration depth given as λ_L which can be sketched as shown in Fig. 11.

Exercise 8

For a superconducting slab with thickness d in a parallel magnetic field, calculate the magnetic flux density inside the slab, and find the approximation for $d \ll \lambda$

Exercise 9

Show that the currents also flow on the surface of a superconductor within the penetration depth.

You may trace back where to substitute $\partial\mathbf{B}/\partial t$ with $\mathbf{B}$, and reach to Eq. (3.5). By removing the time derivative, one can get the other remaining London equation,

$$\nabla \times \mathbf{j} = -\frac{ne^2}{m_e}\mathbf{B}. \tag{3.14}$$

To sum, using λ_L, the two London equations are neatly written as

$$\nabla \times \mathbf{j} = -\frac{1}{\mu_0 \lambda_L^2}\mathbf{B}, \tag{3.15}$$

$$\frac{\partial \mathbf{j}}{\partial t} = \frac{1}{\mu_0 \lambda_L^2}\mathbf{E}, \tag{3.16}$$

where the former represents the Meissner effect and the latter perfect conductivity. An equivalent of the former in terms of the magnetic vector potential,

$$\mathbf{B} = \nabla \times \mathbf{A}, \tag{3.17}$$

can be written as

$$\mathbf{j} = -\frac{1}{\mu_0 \lambda_L^2}\mathbf{A}. \tag{3.18}$$

Exercise 10

Calculate λ_L using a typical value of n for metals, and compare with the measured penetration depth as reported in Fig. 12. Obtain temperature dependence of n by combining an empirical λ from Eq. (2.5) and the λ_L as defined in Eq. (3.8), and guess its implication.

3.3 Nonlocal response and alternating currents

So far we have implicitly assumed two things. One is that a response of an electron to a field is local, and the other is to consider a constant or DC currents only. Here let us have a quick look into the other cases of nonlocal response and AC currents.

It was Pippard who generalised the London equation to take into account the effect that a current at position $\mathbf{r}$ is affected by the field nearby at $\mathbf{r}'$:

$$\mathbf{j}(\mathbf{r}) = -\frac{ne^2}{m_e}\frac{3}{4\pi\xi_0}\int \frac{\mathbf{R}\left(\mathbf{R}\cdot\mathbf{A}\left(\mathbf{r}'\right)\right)}{R^4} e^{-R/\xi} d\mathbf{r}' \tag{3.19}$$

where $\mathbf{R} = \mathbf{r} - \mathbf{r}'$ and

$$\frac{1}{\xi} = \frac{1}{\xi_0} + \frac{1}{l}. \tag{3.20}$$

This equation tells that a current at one position depends on $\mathbf{A}$ throughout a volume of radius $\sim \xi$, where ξ_0 is called the Pippard coherence length and l is the mean free path of the electrons at the Fermi surface; for clearn superconductors $\xi = \xi_0$. Pippard argued that the characteristic dimension ξ_0, or a spatial extent of the superconducting electron wavepacket, could be estimated from the uncertainty principle, $\Delta x \Delta p \gtrsim \hbar$. We may consider that the electrons within $k_B T_c$ of the Fermi energy would play a major role, from which we estimate the spread in momentum as $\Delta p \approx k_B T_c / v_F$, where V_F is the Fermi velocity. Taking the spread in position as the characteristic dimension, we get

$$\xi_0 \sim \frac{\hbar v_F}{k_B T_c}. \tag{3.21}$$

We will not make any direct use of the Pippard's nonlocal equation in this module, but the effects will be incorporated into the microscopic description later in Part III where the meaning of ξ_0 will become clearer.

Exercise 11

Estimate some typical values from Eq. (3.21) and compare with the other lengths scales such as λ, l and interatomic distance.

Now let us turn our attention to the issue of DC vs AC currents. To understand their difference, it is useful to suppose that in a superconductor there are *super*-electrons and *normal* electrons, where the former experience no scattering and is responsible for superconducting properties. The motivation for such division comes from the experimental observations of i) the T dependence of λ and ii) a much sharper drop of ρ when cooled below T_c. As λ_L depends on n, the former implies that the charge density responsible for carrying currents is T dependent; n increases as temperature cools below T_c. Since the total available conduction electron density wouldn't change with temperature, this suggests that there might be a division of electrons which participate in super-conduction and which do not. It could be consistent with the latter as well. Just below T_c, where there are only a small number of sueprelectrons, the currents would be dominated by those resistanceless superelectrons rather than the remaining resistive normal electrons, which results in $\rho = 0$; analogous to a short-circuit. This divisional picture is called the *two fluid model*, which is simple and useful (more exact picture in Part III), and will be supported by other experiments in later sections.

London's Eq. (3.15) tells us that $E = 0$ for constant current carried by superelectrons. As there is no E-field, no drive for normal electrons to move, which reaffirms the currents are carried by superelectrons only. The story goes somewhat differently with AC. We notice that Eq. (3.15) also tells us that changing currents results in finite E-field. This now drives normal electrons also carry the currents. The superelectrons would give rise to inductive impedance by its lagging effects due to inertial mass. However, theses effects, which are frequency dependent, are quite small, accounting for 10^{-8} of total at 1kHz.

4 Critical Fields

We discuss thermodynamics of superconductors in this and the next section. A useful approach to the problem could be to discuss *instability* of thermodynamic state or phase. Indeed you may recall that we have briefly mentioned that superconductivity is destroyed (i.e., the sample restores its resistance in normal metallic state) by

- raising temperature above T_c,
- flowing current larger than j_c or
- applying a magnetic field stronger than H_c.

The magnetic field-temperature $(H - T)$ phase diagram for a few elemental metals as shown in Fig. 15 marks as an example.

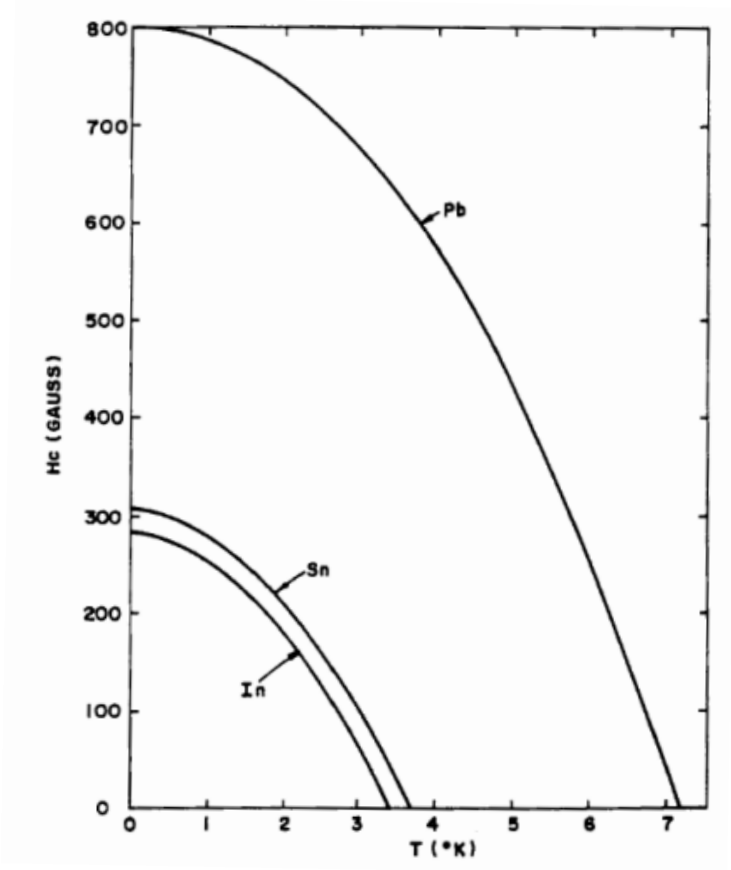


Figure 15. Magnetic field-temperature phase diagram of several elemental metals, Pb, Sn and In [5].

So a given material seems to superconduct only under certain set of conditions being satisfied, beyond which it becomes normal metallic. A natural question would if these conditions are interrelated or independent from one another. Figure 15 already suggest that they are related, then we may ask how.

4.1 Free energy

When one deals with the stability of matter, the *free energy minimum* is likely the first thing to consider. As normal metals support Pauli paramagnetism which is very weak, Gibbs free energy in normal metallic state, $G_n(T, H)$, is presumed to be nearly field independent. On the other hand, we already know that a superconductor does not like a field, it ceases to remain superconducting for a sufficiently large field. So we presume that the Gibbs free energy in superconducting state, $G_s(T, H)$, increases with the increasing field. We may draw tentatively a diagram as shown in Fig. 16, where G_s remains lower than G_n up to the field H_c such that the equilibirum state is superconducting. For $H > H_c$, however, G_n takes over G_s for having lower energy and the equilibrium state becomes a normal metal.

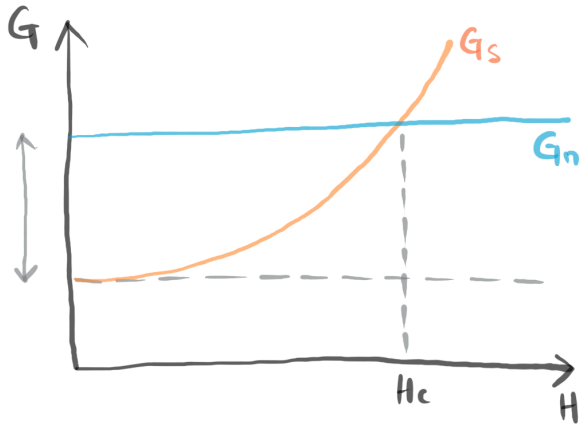


Figure 16. Schematic drawing of Gibbs free energy as a function of applied magnetic field, for normal metallic and superconducting states. Arrow represents the energy gain of the superconducting state relative to the normal state.

Now let's find out an actual functional form of G_s shown in Fig. 16. For a given temperature ($dT = 0$),

$$G_s(T,H) - G_s(T,0) = \int_0^H dG \tag{4.1}$$

$$= -\int_0^H \mu_0 \mathbf{M} \cdot d\mathbf{H} \tag{4.2}$$

$$= \int_0^H \mu_0 H dH \tag{4.3}$$

$$= \frac{\mu_0}{2} H^2, \tag{4.4}$$

where to reach the third line the Meissner effect was used[8]. For a metal at given tempera-

[8]For the total Gibbs energy G, one should multiply $\mathbf{M}$ by the volume V. In this lecture notes, the volume was set to unity for brevity.

ture,

$$G_n(T,H) - G_n(T,0) = -\int_0^H \mu_0 \mathbf{M} \cdot d\mathbf{H} \tag{4.5}$$

$$= -\frac{\mu_0}{2}\chi_P H^2, \tag{4.6}$$

where χ_P is the Pauli paramagnetic susceptibility, i.e., $\mathbf{M} = \chi_P \mathbf{H}$.

We know that $G_s = G_n$ at $H = H_c$. The free energy difference at $H = H_c$ can be written as

$$G_s(T,H_c) - G_s(T,0) - [G_n(T,H_c) - G_n(T,0)] = \frac{\mu_0}{2}H_c^2 + \frac{\mu_0}{2}\chi_P H_c^2. \tag{4.7}$$

After cancelling the equal terms, one obtains the free energy difference at $H = 0$,

$$G_s(T,0) - G_n(T,0) = -\frac{\mu_0}{2}\left(1+\chi_P\right)H_c^2 \tag{4.8}$$

$$\simeq -\frac{\mu_0}{2}H_c^2, \tag{4.9}$$

since $\chi_P \simeq 0$. The free energy difference is negative and thus indicates that the equilibrium state is superconducting. The size of the free energy difference (as shown by an arrow in Fig. 16) represents the stability of the superconducting state. Note that, to reach here, we have only made use of two experimental observations of $\mathbf{M} = -\mathbf{H}$ and $\chi_P \simeq 0$ other than basic thermodynamics. As the free energy form is obtained, other interesting quantities such as entropy and heat capacity can be derived, which is reserved for the next lecture.

Exercise 12

Equation (4.8) expresses the Gibbs free energy difference at zero field. Obtain the Gibbs free energy difference for a finite applied field.

Exercise 13

Draw graphs showing the applied field dependence of magnetic induction and magnetisation by plotting i) B against H_a and ii) M against H_a, respectively. Take into account the fact that a superconducting state undergoes a phase transition into a normal metallic state if the applied field strength exceeds H_c.

4.2 Critical currents

One can easily presume the existence of critical currents from the existence of critical fields, because the flowing currents would accompany a magnetic field surrounding. Let's consider a long wire with the diameter a carrying a supercurrent I. If no other field is applied, the magnetic field due to the current arises along the circular direction to the wire, and its intensity is given by $H = I/2\pi a$ (left in Fig. 17), from which a critical current can be obtained in terms of the critical field.

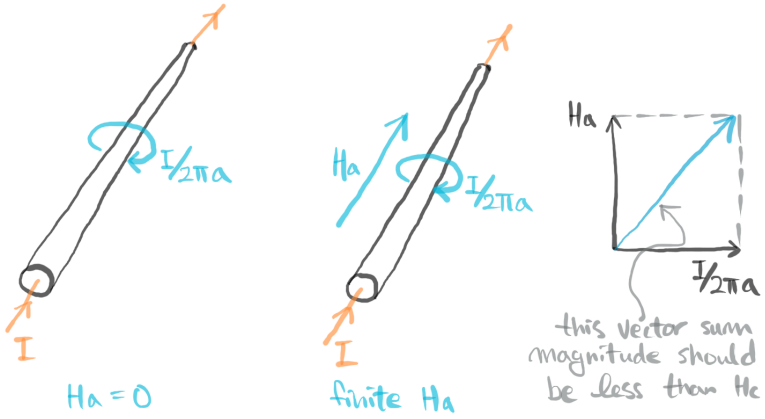


Figure 17. Currents and fields for a superconducting wire.

One can imagine that I_c will reduce by applying a magnetic field. For instance, if we apply the field along the wire direction, the critical current will follow $H_c^2 = H_a^2 + (I_c/2\pi a)^2$ (Fig. 17 on the right). This is consistent with that no supercurrent would flow at H_c, as one can draw the $I_c - H_a$ diagram as shown in Fig. 18.

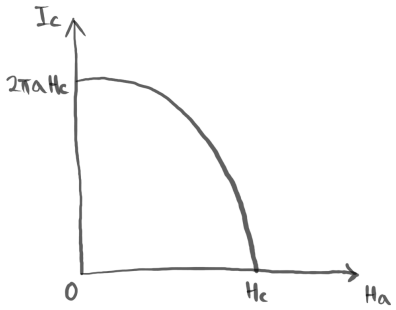


Figure 18. Critical current vs critical field diagram.

5 Transition

We have obtained the expression for the free energy G_s of superconductors in terms of the critical magnetic field H_c that is an experimentally observable quantity. For instance, you cool the sample to the lowest possible temperature, close to the absolute zero, and then increase the applied magnetic field strength H_a until the signatures of superconducting state, that is zero resistance and the Meissner effect, disappear; the sample becomes normal metallic[9]. The corresponding field strength is called the critical field strength H_c.

As you may recall, H_c is a function of temperature, *i.e.*, $H_c(T)$. The trend is observed from the field-temperature $(H-T)$ phase diagram, as shown in Fig. 15, for example; here lines correspond to the $H_c(T)$ curve that separates the low-T low-H superconducting state from the high-T high-H normal state. As temperature is raised from zero, H_c is decreased. And eventually, as temperature reaches T_c, H_c becomes zero; at T_c, one does not need an external field to drive the sample into the normal state.

In this section, starting from the free energy expression, we will deduce the expressions for the entropy S and heat capacity C of a superconductor. Note that the latter is particularly useful since measuring heat capacity is a common practice in any solid state physics laboratory. Measuring heat capacity while varying T and/or H_a, *i.e.*, $C(T, H_a)$, and its comparison with theoretical development, will give us insights into the thermodynamic nature of the superconducting states.

The entropy and heat capacity expressions can be obtained by making use of the following thermodynamic relations,

$$S = -\frac{dG}{dT}, \tag{5.1}$$

$$C = T\frac{dS}{dT}. \tag{5.2}$$

One may notice from the above expressions that the entropy can also be determined experimentally by integrating the measured C/T.

5.1 Entropy

It is straightforward to derive the entropy for a superconducting state[10]:

$$S_s(T, H_a) = -\frac{dG_s(T, H_a)}{dT} \tag{5.3}$$

$$= -\frac{d}{dT}\left[G_n(T, H_a) - \frac{\mu_0}{2}\left(H_c^2 - H_a^2\right)\right] \tag{5.4}$$

$$\simeq -\frac{dG_n(T,0)}{dT} + \frac{d}{dT}\left(\frac{\mu_0}{2}H_c^2\right) - \cancelto{0}{\frac{d}{dT}\left(\frac{\mu_0}{2}H_a^2\right)} \tag{5.5}$$

$$= S_n(T,0) + \mu_0 H_c\left(\frac{dH_c}{dT}\right). \tag{5.6}$$

[9] For brevity, by saying *normal* from now on, we refer to normal metallic state of a superconductor. By saying a superconductor, we refer to a material that would enter a superconducting state for $T < T_c$ such that 'a normal state of a superconductor' could make sense.

[10] The second equality in the following can be obtained from solving Exercise 12.

In the above, we have used the facts that G_n is almost field independent, *i.e.*, $G_n(H_a) \simeq G_n(H_a = 0)$, and that the field we apply should be temperature independent, *i.e.*, $dH_a/dT = 0$.

This expression immediately tells us a very important fact about the superconducting state, that is, the superconducting state is more *ordered* than the normal one. How is it so? Because the entropy of a superconducting state is always lower than (if not equal) that of a normal state. This can be seen by referring to the experimental observation. Look at the last term from the above: H_c should be positive (if not zero), but dH_c/dT is always negative (if not zero) as can be seen from the $H-T$ phase diagram, *e.g.*, Fig. 15. To recap, $S_s \leq S_n$ as $dH_c/dT \leq 0$.

Exercise 14

What would you expect for thermal transport, rather than electrical, to be like in superconducting and normal metallic states?

We can draw some further insights like the followings:

- at $T = T_c$, $S_s = S_n$. This can be seen from $H_c(T = T_c) = 0$,
- at $T = 0$, also $S_s = S_n$ because of the 3rd law of thermodynamics. This leads to $dH_c/dT|_{T=0} = 0$, which agrees with the experimental observation in the phase diagram (Fig. 15).

These already allow us to draw the $S(T)$ graph as shown in Fig. 19 (upper). That the entropy changes continuously across the transition implies no latent heat as $L = T\mathrm{d}S$. Thus a zero-field phase transition at $T_c(H_a = 0)$ is second-order. On the other hand, when a magnetic field is applied, the entropy graph as shown in Fig. 19 remains unchanged[11] while $T_c(H_a)$ gets lowered. Therefore, any normal-to-superconducting transitions occurring in the presence of applied magnetic field, *i.e.*, $T_c(H_a \neq 0) < T_c(H_a = 0)$ accompany a finite change in the entropy and thus latent heat: therefore they are first-order transitions.

5.2 Heat capacity

Before deriving the actual expression, one can already tell something about the heat capacity from the entropy results of the previous subsection. Looking into Fig. 19 (upper), that there is a slope change of $S(T)$ at T_c implies a jump in the heat capacity across that temperature. To this end, one would expect the $C(T)$ graph as shown in Fig. 19 (lower). Again, it is

[11] Note that the entropy for both a superconducting and a normal state do not have explicit dependence on H_a.

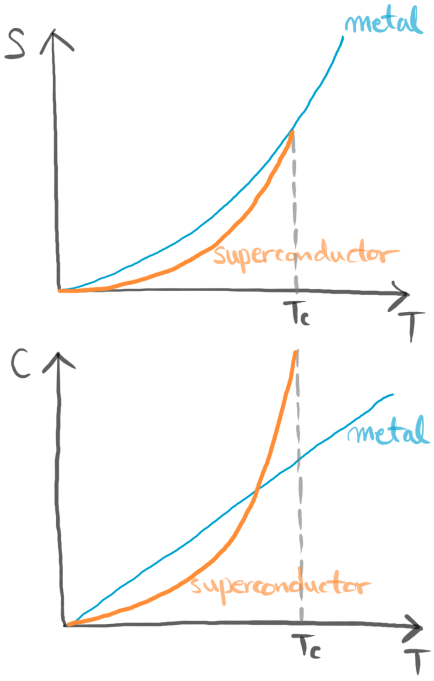


Figure 19. Entropy (upper) and corresponding heat capacity (lower) as a function of temperature.

straightforward to deduce the heat capacity expression.

$$\frac{C_s}{T} - \frac{C_n}{T} = \frac{d}{dT}(S_s - S_n) \tag{5.7}$$

$$= \frac{d}{dT}\left(\mu_0 H_c \frac{dH_c}{dT}\right) \tag{5.8}$$

$$= \mu_0 \left(\frac{dH_c}{dT}\right)^2 + \mu_0 H_c \frac{d^2 H_c}{dT^2}. \tag{5.9}$$

Let's check what this equation gives us at the transition. Since $H_c = 0$ at $T = T_c$, the second term on the right hand side of the above equation vanishes, *i.e.*,

$$C_s(T_c) - C_n(T_c) = \mu_0 T_c \left(\frac{dH_c}{dT}\right)^2. \tag{5.10}$$

This agrees with our expectation for the heat capacity jump at $T = T_c$ since if you look into the right hand side of the above equation no term equals zero thus remains finite.

We can already make comparisons of the above expectations with the observations, whatever comes first. However, before doing that, let us take a step further to make things more quantitative. To do so, we will make another use of the experimental observation of the $H - T$ phase diagram. To be quantitative, we adopt the empirical formula

$$H_c(T) = H_c(0)\left[1 - \left(\frac{T}{T_c}\right)^2\right], \tag{5.11}$$

which is known to match well with the experimental phase diagram. This expression allows us to obtain dH_c/dT which can be plugged back into Eq. (5.10). Then we obtain

$$C_s(T_c) - C_n(T_c) = 4T_c\mu_0 \left(\frac{H_c(0)}{T_c}\right)^2 \tag{5.12}$$

$$\equiv 2\gamma T_c. \tag{5.13}$$

This is very informative. Note that the terms on the right hand side, T_c and $H_c(0)$, are standard measurable quantities. So we can readily check if our thermodynamic description of the superconductors is correct or not from the experiments, which will come shortly after sorting out the followings.

You may notice by now that we could have used Eq. (5.11) from the very beginning so that our descriptions of the entropy and heat capacity be made more quantitative and thus with more predictive power. This will be left for you as exercise, but you may easily obtain

$$S_s - S_n = -2\mu_0 \left(\frac{H_c(0)}{T_c}\right)^2 \left[1 - \left(\frac{T}{T_c}\right)^2\right] T, \tag{5.14}$$

and

$$\frac{C_s}{T} - \frac{C_n}{T} = -2\mu_0 \left(\frac{H_c(0)}{T_c}\right)^2 \left[1 - 3\left(\frac{T}{T_c}\right)^2\right], \tag{5.15}$$

which directly reveals the temperature dependence of S and C (rather than at a single temperature T_c as done before) along with fixed parameters that can be directly obtained by experimental measurements, the critical field $H_c(0)$ and the critical temperature T_c. This would allow direct comparison with the measured experimental data.

Exercise 15

Derive the expressions in Eqs. (5.14) and (5.15).

Lastly, one more input from experiments would help. It is that $C_s/T \to 0$ as $T \to 0$. See, for example, Fig. 21. This observation allows us to separate out the superconducting and normal electron contributions. One can write

$$\frac{C_n}{T} = 2\mu_0 \left(\frac{H_c(0)}{T_c}\right)^2 \tag{5.16}$$

$$= \gamma \tag{5.17}$$

and likewise

$$\frac{C_s}{T} = 3 \times 2\mu_0 \left(\frac{H_c(0)}{T_c}\right)^2 \left(\frac{T}{T_c}\right)^2 \tag{5.18}$$

$$= 3\gamma \left(\frac{T}{T_c}\right)^2. \tag{5.19}$$

By carefully comparing the above two equations, you may notice that it would be helpful to plot $C/\gamma T$ against $(T/T_c)^2$ as shown in Fig. 20. In such a plot, the experimental data

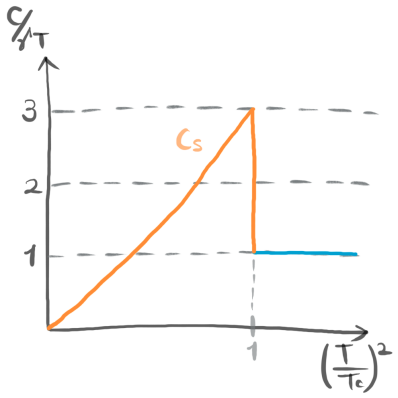


Figure 20. Schematic drawing of the temperature dependence of the heat capacity when making a transition from normal metal to superconductor.

should sit on straight lines which makes comparison between the theory and experiment a lot easier.

Now, finally, let's compare our derivation with the actual experimental data. Figure 21 shows the molar heat capacity of Aluminum (left) Gallium (right) as a function of temperature in zero and some finite fields[12]. In zero field, both materials are superconductors below their respective T_c. The applied field of 300 Gauss for Al and 0.02 T of Ga are strong enough to destroy the superconductivity and restores a normal state. For Al, the T-linear behaviour of C in the field is as expected for a normal metal. Upon cooling in zero field, the heat capacity of Al shows a jump across the transition, and then strongly suppressed as temperature is further lowered. A more quantitative comparison is made possible with the C/T vs T^2 plots for Ga. A near flat line for Ga in the field indicates normal metallic behaviour, and a near linear behaviour of C/T vs T^2 indicates a superconducting state. Moreover, check that the jump of the heat capacity is about twice the value of the normal state heat capacity, as we predicted!

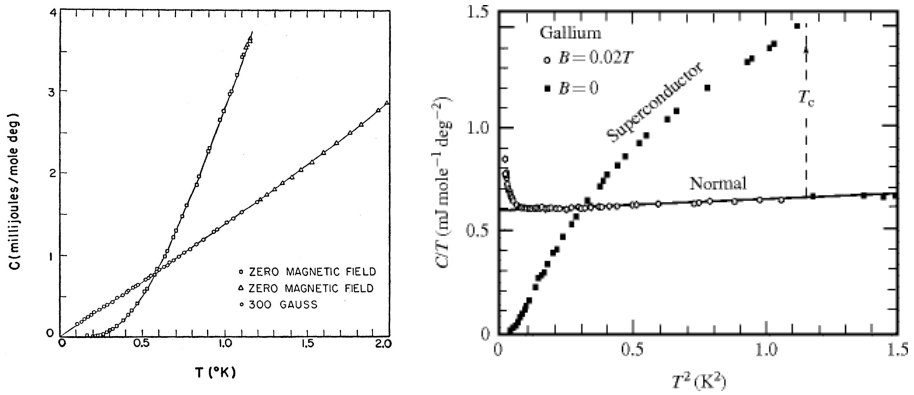


Figure 21. Temperature dependence of specific heat of Al (left [6]) and Ga (right [7]) in zero and finite fields.

[12]Please note that the lattice contribution to the heat capacity is already subtracted out.

So, this was a quite satisfactory (I hope) journey today. Of course, careful students might have noticed some deviation of the data from our expectation. A small upturn at very low T for C_n of Gallium is due to small amounts of magnetic impurities. On the other hand, a stronger than expected suppression of C_s/T vs T^2 points at something more significant, and there are other kinds of measurements (e.g., microwave absorption) that align with this observation which we will go through later. In any case, we should wait until the development of microscopic description for their explanation.

6 Geometry Matters

Over the last few lectures, we have developed electrodynamic and thermodynamic descriptions of a superconductor. You may agree that the procedure was quite simple as we based our development on a few key experimental observations along with a handful of Maxwell's equations and thermodynamic relations.

Here we will touch upon a bit different aspect related to the superconductivity, which may look like adding some bits of complication (which is true). I should confess that so far we have neglected an important aspect of magnetism in condensed matter, that is, demagnetization. You may recall this from your Y2 electromagnetism module, where concepts of demagnetisation field was introduced following a parallel discussion of depolarisation in dielectric materials. If you don't recall, that's fine, we'll quickly go through it. The bottom line is that a magnetic field strength inside a magnet is reduced from the value applied.

Have a look at the schematic drawing of the circumstance in Fig. 22. The phenomenon is quite analogous to the electric dipoles of a dielectric in an electric field. For a paramagnetic substance, as one applies a magnetic field $\mathbf{H}_a$, the material becomes magnetised meaning that the magnetic dipoles are aligned along the field direction,. This gives rise to the internal demagnetising field $\mathbf{H}'$ in the opposite direction to the applied field,

$$\mathbf{H}_i = \mathbf{H}_a - \mathbf{H}'. \tag{6.1}$$

Thus, for a paramagnet, the internal field strength $\mathbf{H}_i$ is reduced. This effect should clearly depend on the shape and geometry of the sample. For instance, for a infinitely long rod, the demagnetising field becomes zero as magnetic charges are separated in infinite distance. On the other hand, for a very thin sheet of material with the field applied to the normal direction, the demagnetising field would become largest due to to closely placed magnetic charges.

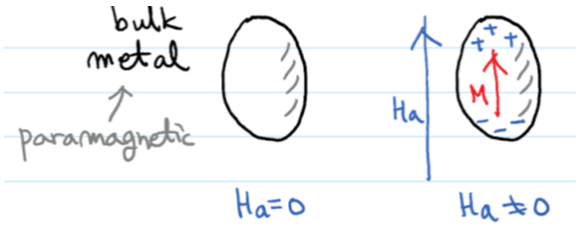


Figure 22. Material response to a magnetic field.

The demagnetising field should be proportional to the magnetisation, $H' \propto M$, and it arises from the aligned magnetic dipoles inside the material. The effect should be therefore very small for ordinary metals, as the Pauli paramagnetic susceptibiltiy is very small, $M \simeq 0$ as $\chi_P \simeq 0$. On the other hand, the effect should be large for strongly magnetic materials like a ferromagnet. We know that a superconductor is perfectly diamagnetic. So one would expect the largest possible demagnetising field, but in the way of increasing the applied field strength unlike the paramagnet. Apparently, such a positive demagnetising field will help a superconductor make a transition to a normal state with an applied field strength

(much) less than that of the critical field. Put otherwise, the critical field strength would be generally reduced from the value determined with the infinite long rod-shaped sample. Since the demagnetizing field is sample shape dependent, so is the critical field. That we have not been bothered by these things implies that we have so far implicitly assumed an infinitely long rod-shaped sample to ignore the demagnetisation effects.

These may all look like some added complication with little fundamental importance. However, by the end of the module we will be introduced to a state where superconducting and normal states coexist hands in hands forming a diversity of geometric patterns in micron scales. The reason behind such pattern formation relates to the interface energy between the superconducting and normal states, which in turn relates to the length scales in a superconductor. So, there lies certain fundamental aspects in all these things, which will be revealed in later sections.

In the followings, let's first examine the demagnetization effects in general solids, then move on to the case with superconductors. We will then look into a coexistence state.

6.1 Demagnetisation in solids

The demagnetizing field $\mathbf{H}'$ is generally position dependent in terms of both intensity and direction. However, such complexity does not exist for an ellipsoid described by

$$\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 + \left(\frac{z}{c}\right)^2 = 1, \tag{6.2}$$

where a, b and c represent the sample dimension along the principal axes. For an ellipsoid, $\mathbf{H}'$ is uniform and its direction is parallel to the applied field. An additional advantage is that it can be easily generalised to mimic useful cases such as an infinite rod ($c \to \infty$) or a plate ($a, b \to \infty$).

For an ellipsoid, then, the vector Eq. (6.1) is reduced to a scalar expression as,

$$H_i = H_a - nM, \tag{6.3}$$

where n is the so-called demagnetisation factor, $0 \leq n \leq 1$. Figure 23 shows how n changes depending on the parameters for an ellipsoid.

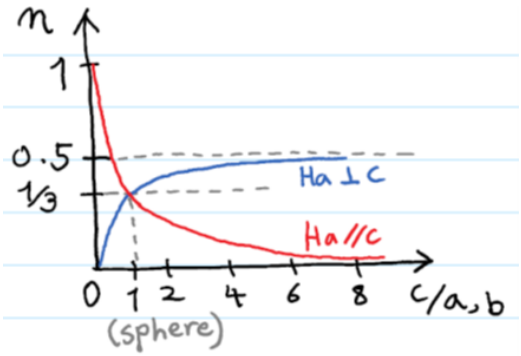


Figure 23. Demagnetisation factor of an ellipsoid as a function of the ratio of the principal axes lengths.

Red line in the figure represents the case where a magnetic field is applied along the z axis, while blue line represent the field transverse to the z axis. You can see from the red line that n is maximum of 1 as $c/a \to 0$ or $c/b \to 0$ which applies to the limit of thin films. The value decreases toward zero to the limit of $c/a \to \infty$ or $c/b \to \infty$ which corresponds to an infinitely long thin wire. For the blue line, n vanishes for $c/a \to 0$ or $c/b \to 0$ which is equivalent to the case where the field is applied along an infinitely long thin wire. As c/a or c/b increases such that the field is applied transverse to an infinitely long wire, n saturates toward 0.5. A special point at $a = b = c$ corresponding to a sphere has $n = 1/3$.

6.2 Demagnetisation in superconductors

For a superconductor, Eq. (6.3) becomes

$$H_i = H_a - nM \tag{6.4}$$
$$= H_a + nH_i, \tag{6.5}$$

owing to the Meissner effect $M = -H_i$. By rearranging, one obtains

$$H_i = \frac{1}{1-n} H_a. \tag{6.6}$$

It is clear that $H_i \geq H_a$ as $0 \leq n \leq 1$ By looking into Fig. 23, $H_i = 1.5H_a$ for a sphere while even $H_i = 2H_a$ for a long rod in a transverse field. What does this imply? It implies that internally the critical field strength could be reached even though the externally applied field is still far less than H_c. Again, this may look like a simple numerical adjustment, but actually the consequence is quite profound as we will see shortly.

Exercise 16

Draw the I_c vs. H_a graph as shown in Fig. 18 for the case where a field is applied in transverse direction to the wire.

6.3 Coexistence

It seems logical to presume that once the internal field reaches the critical field strength $H_i = H_c$ though $H_a = (1-n)H_c < H_c$, a superconductor would make a transition to a normal state . However, this leads to a contradictory situation:

- As soon as the superconducting state becomes normal metallic, the material loses its perfect diamangetism and becomes a Pauli paramagnet such that $H_i \simeq H_a$. Meanwhile, since the applied field $H_a = (1-n)H_c$ is stil smaller than H_c, one is led to the situation of $H_i < H_c$. That is, the internal field has not yet reached to the critical value, thus superconductivity should have persisted.

- Of course, this again restores perfect diamagnetism such that the internal field reaches the critical value destroying superconductivity again. So we're stuck.

How can we settle with this frustrating situation? The resolution comes by allowing coexistence of superconducting and normal states for $H_c' \leq H_a \leq H_c$, where $H_c' = (1-n)H_c$ which also accompanies emergence of the interfaces between the superconducting and normal states.

For the two states to coexist, the first question arising is how their spatial configuration would be. This is not at all easy to answer. However, still we may be able to look into some basic aspects such as how the interface between the two states would orient with respect to the applied field. We can learn about this by imposing the boundary conditions known from electromagnetism:

1. $B_\perp$ be continuous across the interface: Since $B = 0$ inside a superconductor, the interface should align parallel to the applied field for $B_\perp$ remain zero on the metallic side of the interface.

2. $H_\parallel$ be continuous across the interface: At the interface, $H_i = H_c$ to maintain equilibrium, since otherwise one region would grow over the other. For instance, if $H_i > H_c$ having normal states more would gain energy, while superconducting states would be favoured for $H_i < H_c$. Thus, at the very interface and throughout the sample, $H_i = H_c$.

Taking into account the above arguments, one can draw magnetic flux density for a superconducting slab as shown in Fig. 24.

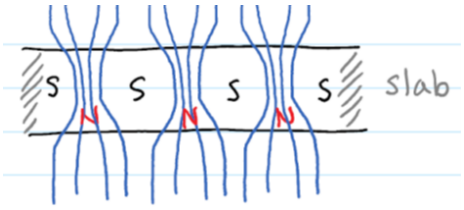


Figure 24. Schematic drawing of magnetic fields inside a superconducting slab.

Exercise 17

Draw schematically four graphs each plotting average values of internal field strength, normal metal volume fraction, magnetic flux density and magnetic moments as a function of the applied field strength.

The next questions would be how these interfaces should arrange themselves spatially, what are the respective volume fraction or how densely they would form, etc. As these are the question of minimising free energy, the result would depend on the applied field. This is generally a very complicated problem, and we will not attempt to solve for any particular model. Instead, we can get some flavour from the experimental observation. Figure 25 presents photographs taken for the surface of bulk superconductors in different applied magnetic fields. Notice typical size of the structure as large as millimeters, which is huge compared to the length scales in superconductors such as λ or ξ. What shall determine the size of the superconducting-normal structure?

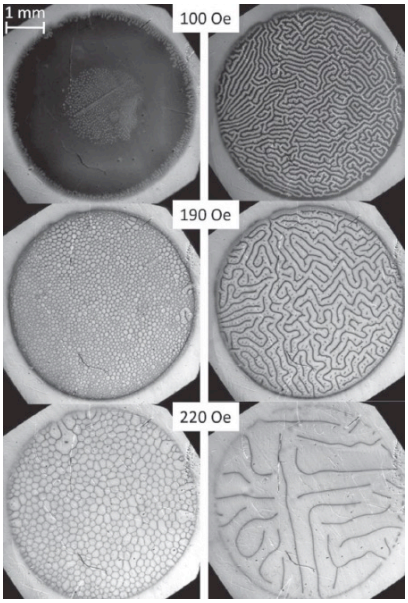


Figure 25. Magneto-optical images (using Faraday rotation) of an intermediate state of a Pb single crystal at 5 K in different magnetic fields [8]. Bright regions correspond to the normal state and dark regions to the superconducting state

6.4 Interface energy

One further element to naturally enter the problem is the interface or surface energy that arises due to the emergence of superconducting-normal interface. Should it be positive or negative? One hint may come from the mm sized structure of the interfaces. Suppose the interface energy is negative. Then, forming as many interfaces as allowed would decrease the total energy of the system. If the interface energy is positive, the system would favour less density of the interfaces with larger structure as observed, e.g., Fig. 25.

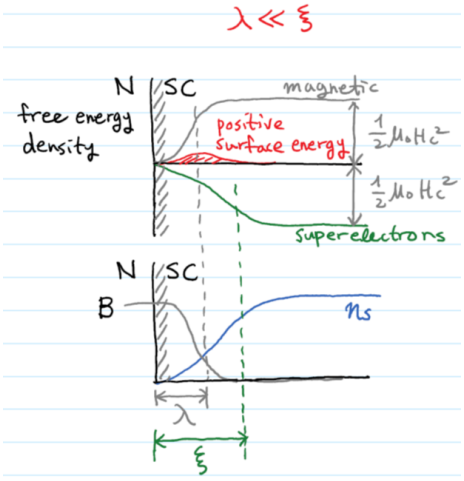


Figure 26. Magnetic flux and superconducting electron density profile across the normal metal-superconductor boundary

So, in which case would the surface energy become positive? See Fig. 26 where two different contributions to the free energy are drawn. One contribution comes from the energy gain due to superelectrons. The other comes from the energy cost to expel the magnetic flux. They may well follow different profile in space. The flux expulsion occurs beyond the length scale of the penetration depth λ. The density of superelectrons may follow a different length scale, the coherence length ξ. If $\lambda \ll \xi$, one could expect the positive surface energy. This may not be a totally satisfying explanation, but remember that we don't yet have a microscopic description. Still, we will elaborate on this idea further in the following section, to be able to distinguish superconductors into two totally different classes.

7 Type II

In the last section, we learned about the intermediate state, a form of inhomogeneous state of a superconductor where superconducting and normal states coexist. We have seen that such an inhomogeneous state arises due to a demagnetising field which is sample shape dependant. The coexisting normal-superconducting structure turned out to be quite large scaled compared to the characteristic length λ or ξ. To explain this, we have introduced the idea of positive surface energy and the corresponding condition $\lambda \ll \xi$.

In fact, there exists another type of inhomogenous state of a superconductor, called a mixed state, which is quite distinct from the intermediate state. For instance, it arises in the absence of demagnetising field, like in a infinitely long rod. The coexisting normal-superconducting structure has the same scale as λ and or ξ. Such differences can be explained by allowing a *negative* surface energy under the condition $\lambda \gg \xi$.

You can easily see the point by drawing a parallel graph as done in the last lecture. See Fig. 27) where the condensation energy gain and the magnetic energy cost are drawn for the $\lambda \gg \xi$ and $\lambda \ll \xi$ cases on the left and right, respectively.

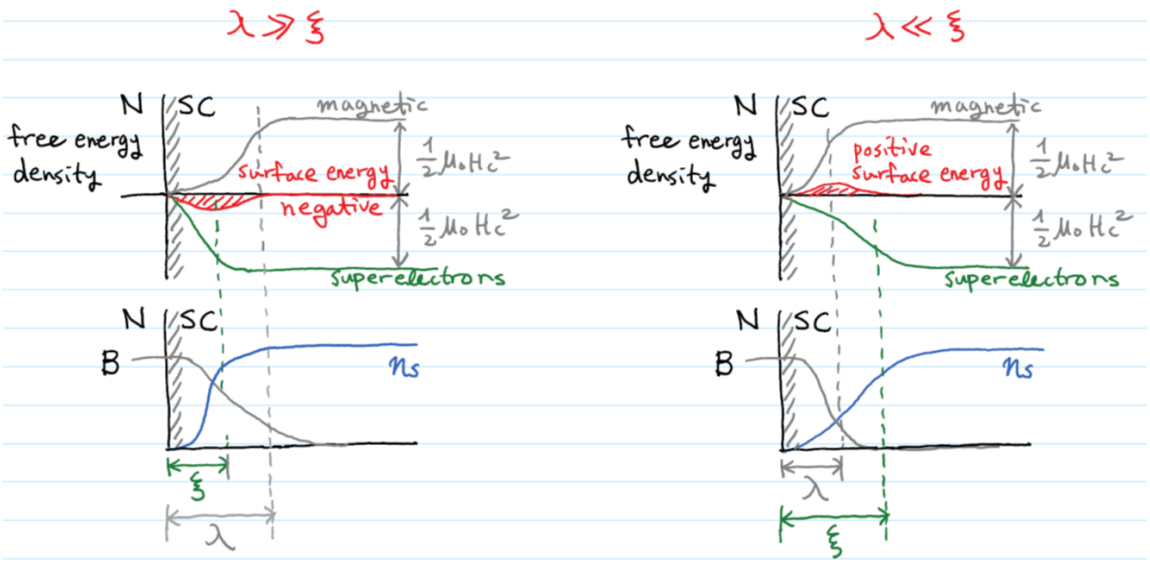


Figure 27. Magnetic flux and superconducting electron density profile across the normal metal-superconductor boundary

From now on, for simplicity, let's forget about the demagnetising field effect and an intermediate state. Then, all superconductors fall into one of the two categories of either hosting a mixed state or not. Such distinction could be made from early days by comparing their responses to an applied magnetic field. See for instance Fig. 28. In the left column, you find a familiar graphs of the $H - T$ phase diagram, B vs. H and M vs. H curves. That the magnetic flux B remains zero and the magnetisation M is linear with an applied magnetic field up to a critical strength of H_c represent perfect diamagnetism or Meissner effect. In the next column, on the other hand, you observe that a single crtical field H_c is

replaced by two critical fields of H_{c1} and H_{c2}. Looking into the B vs. H curve in the middle, you observe that B start increasing rather gradually from H_{c1} and follows the normal state behaviour from H_{c2} upward. So, the mixed state is characterised by imperfect Meissner effect. The behaviour shown in the first column is observed in pure elemental metals, while those in the second column observed in alloys or impure metals. So, in the early days, the mixed state behaviour was attributed to some dirty effects.

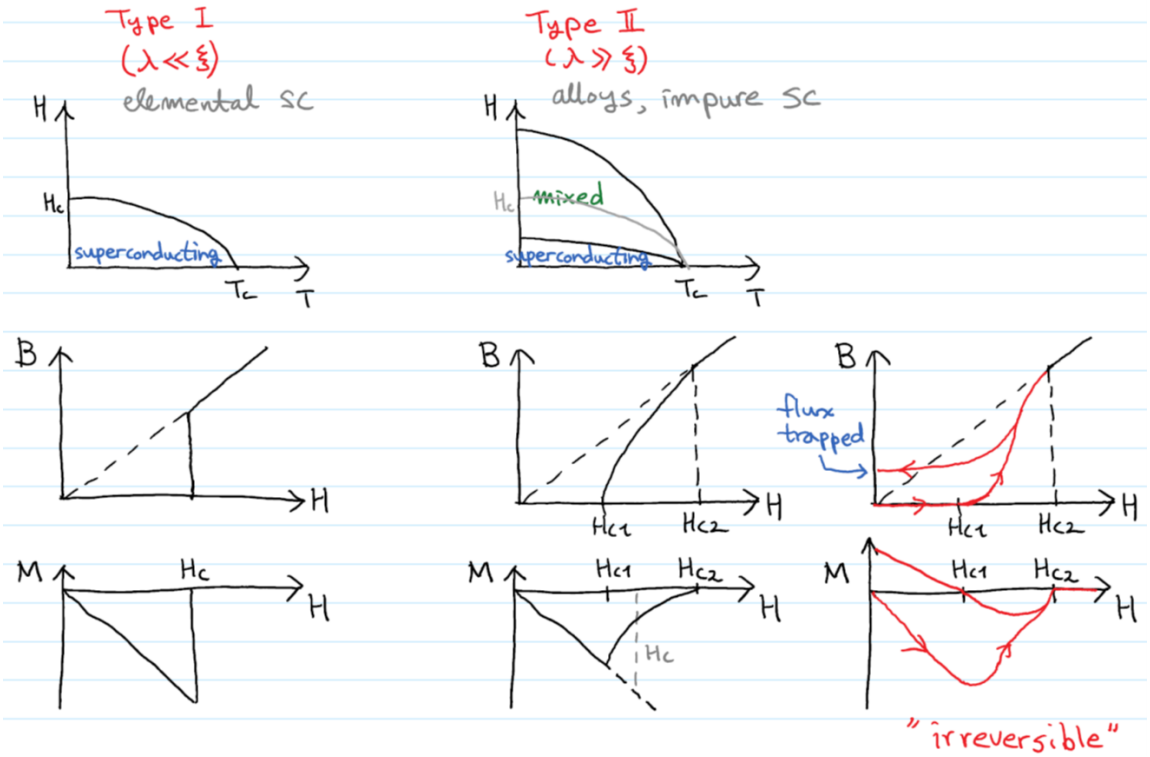


Figure 28. Magnetic field-temperature phase diagram and field responses of two different types of superconductors.

It was only much later when the idea of negative surface energy was put forward that people started formally distinguish between the two cases by calling one type-I (no mixed state; Meissner effect persists) and the other type-II (with mixed state) superconductors. Now let's look into the problem more microscopically as to the outcome of the negative surface energy.

7.1 Vortex

With having negative surface energy, the system would prefer to have as many normal-superconductor interfaces as possible as they would lower the total energy. In turn, to make the surface area maximal for a given volume respecting electromagnetic boundary conditions, a rod-like structure with the shrinking radius would be preferred. Based on these arguments along with Fig 27, one can draw a cross-section of such rod-like structure as shown in Fig. 29. Each rod-like structure consists of a normal-metallic core and the surrounding superconducting state with certain penetration of magnetic flux. The size of

the normal core is determined by ξ whereas the surround superconducting regions with penetration determined by λ.

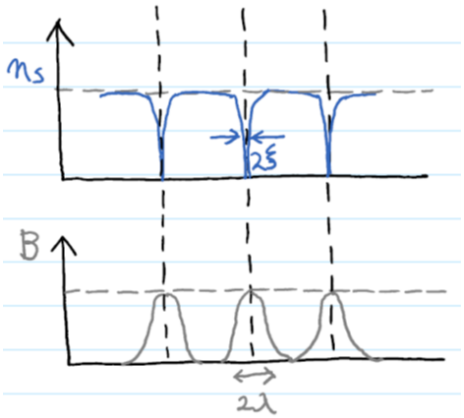


Figure 29. Schematic drawing of a flux lattice.

Figure 30 shows a perspective such structures. To maintain zero flux deep inside the superconductor, a supercurrent would circulate withing λ around the core. Such a structure is called a flux line or simply a vortex.

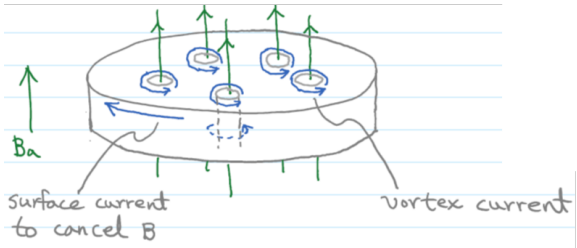


Figure 30. Schematic drawing of a flux lattice.

So, how would these vortices mount up? In fact, there is a repulsion between the vortices. It is evidenced by that the voritces, much like atoms in a crystal, form a periodic arrangement or flux lattice.

7.2 Flux lattice

The repulsive interaction between the vortices would stabilise certain flux lattice formation, and some of their experimental observations are shown in Fig. 31. Note that they show different materials in different environmental conditions.

Exercise 18

Search the papers cited in Fig. 31 and read them through to understand how different techniques are applied to image the mixed states.

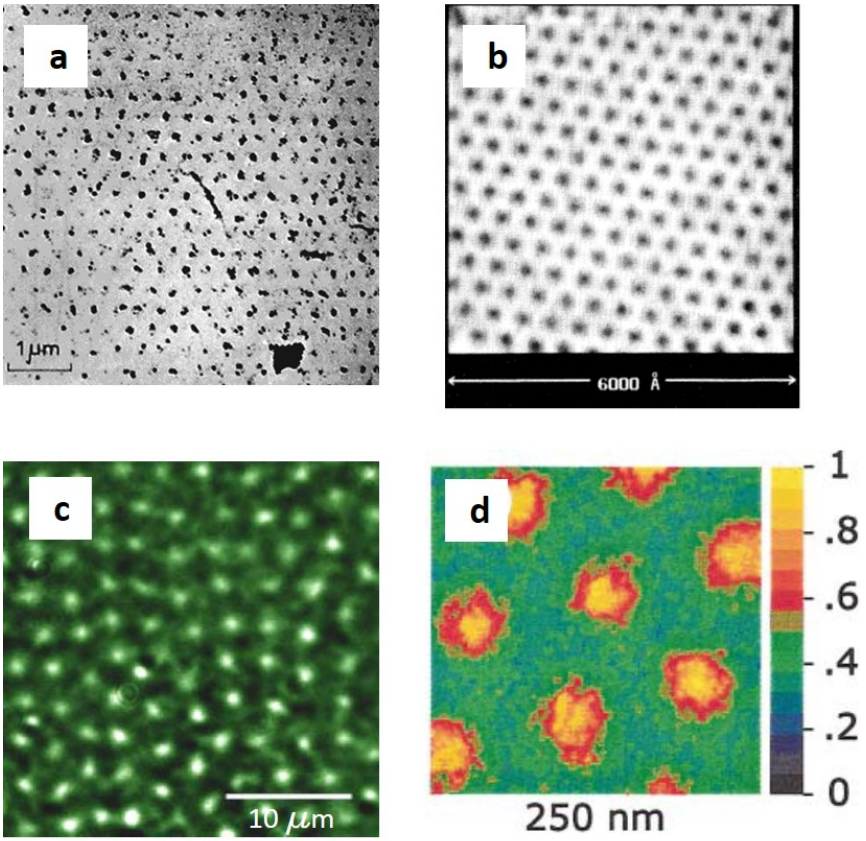


Figure 31. Flux lattices of mixed state in different materials. (a) Electron microscope image of Pb-In at 1.1 K in a field of 3 kOe [9] (b) Scanning tunneling microscopic image of $NbSe_2$ [10]. (c)Magneto-optical image of $NbSe_2$ at 4 K in earth field [11]. (d)Scanning tunneling spectroscopic eimage of MgB_2 at 2 K in a field of 0.05 T [12].

Fig. 31(a) shows one of the earliest observation and Fig. 31(b) the latest. You may notice how much in detail and closely one can observe the vortices by now. For instance, from the latter you can observe a clear triangular lattice and even estimate the size of the metal core and thus ξ.

7.3 Type I vs II

To recap, and to add a few more, for almost all *pure* elemental superconductors, they were observed to show

- a reversible first-order transition with a latent heat across H_c and
- an intermediate state.

These are called the type-I superconductors. On the other hand, superconducting *alloys* or impure metals were observed to

- contain trapped magnetic flux,

- show a large magnetic hysteresis,
- remain superconducting at fields much greater than the thermodynamic H_c one would expect from the heat capacity.

These are called the type-II superconductors. The large magnetic hysteresis is shown as red lines in the last column in Fig. 28, which are observed when there are defects in materials. The reason is because the vortices could be pinned to imperfections (e.g., dislocation or chemical impurities) such that they persist even after removing the externally applied field.

To properly discuss such a highly inhomogeneous state such as a mixed state, one needs to go beyond the framework we have developed so far in Part I which essentially applies to a homogeneous superconductor. We will instead need a theory that can accommodate spatially varying superelectron density. We will explore such a theory, called the Ginzburg-Landau theory[13], which is the subject of Part II, starting from the next section.

[13]Indeed, Abrikosov first worked out with the negative surface energy within the Ginzburg-Landau theory to show the existence of two types of superconductors along with detailed description of mixed state, vortices and flux lattice.

Part II
Ginzburg-Landau Theory

We develop a phenomenological theory of superconductivity using thermodynamic arguments and ideas from phase transitions.

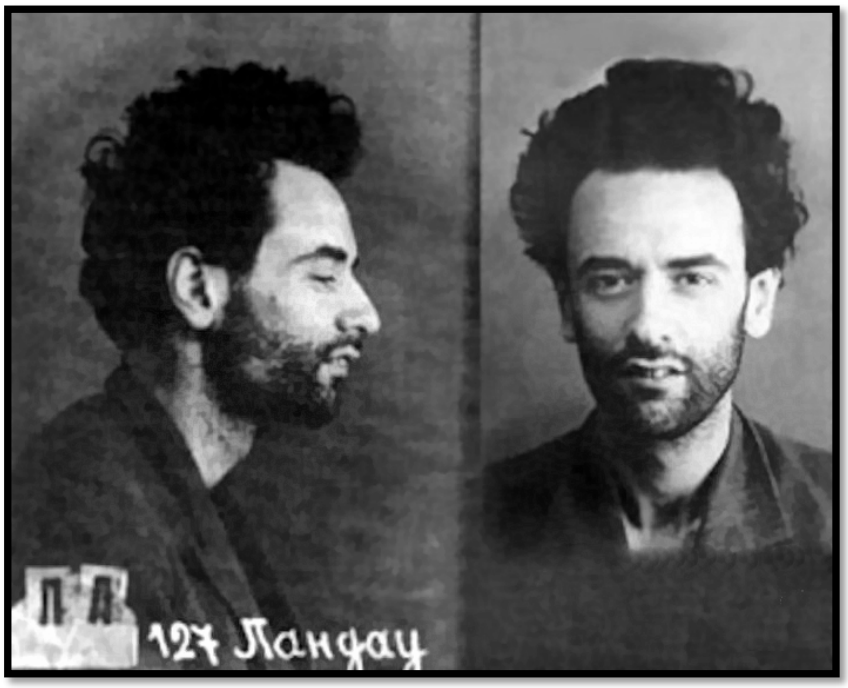


Figure 32. A mug shot of young Lev D. Landau in prison, 1938-9. What has happened? Landau developed masterful theories of metal, diamagnetism, superfluid, superconductor and phase transition, to name a few! (Photo from Wikipedia)

8 Order!

We learned from thermodynamic arguments that a superconducting state is a *ordered* state relative to its normal-metal counterpart. We also learned that a second-order phase transition separates the low-temperature superconducting state from high-temperature metal one, in the absence of applied magnetic field. Indeed, a second-order transition accompanies a continuous development from zero of what is called the order parameter as temperature is decreased across the transition temperature. One may then hope that a theory of phase transition could provide further insights into the superconducting phase.

Let's start from a familiar example of phase transitions. For example, recall a phase transition of a ferromagnet whose schematic phase diagram is shown in Fig. 33. At high temperatures, magnetic moments or electron spin moments located at ionic lattice would randomly fluctuate due to thermal agitations. Even though the neighbouring spins may interact with each other through magnetic interactions, they may be outweighed by thermal fluctuations at sufficiently high temperatures.

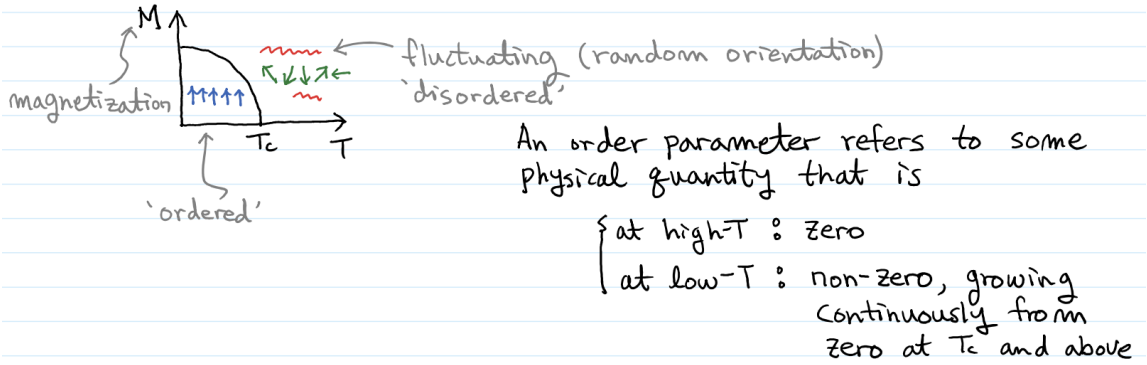


Figure 33. Schematic drawing of the temperature evolution of magnetization in a ferromagnet.

However, as temperature is lowered, the interaction energy between the spins may become comparable in scale to the thermal energy. These interactions may favour the neighbouring spins align along the same direction. As temperature is decreased further, at some point, the magnetic interaction could dominate over the thermal fluctuations such that all the spins cooperate to align along the same direction; the transition has occurred. This long-ranged alignment, which we may simply refer to as an order, manifests itself as the development of both local and bulk magnetization; such static magnetization wouldn't exist in the high-temperature *disordered* phase, where things are simple averaged out.

The story with a ferromagnet motivates us to define the quantity called an order parameter, the magnetization in the case of a ferromagnet, which is zero at high-temperature disordered phase while non-zero at low-temperature ordered phase. As shown Fig. 33, one may further constrain the definition of the order parameter which is zero at T_c and grows continuously as temperature is lowered. In an advanced term, such transition could also be said to have broken the symmetry across T_c. The broken symmetry in this particular case is a rotational one as at high temperature you wouldn't able to see a difference when you

rotate the material around, while at low temperature they all point along one direction so you can tell a difference when rotating the sample.

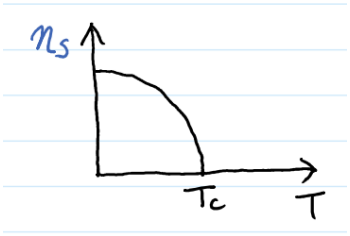


Figure 34. Schematic drawing of the temperature evolution of the superconducting electron density in a superconductor.

Now look at the Fig. 34 . It plots the superconducting electron density of a superconductor as a function of temperature. You can immediately notice a clear resemblance to the ferromagnet case, and should naturally ask if n_s could be taken as an order parameter. And, if so, what is the broken symmetry? The original inventor of the theory of second-order phase transition[14], Lev D. Landau, investigated this problem with his colleague Ginzburg.

8.1 Constructing free energy

Ginzburg and Landau considered the problem on a very general ground. They wanted to construct a whatever free energy so that, by minimising the free energy, they can deduce some useful thermodynamic quantities such as heat capacity. The key element to do so, they presumed, is to write down the free energy in terms of an order parameter. Of course, we do not have yet a clear idea what it should be or look like. But, that's fine. Recall that Schrodinger has written down his famous wave equation without understanding the meaning of the wavefunction; yet its solution for the hydrogen atom provided the correct energy spectrum and thus guided a full formulation of quantum mechanics.

So, we naively have an idea that the superconducting electron density n_s may represent, or at least related to, an order parameter. This motivates us to define a quantity that directly represents an order parameter, let it be refer to as ψ, whose squared value could be interpreted as n_s. And yes, this somewhat mimics the Born's interpretation of the Schrodinger wavefunction for a particle[15]. We may take a similar step forward by assuming a *complex* order parameter and write it down as $\psi = |\psi|e^{i\theta}$. To be considered as an order parameter $\psi(T) = 0$ for $T > T_c$ and $\psi(T) \neq 0$ for $T < T_c$. Nothing really prevents us from defining a complex order parameter, and after all such phase term attached used to play an important role in quantum mechanical problems.

Next, we assume that the free energy F depends on ψ smoothly. Or it may be better to use $|\psi|$ since the free energy itself must be real. Then, for $T \simeq T_c$ where $|\psi|$ should remains small, the difference in free energy density f between superconducting and normal

[14] Also called a continuous phase transition.
[15] Bear in mind that we are dealing with macroscopic number of particles.

state can be written with the help of Taylor series expansion as,

$$f_s - f_n = a_0 + a_1|\psi| + a_2|\psi|^2 + a_3|\psi|^3 + a_4|\psi|^4 + ..., \tag{8.1}$$

where f_s is the free energy density of a superconducting state and f_n that of a normal metal state and a_i's the coefficients. In writing the above expression, we have assumed the simplest cases, that is, we have ignored possibility of spatial variation of $\psi(x)$ and we have not considered an applied magnetic field. To restore these effects in the free energy form will be the subject of the next two lectures.

So, here we consider the above uniform case in the absence of an applied field. Since we'd expect to find the free energy minimum solution, we may remove the odd terms from the above as they will give an infinity. Neglecting higher order terms (because $|\psi|$ is small), we can rewrite the above as

$$f_s - f_n = a|\psi|^2 + \frac{1}{2}b|\psi|^4 + ... \tag{8.2}$$

Note that all quantities in the expression, f_s, f_n, ψ, a and b, could be temperature dependent.

Now we can implement our expectation with the order parameter that it is zero for $T > T_c$ while it grows as T decreases below T_c. For instance, see Fig. 35. For $T > T_c$, we'd expect that $f_s - f_n$ has a minimum at $|\psi| = 0$. That is, $f_s - f_n$ should be a parabolic function of $|\psi|$ with the origin at $|\psi| = 0$. For $T < T_c$, we expect the minimums of the wells are at some finite $|\psi|$. The simplest may be a double well form of $f_s - f_n$ as a function of $|\psi|$ as shown in Fig. 35.

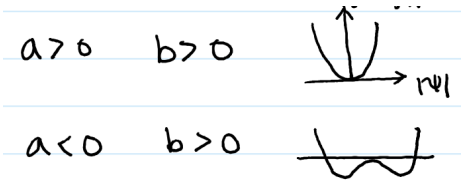


Figure 35. Free energy density difference expected for $T > T_c$ (left) and for $T < T_c$ (right).

The above expectation already constrains the signs of a and b, as shown in Fig. 35. To sum, one expects $b(T) > 0$ always. On the other hand, $a(T) > 0$ in disordered state with $T > T_c$ while $a(T) < 0$ in ordered state with $T < T_c$. If we really limit our attention close to the transition, since $a(T) = 0$ across T_c, one gets

$$a(T) \simeq \cancel{a} + \frac{da}{dT}(T - T_c) + ... \tag{8.3}$$

$$b(T) \simeq b + ... \tag{8.4}$$

that is, a is linear in temperature while b remains a constant. It may not be clear yet where all these efforts would lead us to, but let's continue to push to the limit if we can come up with more useful stuff in the followings.

8.2 Finding minimum

Let's first see what we can get when we try to minimise the free energy difference. Taking derivative with respect to $|\psi|$ and equating to zero,

$$\frac{d}{d|\psi|}\left(f_s - f_n\right) = 2a|\psi| + 2b|\psi|^3 \tag{8.5}$$

$$= 0, \tag{8.6}$$

gives us the condition for $|\psi|$ to get the minimum free energy difference. The solutions are $|\psi|^2 = -a/b$ or $|\psi| = 0$. Using Eq. (8.3), these solutions can be rewritten as

$$|\psi| = 0 \text{ for } T > T_c \tag{8.7}$$

$$|\psi| = \sqrt{\frac{1}{b}\frac{da}{dT}\left(T_c - T\right)} \text{ for } T < T_c. \tag{8.8}$$

Now we can plug the solutions back to Eq. (8.2) to obtain the expression for the minimum free energy difference. When in equilibrium, the free energy difference becomes

$$f_s - f_n|_{|\psi|^2 = -a/b} = -\frac{1}{2b}\left(\frac{da}{dT}\right)^2 \left(T_c - T\right)^2 . \tag{8.9}$$

We can summarise the results as in Fig. 36.

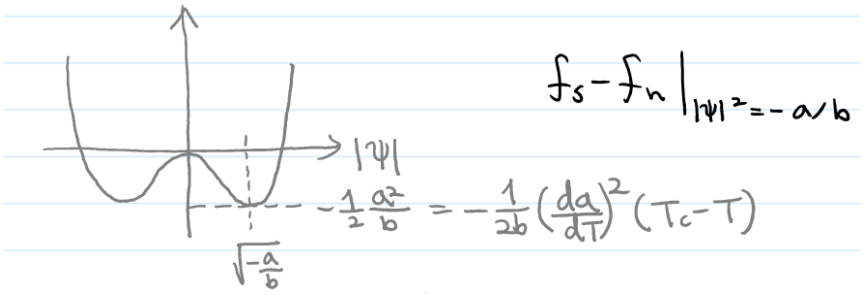


Figure 36. Schematic drawings for the solutions to finding minimum of the free energy difference.

8.3 Thermodynamics

Given the free energy expression, one can obtain the respective expressions for the entropy and heat capacity as well[16],

$$s_s - s_n = -\frac{d}{dT}\left(f_s - f_n\right) \tag{8.10}$$

$$= -\frac{1}{b}\left(\frac{da}{dT}\right)^2 \left(T_c - T\right), \tag{8.11}$$

[16] Since we have used the free energy density, we calculate here the entropy per volume and heat capacity per volume, denoted by lower case symbols

and

$$c_s - c_n = T\frac{d}{dT}(s_s - s_n) \tag{8.12}$$

$$= T\left(\frac{1}{b}\right)\left(\frac{da}{dT}\right)^2 \text{ for } T < T_c, \tag{8.13}$$

while $c_s - c_n = 0$ for $T > T_c$.

Exercise 19

Find the expression for $H_c(T)$ in terms of $a(T)$ and b, and argue for $(da/dT) > 0$.

All those expressions derived in this section may not appear very useful at first sight because we don't know what is a and b. However, note that the experimentally observable quantities are expressed in terms of the parameters a and b. If we find more and more observable quantities that can be expressed in terms of a and b, then we will be able to check the consistency of our theoretical framework by making comparisons between different quantities and also to determine a and b

9 Gradient

The Ginzburg-Landau free energy expression that we obtained in the last lecture is concerned with a sort of the simplest case: homogeneous and uniform in the absence of an applied magnetic field. We can easily extend the expression, however, to incorporate the situations where the superconducting electron density varies in space (e.g., near the metal-superconductor interface) or is affected by the applied magnetic field. The latter will be the subject of the next lecture, and here we focus on the former case of spatial inhomogeneity.

9.1 Spatial inhomogeneity

We have written down the following free energy density difference,

$$f_s - f_n = a|\psi|^2 + \frac{b}{2}|\psi|^4. \tag{9.1}$$

In order to account for spatial inhomogeneity, one may first wish to replace ψ with $\psi(\mathbf{r})$. Then one can add a term to represent the effects of the spatial inhomogeneity as

$$f_s - f_n = a|\psi(\mathbf{r})|^2 + \frac{b}{2}|\psi(\mathbf{r})|^4 + \frac{\hbar^2}{2m^*}|\nabla\psi(\mathbf{r})|^2, \tag{9.2}$$

where the coefficient $\hbar^2/2m^*$ was chosen to mimic the kinetic energy term in the Schrodinger equation and we will indeed consider the last term to represent the kinetic energy. By the way, nothing we kept only the squared term of $\nabla\psi$ for similar reason as before, that is, the free energy must be real and the higher order terms are presumed not important close to T_c where ψ is small.

Again, the free energy must be minimised to find the thermodynamic equilibrium state. However, unlike the uniform case, we now have a spatial variation of ψ(r), so the free energy density $f_s(T)$ must be first integrated over the whole space to find the total energy,

$$F_s(T) = F_n(T) + \int \left(a|\psi(\mathbf{r})|^2 + \frac{b}{2}|\psi(\mathbf{r})|^4 + \frac{\hbar^2}{2m^*}|\nabla\psi(\mathbf{r})|^2 \right) d^3r. \tag{9.3}$$

Now we can proceed to find the condition for having minimum free energy, i.e., $\delta F_s(T) = 0$, by imposing infinitesimal variation of $\psi(\mathbf{r})$, i.e., $\psi(\mathbf{r}) \to \psi(\mathbf{r}) + \delta\psi(\mathbf{r})$. That is, we are looking to the condition where $\delta F_s = 0$ for any $\delta\psi$(r). One then obtains the condition,

$$-\frac{\hbar^2}{2m^*}\nabla^2\psi + a\psi + b\psi|\psi|^2 = 0, \tag{9.4}$$

which indeed looks similar to the Schrodinger equation but only with a non-linear term. This is called the Ginzburg-Landau (GL) equation.

Exercise 20

Derive Eq. (9.4). While deriving, you may keep terms only linear in $\delta\psi$(r) ignoring higher order contribution.

9.2 Normal-superconductor interface: coherence length

Since the GL equation was motivated by having a situation where ψ varies in space, let's try to the equation in such a case. The simplest case we can consider is a metal-superconductor interface in one dimension. Suppose a normal metal for $x < 0$ and a superconductor for $x > 0$. The GL equation is rewritten as

$$-\frac{\hbar^2}{2m^*}\frac{d^2\psi(x)}{dx^2} + a\psi(x) + b\psi(x)|\psi(x)|^2 = 0 \text{ for } x > 0, \tag{9.5}$$

where $\psi(0) = 0$. In order to simplify the look of the equation so as to make it easier to solve, we may define the followings:

$$\xi(T) \equiv \left(\frac{\hbar^2}{2m^*|a(T)|}\right)^{1/2}, \tag{9.6}$$

$$y \equiv x/\xi, \tag{9.7}$$

$$\psi_\infty \equiv \sqrt{\frac{|a|}{b}} \tag{9.8}$$

and

$$\psi(x) = \psi_\infty f(y). \tag{9.9}$$

Note that Eq. (9.8) refers to deep inside the superconductor where no spatial variation could exist, so that the solution indeed becomes identical to the uniform case we have seen in the last section.

By plugging in the definitions, we can rewrite Eq. (9.5) as

$$-\frac{d^2}{dy^2}f(y) - f(y) + f^3(y) = 0. \tag{9.10}$$

Indeed, this looks much simpler than the original, and we can immediately work out for the solution. For instance, you can easily confirm that

$$f(y) = \tanh(y/\sqrt{2}). \tag{9.11}$$

could be a solution. Recovering original variables, the solution can be written as

$$\psi(x) = \psi_\infty \tanh\left(\frac{x}{\sqrt{2}\xi(T)}\right). \tag{9.12}$$

The order parameter ψ is zero at the interface, but increases fast as entering into the superconductor and reaches the value expected for a uniform superconductor.

Exercise 21

Derive Eq. (9.10) and show that Eq. (9.11) is its solution.

One can notice that $\xi(T)$ defines the length scale for the spatial evolution of the order parameter at the metal-superconductor interface. The length scale $\xi(T)$ is called the

Ginzburg-Landau (GL) coherence length. In the limit of zero temperature, this is equivalent to the Pippard coherence length. However, they are different quantities and indeed the GL coherence length is temperature dependent as $a(T)$ is temperature dependent. By inserting $a(T) = \frac{da}{dT}(T - T_c)$ into Eq. (9.6), one obtains

$$\xi(T) = \left(\frac{\hbar^2}{2m^* \frac{da}{dT} (T - T_c)} \right)^{1/2} \tag{9.13}$$

$$= \left(\frac{\hbar^2}{-2m^* \frac{da}{dT} T_c} \frac{T_c}{T_c - T} \right)^{1/2} \tag{9.14}$$

$$= \xi(0) |\frac{T - T_c}{T_c}|^{-1/2} \tag{9.15}$$

$$= \xi(0)|t|^{-1/2}, \tag{9.16}$$

where t is called the reduced temperature to represent the relative distance in temperature from the critical temperature.

Exercise 22

If the GL coherence length corresponds to the coherence we have defined in terms of the superconducting electron densities when distinguishing type-I and II superconductors, what would you expect for the temperature dependence of the penetration depth?

10 Fields

In the last lecture, we have introduced a gradient term into the GL free energy, and obtained the GL coherence length $\xi(T)$ which characterises the length scale over which the order parameter $\psi(\mathbf{r})$ varies at the normal-superconductor interface.

In this lecture, we will further add on a magnetic field term, and we will make use of another length scale. Guess what, yes, that is the penetration depth λ which characterise the length scale over which a magnetic flux penetrates into a superconductor. There could be different ways of introducing the penetration depth, but for simplicity we may adopt an approach of presuming $|\psi|^2 = n_s = -a/b$ and relating with the definition of $\lambda = \sqrt{m^*/\mu_0 n_s e^{*2}}$ from London's. So, one can write

$$\lambda(T) = \sqrt{\frac{m^* b}{\mu_0 e^{*2} |a(T)|}}. \tag{10.1}$$

After obtaining the GL free energy and equation, we will look into the two simple cases where we may ignore the spatial gradient term such as a thin film or a thin wire. If the thickness or radius of them, say d, is much smaller than ξ, there would be only little or negligible contribution to the free energy from the spatial inhomogeneity of $|\psi|$. We will evaluate how the critical current and critical field changes in such cases.

10.1 Magnetic vector potential

One can follow the standard procedure in quantum mechanics of introducing a magnetic field into the problem, namely, replacing the momentum with the help of vector potential as

$$-i\hbar\nabla \Rightarrow -i\hbar\nabla - q\mathbf{A}. \tag{10.2}$$

Then, one can write down a further extended the free energy difference form as

$$f_s - f_n = a|\psi(\mathbf{r})|^2 + \frac{b}{2}|\psi(\mathbf{r})|^4 + \frac{1}{2m^*}|\left(-i\hbar\nabla + e^*\mathbf{A}\right)\psi(\mathbf{r})|^2, \tag{10.3}$$

where we introduced $q = -e^*$. Following the same procedure of performing functional derivative of the total free energy with respect to ψ and ψ^*, one obtains a full GL equation,

$$\frac{1}{2m^*}\left(-i\hbar\nabla + e^*\mathbf{A}\right)^2\psi + a\psi + b\psi|\psi|^2 = 0. \tag{10.4}$$

Thus nothing has really changed except the change in the canonical momentum by adding a vector potential in the first term.

Solving the GL equation is generally quite involved, and we reserve its discussion for later sections. In the following, we instead make use of the above form of the free energy difference to gain insights into the critical field and critical current in simple cases.

10.2 Critical field

Suppose that a magnetic field $H_a = B_a/\mu_0$ is applied parallel to the surface of a superconducting slab. Let say x refers to the direction normal to the slab of thickness d. Then one can use the gauge $\mathbf{A} = (0, \mu_0 H_a x, 0)$. By the way, before proceeding, can you guess what would be the result look like? How would the critical field value would change from that of the bulk superconductor? Bear your answer in mind, and let's evaluate.

Exercise 23

Check if the above gauge indeed gives the supposed magnetic field.

Let us consider an infinitely large slab. Then we need to consider only the free energy per unit area F_s[17], that is, the free energy density being integrated over thickness of the slab, which will be later minimised with respect to $|\psi|^2$.

So, the free energy per unit area is written as

$$F_s = \int_{-d/2}^{d/2} f_s \, dx \tag{10.5}$$

$$= \int_{-d/2}^{d/2} \left[f_n + a|\psi|^2 + \frac{b}{2}|\psi|^4 + \frac{1}{2m^*}|\left(e^*\mu_0 H x\right)\psi|^2 \right] dx, \tag{10.6}$$

which you can easily evaluate. Now, by minimizing F_s with respect to $|\psi|^2$,

$$\frac{\partial F_s}{\partial |\psi|^2} = \left(ad + \frac{e^{*2}H^2 d^3}{24 m^*} \right) + bd|\psi|^2 \tag{10.7}$$

$$= 0, \tag{10.8}$$

one would get

$$|\psi|^2 = |\psi_\infty|^2 \left[1 - \frac{1}{24} \left(\frac{d}{\lambda} \right)^2 \left(\frac{H}{H_c} \right)^2 \right]. \tag{10.9}$$

Exercise 24

Derive Eq. 10.9.

Equation (10.9) allows us to find the modified critical field strength H_c' by setting the condition where $|\psi|^2 = 0$. It shows that the critical field H_c' has increased from the bulk value H_c as

$$H_c' = 2\sqrt{6} H_c \left(\frac{\lambda}{d} \right), \tag{10.10}$$

where you could see that the increase could be significant for a very thin slab with large penetration depth. This can be expected as more the field penetrates less energy cost due to diamagnetic term.

[17] Well, I should have find a better notation than this to reserve a lower case f for the energy density and the upper case F for the energy; but there's no middle case, so ... any suggestion?

Exercise 25

You have had considered a similar case in Exercise within London equations before. Compare the results with Eq. (10.10).

10.3 Critical current

Consider a thin wire carrying a supercurrent. As you may recall, there is a maximal or critical current density which generates a magnetic field at the surface reaching the critical field strength. Let us evaluate the critical current density and compare later with the value from London theory.

We may first write the supercurrent density in a Drude-like form as

$$\mathbf{j}_s = e^*|\psi|^2\mathbf{v}_s, \tag{10.11}$$

where $\mathbf{v}_s$ is velocity of the superconducting charge carriers. We may further write down the free energy density difference as

$$f_s - f_n = a|\psi|^2 + \frac{1}{2}b|\psi|^4 + \frac{1}{2}m^*v_s^2|\psi|^2. \tag{10.12}$$

By minimizing the above with respect to $|\psi|^2$,

$$\frac{\partial}{\partial|\psi|^2}(f_s - f_n) = a + b|\psi|^2 + \frac{1}{2}m^*v_s^2 \tag{10.13}$$

$$= 0, \tag{10.14}$$

one obtains

$$|\psi|^2 = -\frac{a}{b}\left(1 - \frac{m^*v_s^2}{2|a|}\right) \tag{10.15}$$

$$= |\psi_\infty|^2\left(1 - \frac{m^*v_s^2}{2|a|}\right) \tag{10.16}$$

$$= |\psi_\infty|^2\left[1 - \left(\frac{\xi m^* v_s}{\hbar}\right)^2\right]. \tag{10.17}$$

Therefore, the supercurrent density can be written as

$$\mathbf{j}_s = e^*|\psi|^2\mathbf{v}_s \tag{10.18}$$

$$= e^*|\psi_\infty|^2\left[1 - \left(\frac{\xi m^* v_s}{\hbar}\right)^2\right]\mathbf{v}_s. \tag{10.19}$$

You can see that the superconducting electron density decreases as the velocity increases. This implies that the current density increases initially with the velocity, but at some point when the reduction of the superelectron density outweighs the velocity increase, the current

density will hit the maximum and then decline. We can find the maximum supercurrent density by evaluating

$$\frac{\partial j_s}{\partial v_s} = e^*|\psi_\infty|^2 - 3e^*|\psi_\infty|^2 \left(\frac{\xi m^* v_s}{\hbar}\right)^2 \tag{10.20}$$

$$= e^*|\psi_\infty|^2 \left[1 - 3\left(\frac{\xi m^* v_s}{\hbar}\right)^2\right] \tag{10.21}$$

$$= 0. \tag{10.22}$$

From the above, the supercurrent velocity when the current is maximum becomes

$$v_s = \frac{1}{\sqrt{3}} \frac{\hbar}{\xi m^*}. \tag{10.23}$$

Plugging this back into Eq. (14.1), we find that the maximal current is obtained when $|\psi|^2 = (2/3)|\psi_\infty|^2$. Now, plugging this back to the expression for j_s gives the maximum value as

$$j_{s,\max} = \frac{2e^*\hbar n_s}{3\sqrt{3}m^*\xi} \propto (1 - T/T_c)^{3/2}, \tag{10.24}$$

which we may consider as the critical current density j_c.

Within the London, the critical current density is expressed as

$$j_c = H_c/\lambda. \tag{10.25}$$

Exercise 26

Show that Eq. (10.24) is 1.84 times smaller than the value from Eq. (10.25), and explain why.

11 Strong Field

Solving a non-linear differential equation like the original Ginzburg-Landau (GL) equation is generally challenging. However, we may simplify the equation, for instance, by linearising under certain circumstances. Suppose one applies a strong enough magnetic field to drive a superconductor to its normal state. Then, the applied field is decreased down to just below a critical field[18]. There one can expect the order parameter ψ small. In this lecture, we will see how a solution to the linearised GL equation can lead us to distinguish quantitatively between type-I and II superconductors.

11.1 Linearised Ginzburg-Landau equation

Let us start by rewriting the original GL equation,

$$\frac{1}{2m^*}\left(-i\hbar\nabla + e^*\mathbf{A}\right)^2\psi + a\psi + b\psi|\psi|^2 = 0. \tag{11.1}$$

Suppose a magnetic field slightly less than H_c is applied along z direction as

$$\begin{aligned}\mathbf{B}_a &= \mu_0\mathbf{H}_a &(11.2)\\ &= (0,0,B_a). &(11.3)\end{aligned}$$

Let us use the Landau gauge

$$\mathbf{A} = (0, xB, 0). \tag{11.4}$$

By plugging these into the GL equation, and dropping higher order terms than ψ, one can obtain

$$\left(-\frac{\hbar^2}{2m^*}\nabla^2 - \frac{\hbar e^* B_a}{m^*}ix\frac{\partial}{\partial y} + \frac{e^{*2}B_a^2}{2m^*}x^2\right)\psi = |a|\psi \tag{11.5}$$

Exercise 27

Derive Eq. (11.5). You may use $\nabla = \hat{e}_x\frac{\partial}{\partial x} + \hat{e}_y\frac{\partial}{\partial y} + \hat{e}_z\frac{\partial}{\partial z}$

By defining a cyclon frequency $\omega_c \equiv eB_a/m^*$, the above equation can be expressed as

$$\left(-\frac{\hbar^2}{2m^*}\nabla^2 - \hbar\omega_c ix\frac{\partial}{\partial y} + \frac{1}{2}m^*\omega_c^2 x^2\right)\psi = |a|\psi \tag{11.6}$$

This equation looks alike the Schrodinger equation for a charged particle in a magnetic field. The resulting Landau level solution can be found in standard Quantum Mechanics textbooks, but we may repeat in the following.

[18] This may be H_c for type-I or H_{c2} for type-II.

11.2 Solution to the linearised equation

We can start with a trial solution,

$$\psi(\mathbf{r}) = e^{i(k_y y + k_z z)} f(x), \tag{11.7}$$

which leads us to

$$\nabla^2 \psi = e^{(k_y y + k_z z)} \frac{d^2 f}{dx^2} - \left(k_y^2 + k_z^2\right) e^{i(k_y y + k_z z} f \tag{11.8}$$

and

$$\frac{\partial}{\partial y}\psi = i k_y e^{i(k_y y + k_z z)} f. \tag{11.9}$$

Using the above, one can rewrite the linearised GL equation as

$$-\frac{\hbar^2}{2m^*}\frac{d^2 f}{dx^2} + \left(\hbar\omega_c k_y x + \frac{1}{2}m^*\omega_c^2 x^2\right) f = \left(|a| - \frac{\hbar^2\left(k_y^2 + k_z^2\right)}{2m^*}\right) f \tag{11.10}$$

This can be further simplified by defining $x_0 \equiv -\hbar k_y / m^* \omega_c$ to

$$-\frac{\hbar^2}{2m^*}\frac{d^2 f}{dx^2} + \frac{m^*\omega_c^2}{2}\left(x - x_0\right)^2 f = \left(|a| - \frac{\hbar^2\left(k_z^2\right)}{2m^*}\right) f. \tag{11.11}$$

You may recognise that this has the same form as the Schrodinger equation for a simple harmonic oscillator. So one can immeidately write down the solution as

$$\left(n + \frac{1}{2}\right)\hbar\omega_c = |a| - \frac{\hbar^2 k_z^2}{2m^*} \tag{11.12}$$

If this were a problem of an electron in a magnetic field, then |a| would've been an energy $E_n(k_z)$. Then the solution becomes (by rearranging the above)

$$E_n(k_z) = \left(n + \frac{1}{2}\right)\hbar\omega_c + \frac{\hbar^2 k_z^2}{2m_e} \tag{11.13}$$

where ω_c should be eB/m_e.

11.3 Implication of the solution

Now let's look into the implication of the solution Eq. (11.12). We will first see how the transition temperature changes with an applied magnetic field of which answer we already know from Part I. Then we will see what happens in type-II superconductors when a field is reduced such that a transition from normal to superconducting state with a vortex.

One may first expand as

$$\left(n + \frac{1}{2}\right)\hbar\omega_c + \frac{\hbar^2 k_z^2}{2m^*} = \frac{da}{dT}(T_c - T). \tag{11.14}$$

For the ground state with $n = 0$ and $k_z = 0$, this reduces to

$$\frac{1}{2}\hbar\omega_c = \frac{da}{dT}(T_c - T). \tag{11.15}$$

Note that here T_c is for $H_a = 0$; let's write explicitly $T_c(0)$. For the boundary, by setting $T \to T_c(H_a)$, one obtains

$$T_c(H_a) = T_c(0) - \frac{1}{2\,(da/dT)}\hbar\omega_c. \tag{11.16}$$

The last term on the right is always positive. So the critical temperature is suppressed with an applied field.

Now, let us rewrite the solution once more as

$$\frac{1}{2}\hbar\frac{e^*\mu_0 H_{c2}}{m^*} = |a| \tag{11.17}$$

where we have restored $\omega_c = e^*B/m^*$. As we bring in the definition of the GL coherence length $\xi(T)$ and a flux quantum $\Phi_0 = h/e^*$, one can obtain

$$\mu_0 H_{c2} = \frac{\Phi_0}{2\pi\xi(T)^2}. \tag{11.18}$$

You may appreciate that H_{c2} is simply defined with Φ_0 and $\xi(T)$. This can be interpreted as one flux quantum in unit area of $2\pi\xi^2$ at H_{c2}.

Exercise 28

Show Eq. (11.18).

From the temperature dependence of $\xi(T)$, you can easily show that

$$\mu_0 H_{c2} = \frac{\Phi_0}{2\pi\xi(0)^2}\frac{T_c - T}{T_c}. \tag{11.19}$$

What does this tell us? It allows us to obtain $\xi(0)$ by measuring temperature dependence of H_{c2}.

11.4 Type I & II criterion

From the previous lectures, you have expressions of $\lambda(T)$ and H_c in terms of the parameters a and b.

$$H_c = \frac{(da/dT)}{(\mu_0 b)^{1/2}}(T_c - T) \tag{11.20}$$

$$= \frac{1}{\mu_0}\frac{\Phi_0}{2\pi}\frac{1}{\xi^2}\frac{\xi}{\sqrt{2}\lambda} \tag{11.21}$$

$$= H_{c2}\frac{1}{\sqrt{2}}\left(\frac{\xi}{\lambda}\right) \tag{11.22}$$

Exercise 29

Show Eq. (11.20).

By defining a temperature independent parameter

$$\kappa = \lambda/\xi \tag{11.23}$$

one can write $H_{c2} = \sqrt{2}\kappa H_c$. So there can be two cases where i) $\kappa > 1/\sqrt{2}$ such that $H_{c2} > H_c$ and ii) $\kappa < 1/\sqrt{2}$ such that $H_{c2} < H_c$. The former corresponds to type-II and the latter type-I superconductors, respectively.

12 Invariance & Quantisation

We haven't made use of the phase in the complex order parameter so far. Now is the time to show what physical consequences that the phase term would bring about. In fact, such phase term is essential in quantum mechanical problems. We will discuss the manifestation of the phase term in this and the next few lectures before entering Part III. This week, we have less materials to cover than usual, and the materials up to this section will be subject to Class Test in Week 8.

12.1 Gauge symmetry

We write the order parameter as

$$\psi(\mathbf{r}) = |\psi(\mathbf{r})|e^{i\theta(\mathbf{r})}. \tag{12.1}$$

Suppose a phase change by ϕ(r) such that

$$\psi(\mathbf{r}) \to \psi(\mathbf{r})e^{i\phi(\mathbf{r})} \tag{12.2}$$

We'd like to see how the free energy should change upon this (we'd hope it not changing). To do so, we may first evaluate an element entering into the free energy expression, the canonical momentum.

$$\hat{p}\left(\psi(\mathbf{r})e^{i\phi(\mathbf{r})}\right) = e^{i\phi(\mathbf{r})}\left(-i\hbar\nabla + e^*(\mathbf{A} + \frac{\hbar}{e^*}\nabla\phi)\right)\psi(\mathbf{r}). \tag{12.3}$$

Thus one can see that the free energy would remain unchanged if

$$\psi(\mathbf{r}) \to \psi(\mathbf{r})e^{i\phi(\mathbf{r})} \tag{12.4}$$

$$\mathbf{A}(\mathbf{r}) \to \mathbf{A}(\mathbf{r}) + \frac{\hbar}{e^*}\nabla\phi \tag{12.5}$$

Note in the latter that the B field from the curl of the vector potential does not change as the curl of gradient is always zero. So, however the phase changes, one can always find the vector potential that keeps the free energy unchanged. This is said that the GL theory satisfies local gauge invariance.

If we say F_s^0 is the free energy for constant θ and $\mathbf{A} = 0$, the total free energy could be written

$$F_s = F_s^0 + \int d^3r \frac{1}{2m^*}\left(e^*(\mathbf{A} + \frac{\hbar}{e^*}\nabla\phi)\right)^2 |\psi|^2 \tag{12.6}$$

$$= F_s^0 + \rho_s \int d^3r \left(\frac{e^*}{\hbar}\mathbf{A} + \nabla\phi\right)^2 \tag{12.7}$$

where a stiffness

$$\rho_s = \frac{\hbar^2}{2m^*}|\psi|^2. \tag{12.8}$$

To minimise energy cost is to have $\nabla\phi$ as small as possible or zero, that is, having the the same phase throughout a superconductor. This phase can take any value as long as it is fixed. So it is much alike a ferromagnet chooses a direction to collectively align. A difference is that the order is set up in the phase of the complex order parameter.

12.2 Flux quantisation

Suppose a superconducting hollow cylinder whose thickness is much larger than λ. We will see that a magnetic flux through this cylinder should be quantised.

To address the above, we need to step back to get an expression for the current first. Let us express a persistent current density starting from its definition with $\psi = |\psi|e^{i\theta}$ treating it as a sort of wavefunction.

$$\mathbf{j_s} = q\mathrm{Re}(\psi^* \mathbf{v} \psi) \tag{12.9}$$

$$= \frac{-e^*|\psi|^2}{m^*}(\hbar\nabla\theta + e^*\mathbf{A}) \tag{12.10}$$

Exercise 30

Derive Eq. (12.10).

Now suppose a path within a superconductor which encloses a hollow hole. If this path is taken beyond λ from the surfaces, there would be no supercurrent so that the integration of the current density becomes zero. Thus from Eq. (12.10),

$$0 = \oint \nabla\theta \cdot d\mathbf{l} + \oint \frac{e^*}{\hbar}\mathbf{A} \cdot d\mathbf{l} = 2n\pi + 2\pi\frac{\Phi}{\Phi_0}, \tag{12.11}$$

as

$$\oint \mathbf{A} \cdot d\mathbf{l} \equiv \Phi \tag{12.12}$$

and

$$\Phi_0 = \frac{h}{e^*} \tag{12.13}$$

Here we set the boundary condition that the cylinder requires the phase to be a multiple of 2π.

Now look into Eq. (12.13). It only consists of a fundamental constant, and e^* which should be naively expected to be another fundamental constant, charge of an electron. Now see Fig. 37.

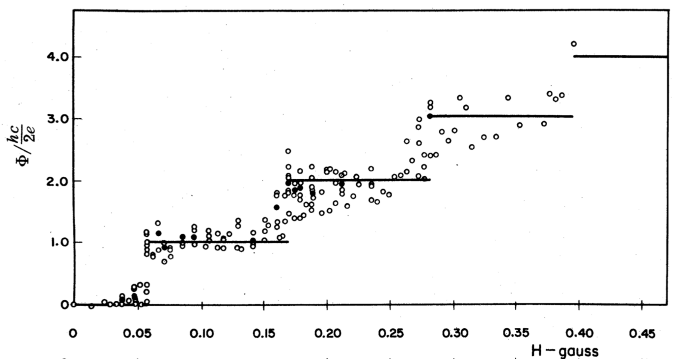


Figure 37. Quantised increase in trapped flux as a function of applied magnetic field [13].

It shows a quantised increase of magnetic flux as a function of an applied magnetic field. You may notice that the step is given as $h/2e$ rather than h/e[19]. This experimental observation ensures that two electrons rather than one are relevant for superconductivity, leaving hints on electron pairing. You can now safely replace e^* with $2e$, and also m^* with $2m_e$.

For more general cases, it is fluxoid that includes persistent current terms in addition to magnetic flux that is conserved, rather than the flux itself. We will discuss later.

[19]Ignore c which enters here due to cgs unit used in that work.

13 Mixed States

In the last two lectures, we learned about magnetic flux lines threading a superconductor which are quantised in unit of the ratio between the Planck constant and electron charge. By comparing with experiments, it turned out that two electrons, rather than one, are involved in the magnetic flux quantum.

Here, let us have a closer look into the mixed states that survive for $H_{c1} < H < H_{c2}$ in type-II superconductors. The mixed states are characterised by the existence of magnetic flux lines or vortices which consist of a normal metallic core and encircling supercurrent. That it allows magnetic flux penetrating through the body implies no more perfect diamagnetism. As the applied magnetic field H_a increases from H_{c1}, magnetisation will increase from $-H_a$ toward zero; at H_{c2}, it reaches to zero, which corresponds to normal state. Figure 38 schematically shows magnetisation as a function of applied magnetic field at a constant temperature.

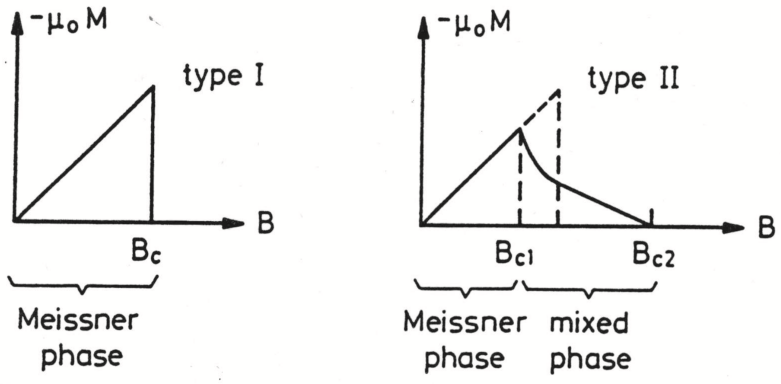


Figure 38. Isothermal magnetisation curve of a type I (left) and type II (right) superconductors.

For type II superconductor on the right, you may notice that the magnetisation (M) increases rather rapidly just above B_{c1} while it slows down as the field further increases. How can we understand this? For this, we need to know a bit more what would happen internally.

Just above H_{c1}, one would expect only a few flux lines, which would be well separated away from one another so that one can consider a single isolated flux line. One the other hand, just below H_{c2}, one would expect a closely packed lattice of flux lines. We may then hand-waive argue for the observed magnetisation curve (Fig. 38, right panel) as follows. Just above H_{c1}, there is not much hurdle to introduce more and more flux lines with increasing applied field. However, if a field strength is somewhat large such that the flux lines start interacting one another, to form a lattice, it would become harder to introduce further flux lines.

To get some flavour of the vortex lattice in different fields, Fig. 39 (upper panel) shows field evolution of vortex lattice, from 1 to 8 T, in a iron-based superconductor LaFeAs crystal. The lower panel shows a FFT of the above images.

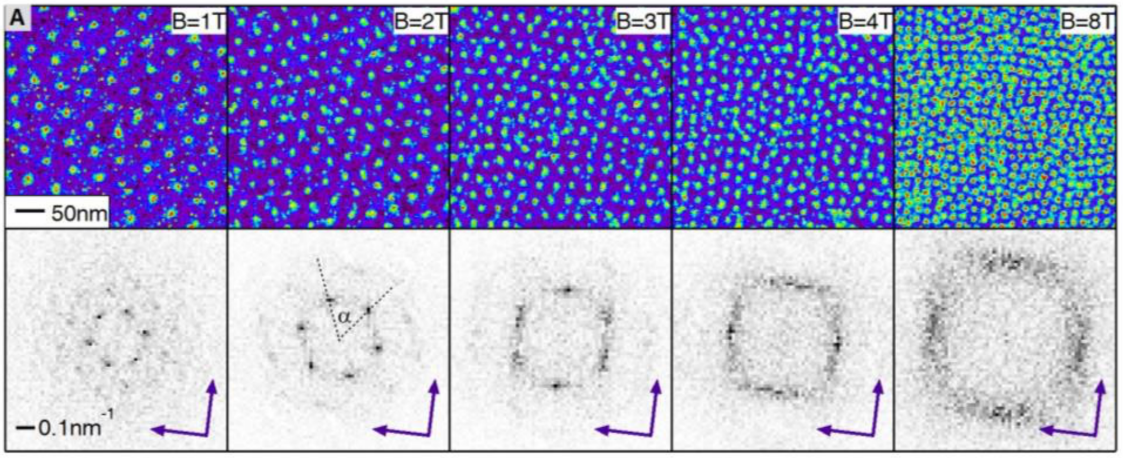


Figure 39. Field evolution of a vortex lattice in LaFeAs superconductor (Ref. J. Yin et al., "Vector field based emergent phase diagram in a correlated superconductor" arXiv 2018.

Exercise 31

What can we learn from the FFT image (lower panel) in Fig. 39?

13.1 A single flux line

We now know that the core of vortex is essentially normal metallic, and the vortex has a diameter of the order of coherence length ξ. For very small ξ relative to λ, most part of the superconductor will have the amplitude of ψ more or less constant in space, except the region around the vortex core within ξ, thus we can apply the London theory. So, let us start from the expression resulting from the London equation, $\nabla \times (\nabla \times \mathbf{B}) = -\mathbf{B}/\lambda^2$. Suppose that the vortex has a cylindrical symmetry that we can exploit, and employ polar coordinate system to handle the problem. Then, we can express a magnetic flux $\mathbf{B} = (0, 0, B_z)$, in cylindrical coordinate, with B_z being a function of r only, i.e., $B_z(r)$. One can evaluate the curl of the flux as

$$\nabla \times \mathbf{B} = \frac{1}{r} \begin{vmatrix} \hat{r} & r\hat{\phi} & \hat{z} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ B_r & rB_\phi & B_z, \end{vmatrix} \tag{13.1}$$

which will leads to $\nabla \times \mathbf{B} = -\hat{\phi}\partial B_z/\partial r$. Evaluating further $\nabla \times (\nabla \times \mathbf{B})$, one obtains

$$-\frac{\hat{z}}{r}\frac{\partial}{\partial r}\left(r\frac{\partial B_z}{\partial r}\right) = -\frac{B_z}{\lambda^2}\hat{z}. \tag{13.2}$$

So, the the magnetic field outside the core is described by the Bessel equation,

$$\frac{1}{r}\frac{d}{dr}\left(r\frac{dB_z}{dr}\right) = \frac{B_z}{\lambda^2}. \tag{13.3}$$

The above Bessel equation may be simplified under certain conditions. For instance, for $r \ll \lambda$, by letting $\lambda \to \infty$, one removes the RHS of the equation such that

$$-\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial B_z}{\partial r}\right) = 0. \tag{13.4}$$

One can obtain the solution of the form $B_z(r) = c_1 \ln(r) + c_2$ with some constants c_1 and c_2. From this, one can also obtain the current using $\mu_0 \mathbf{j} = \nabla \times \mathbf{B}$; one would obtain $\mathbf{j} = -(c_1/\mu_0 r)\hat{\phi}$. Try to draw on paper how the field and current would actually look like surrounding the vortex core.

Similarly, for $r \gg 1$, by letting $r \to \infty$, one obtains,

$$\frac{d^2 B_z}{dr^2} = \frac{B_z}{\lambda^2}, \tag{13.5}$$

of which solution must have a form of exponentially suppressed function of distance as one may naively expect for deep into a superconductor (which, however, needs a correction by $r^{-1/2}$ being multiplied as one can show with a full solution to the Bessel equation).

The proper solution to the Bessel equation can be written as

$$B_z = \frac{\Phi_0}{2\pi\lambda^2} K_0\left(\frac{r}{\lambda}\right), \tag{13.6}$$

where the magnetic flux $\Phi_0 = \int B_z d^2 r$ which we know is given by $h/2e$. $K_0(x)$ is a modified Bessel function which becomes

$$K_0(x) \sim -\ln x \quad (\text{small } x) \tag{13.7}$$

$$\text{or } \sqrt{\frac{\pi}{2x}} e^{-x} \quad (\text{large } x) \tag{13.8}$$

Using the above function, by simply plugging in to the Bessel equation, one can derive the proper expression of B_z and $\mathbf{j}$ in each limit (e.g., small r or large r). The results are:

$$B_z = \frac{\Phi_0}{2\pi\lambda^2} \ln\left(\frac{\lambda}{r}\right) \quad (r \ll \lambda) \tag{13.9}$$

$$\text{or } \frac{\Phi_0}{2\pi\lambda^2}\sqrt{\frac{\pi\lambda}{2r}} e^{-r/\lambda} \quad (\text{large } r) \tag{13.10}$$

Bear in mind that for the above expressions describe superconducting state outside the vortex as ψ is suppressed toward zero within ξ. For instance, when we say $r \ll \lambda$, it actually means $\xi \ll r \ll \lambda$.

Exercise 32

Show that the energy of the vortex line per unit length is given by

$$\frac{\Phi^2}{4\pi\mu_0\lambda^2}\ln\left(\frac{\lambda}{\xi}\right), \tag{13.11}$$

by using $\mathbf{j} = -en_s\mathbf{v}$ and the kinetic energy density of $mv^2n_s/2$. You may use

$$\mathbf{j} = \frac{1}{\mu_0}\frac{\Phi_0}{2\pi\lambda^2}\frac{1}{r}\hat{\phi} \tag{13.12}$$

In the end, you would need to evaluate the integral

$$\int\frac{d^2r}{r^2} = \int_\xi^\lambda \frac{2\pi r dr}{r^2}. \tag{13.13}$$

Given the energy for a single flux line as obtained above, one can obtain the lower critical field B_{c1} which may correspond to a first single flux line enters a superconductor. It is left as an exercise to derive

$$B_{c1} \simeq \frac{\Phi_0}{4\pi\lambda^2}(\ln\kappa + 0.08) \tag{13.14}$$

where 0.08 was added by hands here for taking better account of the neighbourhood of the core region (beyond the scope of this module).

13.2 Interacting flux lines

Interacting vortices or flux lines ...

13.3 Flux lattices

and the formation of flux lattice could be a fascinating subject, but we will not cover them in this module, and I only encourage you to read the later part of the chapter in Annett's book (via ResoureList) to get a flavour if interested.

Finally, it may be worth collating here the expressions for the other critical fields for comparison.

$$B_{c2} = \frac{\Phi_0}{2\pi\xi^2}. \tag{13.15}$$

$$B_c = \frac{\Phi_0}{2\sqrt{2}\pi\lambda\xi}. \tag{13.16}$$

14 Josephson Junction

For an atom, electrons orbiting around the nucleus does not radiate electromagnetic energy despite the acceleration. For superconductor, electrons are not perturned when flowing over a macroscopic scale despite defects. Both point to a stationary quantum state. For an atom, it was a single electron state that we bothered. For a superconductor, it is a collective quantum state of many electrons. Here we will see the aspects and consequences of taking the complex order parameters as a sort of macroscopic wavefunction.

14.1 Macroscopic wavefunction & fluxoid quantisation

The persistent current can be written as

$$\mathbf{j_s}(\mathbf{r}) = \frac{-e|\psi|^2}{m_e}\left(\hbar\theta + 2e\mathbf{A}\right) \tag{14.1}$$

as shown in Lecture 12. In the support session we also showed the same starting from the definition of a current from Quantum Mechanics. Another method could be to differentiate the total free energy with respect to a vector potential, i.e.,

$$\mathbf{j_s}(\mathbf{r}) = -\frac{\partial F_s[\mathbf{A}]}{\partial \mathbf{A}(\mathbf{r})}. \tag{14.2}$$

We have shown in Lecture 12 that for a superconducting ring Eq. (14.1) leads to the quantisation of a flux threading the hole.

Now let's consider a superconductor and suppose to draw a contour anywhere inside the superconductor. For the continuity of ψ, one can write

$$\oint_c \nabla\left(\theta(\mathbf{r})\right)\cdot d\mathbf{r} = 2n\pi. \tag{14.3}$$

If we now integrate Eq. (14.1) along the contour,

$$\oint_c \left(\frac{-m_e}{e|\psi|^2}\mathbf{j} - 2e\mathbf{A}\right)\cdot d\mathbf{l} = nh \tag{14.4}$$

Thus,

$$\frac{m_e}{2e^2|\psi|^2}\oint_c \mathbf{j}\cdot d\mathbf{l} + \Phi = n\Phi_0, \tag{14.5}$$

where $\Phi_0 = h/2e$. Therefore, generally, it is the left hand side of Eq. (14.5) that is quantised, which is called a fluxoid.

For a contour encircling a hole, as we have seen in Lecture 12, $j = 0$, so

$$\Phi = n\Phi_0. \tag{14.6}$$

For any contour deep inside a bulk superconductor, beyond the penetration depth, $B = 0$ and $j = 0$, so $n = 0$, that is,

$$\oint_c \nabla\left(\theta(\mathbf{r})\right)\cdot d\mathbf{r} = 0. \tag{14.7}$$

This tells that in a bulk superconductor, θ is uniform, i.e., $\nabla\theta = 0$.

The fluxoid quantisation would be apparent when the contour is made within the penetration depth where supercurrent flows, e.g., thin cylinder. This has been shown in the so-called Little-Park experiment, of which consequence is profound than naively expected[20].

By now, you can take a curl of the supercurrent. Then, you can directly compare with the London equation. By doing so, essentially due to the experimental input of $e^* = 2e$ and $m^* = 2m_e$ one can realise that $|\psi|^2 = n_s/2$. So, the order parameter squared should be considered more properly the density of pairs of superelectrons.

14.2 Mind the gap

One of the striking consequences of quantum mechanics comes from a particle confined in a potential well, where you find the probability of finding the particle outside is finite contrary to classical expectation.

Suppose you have two superconductors separated by a small gap (e.g., an insulating layer). Let say one has ψ_1 and the other ψ_2, and express as

$$\psi_1 = |\psi_1|e^{i\theta_1} = \sqrt{n_{s1}/2}e^{i\theta_1}, \tag{14.8}$$

$$\psi_2 = |\psi_2|e^{i\theta_2} = \sqrt{n_{s2}/2}e^{i\theta_2}. \tag{14.9}$$

Suppose a coupling 'K' between the two superconductors and set potential difference V across the superconductors. Then we can write down coupled equations

$$i\hbar\frac{\partial}{\partial t}\psi_1 = E_0\psi_1 + K\psi_2, \tag{14.10}$$

$$i\hbar\frac{\partial}{\partial t}\psi_2 = (E_0 - 2eV)\,\psi_2 + K\psi_1. \tag{14.11}$$

You can plug Eqs. (14.8) into Eqs. (14.10). This is nothing but starting from evaluating $\partial\psi/\partial t$. For instance, one can show

$$\frac{\partial}{\partial t}\psi = \frac{\partial}{\partial t}\left(|\psi|e^{i\theta}\right) \tag{14.12}$$

$$= \left(\frac{1}{2n_s}\frac{\partial n_s}{\partial t} + i\frac{\partial\theta}{\partial t}\right)\psi. \tag{14.13}$$

Then one can rewrite the above coupled equations as

$$\left(i\hbar\left(\frac{1}{2n_{s1}}\frac{\partial n_{s1}}{\partial t} + i\frac{\partial\theta_1}{\partial t}\right) - E_0\right)\sqrt{n_{s1}}e^{i\theta_1} = K\sqrt{n_{s2}}e^{i\theta_2}, \tag{14.14}$$

$$\left(i\hbar\left(\frac{1}{2n_{s2}}\frac{\partial n_{s2}}{\partial t} + i\frac{\partial\theta_2}{\partial t}\right) - E_0 + 2eV\right)\sqrt{n_{s2}}e^{i\theta_2} = K\sqrt{n_{s1}}e^{i\theta_1} \tag{14.15}$$

This can be easily solved by taking the real and imaginary parts separately. For instance, by equating the imaginary parts, one can show

$$\frac{dn_{s1}}{dt} = -\frac{dn_{s2}}{dt} \tag{14.16}$$

$$= \frac{2K}{\hbar}\sqrt{n_{s1}n_{s2}}\sin(\theta_2 - \theta_1). \tag{14.17}$$

[20] We will cover this in a problem sheet by the end of the term.

Similarly, by equating the real parts, one obtains

$$\frac{d}{dt}\left(\theta_2 - \theta_1\right) = \frac{2eV}{\hbar}. \tag{14.18}$$

Exercise 33

Derive Eqs. (14.12), (14.16) and (14.18).

So, what do these equations tell us? Firstly, from Eq. 14.16, one can write the current density as

$$j = -e\left(\frac{dn_{s2}}{dt} - \frac{dn_{s1}}{dt}\right) \tag{14.19}$$

$$= \frac{4Ke}{\hbar}\sqrt{n_{s1}n_{s2}}\sin\left(\theta_2 - \theta_1\right) \tag{14.20}$$

$$= j_{\max}\sin\left(\theta_2 - \theta_1\right). \tag{14.21}$$

This equation shows that the current can flow in the absence of a voltage, and what drives the current is the difference in the phase between the two superconductor.

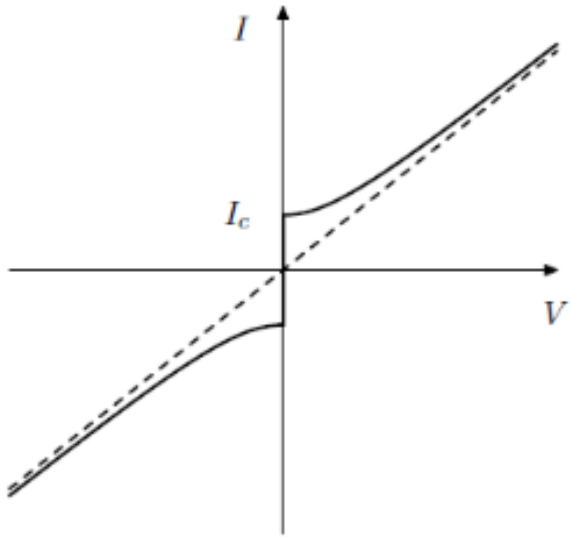


Figure 40. $I - V$ characteristic expected for a Josephson junction [14].

Figure 40 shows $I - V$ characteristic expected for a Josephson junction according to Eq. (14.16). When a current is set up to flow across the junction, it does not develop a voltage difference up to a maximum or critical current. However, when the current exceeds a critical value, the voltage develops and the characteristic for a ohmic metal is restored (linear dashed line).

From Eq. (14.18), one can tell a current is constant in the absence of voltage. On the other hand, if a voltage is applied, the phase difference changes with time, and thus an

alternating current sets in with the frequency of $2eV/\hbar$. This is called the AC Josephson effect. This is another example of two electrons involved in the process. Since the frequency can be measured to a high precision and it is expressed in terms of the ratio of two fundamental constants, the AC Josephson effect can be used as a voltage standard.

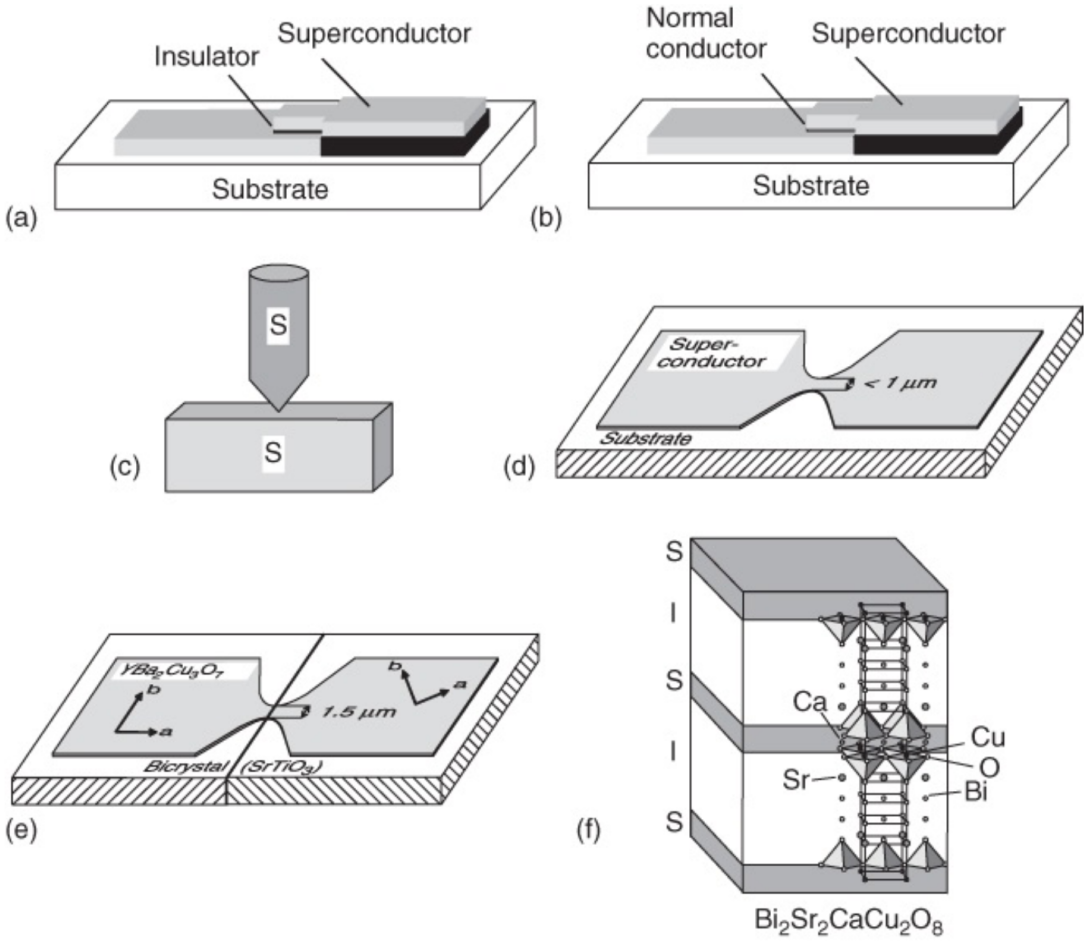


Figure 41. Examples of physical realisation of Josephson junctions. (a) Superconductor-insulator-superconductor junction with an oxide layer as a barrier (b) Superconductor-normal metal-superconductor junction with a normal conducting barrier (c) point contact (d) microbridge (e) grain boundary junction and (f) intrinsic Josephson junction in a cuprate superconductor (from R. Kleiner et. al., Superconductivity: Introduction).

So how do we actually construct the Josephson junctions? Figure 41 shows a variety of examples.

15 SQUID

We have seen some fascinating phenomena with the Josephson junction, that is, i) phase difference-driven supercurrent in the absence of voltage and ii) AC current set up by a DC voltage. As much as they are manifested by a phase term in the macroscopic wavefunction, one can imagine a situation where we put two such junctions and interfere. Moreover, this can be naturally arranged in a ring geometry, which will have further effects when imposed by a magnetic field (recall flux quantisation). This is called a SQUID (Superconducting Quantum Interference Device) (see Fig. 42, where the reason for the naming will become clear shortly[21].

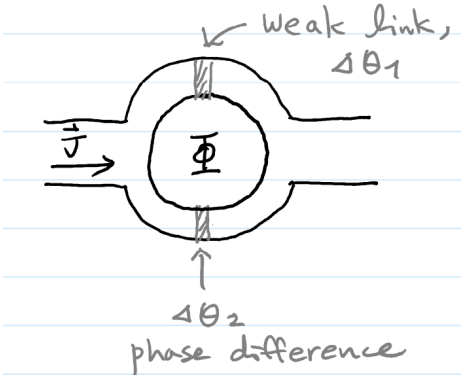


Figure 42. Schematic drawing of a SQUID

The corresponding circuit diagram of the SQUID is also shown in Fig. 43.

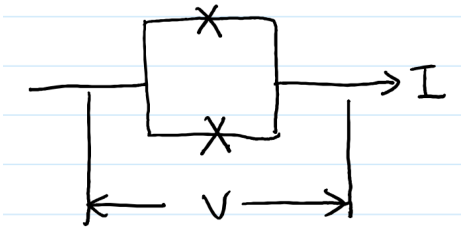


Figure 43. Circuit drawing of a SQUID

Before proceeding, however, for completeness with our study of the Josephson junction, we'll start from a few more thoughts on the Josephson junction, followed by the SQUID (Superconducting Quantum Interferometric Device).

15.1 The RCSJ model

The physical (real than ideal) Josephson junction is often described by the so-called RCSJ (resistively and capacitively shunted junction) model with a parallel resistance and capac-

[21] Apologies for mixed notation here and there; current is equivalently designated by I or J. The Josephson maximum current is also called the Josephson critical current.

itance. The total current in such a model is a sum of the current across each component such that $I = I_R + I_C + I_J$ where I_J is the Josephson current, $I_J = I_c \sin\phi$. By equating this with the AC Josephson equation, one can obtain

$$\frac{\hbar C}{2e}\frac{d^2\phi}{dt^2} + \frac{\hbar}{2eR}\frac{d\phi}{dt} + I_c \sin\phi = I \tag{15.1}$$

In order to obtain some qualitative insight into the above equation, let us rewrite it as

$$\left[\left(\frac{\hbar}{2e}\right)^2 C\right]\frac{d^2\phi}{dt^2} = -\left[\left(\frac{\hbar}{2e}\right)^2 \frac{1}{R}\right]\frac{d\phi}{dt} - \frac{d}{d\phi}\left[-\frac{\hbar I_c}{2e}\cos\phi - \frac{\hbar I}{2e}\phi\right] \tag{15.2}$$

This equation has a mechanical analogue if one considers a particle of mass $(\hbar/2e)^2 C$ with its position at ϕ moving in a potential given by the term in the last bracket, and finally subject to a damping force of $-k(d\phi/dt)$ where $k = (\hbar/2e)^2/R$. Its graphical representation is shown in Fig. ig. 44 which may help understand better the meaning of the equation (for general case, see the middle one).

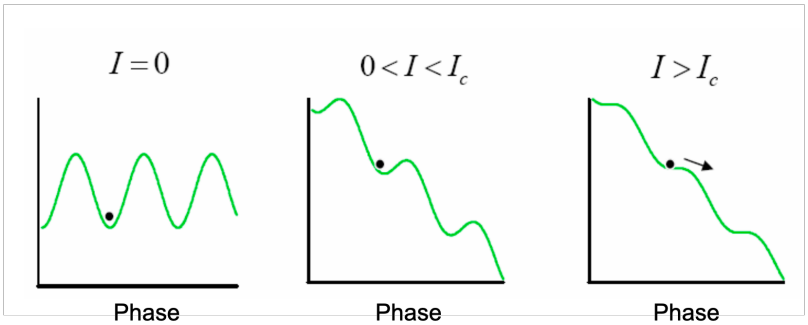


Figure 44. Tilted washboard model (figure credit: Wen Zhong, UC Riverside thesis.)

The above is called the tilted washboard model following the curve shape. One may notice that for $I > I_c$ there exist no more local minima.

One can take certain limits to the RCSJ model, for instance, by considering C is very small or very large, which correspond to overdamped and underdamped junctions, respectively. In particular, for the overdamped junction, time average voltage current relation follow

$$V = R(I^2 - I_c^2)^{1/2} \tag{15.3}$$

which connects the zero-voltage current spike for $I < I_c$ to the Ohmic behaviour of $V = IR$ for $I \gg I_c$, as sketched in Fig. 40.

15.2 Josephson in oscillating voltage and current

Back to the ideal model of a Josephson junction, why don't we consider a case where alternating voltage rather than DC is applied? One may consider a case where both AC and DC voltages, or a case where currents are applied. The results somewhat differ between the cases where oscillating voltage or current are applied. It is easier to formulate with the

former, while the latter is better feasible experimentally. For instance, one can easily substitute the voltage in the Josephson equation with $V_0 + V_1 \cos \omega t$ where V_0 is the DC voltage, V_1 the oscillating voltage amplitude and ω the angular frequency.

The experimental results for the case where the oscillating current is applied are shown in Fig. 45 where a step-wise increase in voltage is observed. This is called the Shapiro step and occurs at the voltages where the Josephson angular frequency, $2eV_0/\hbar$, becomes multiple of the angular frequency ($n\omega$) of the bias AC current.

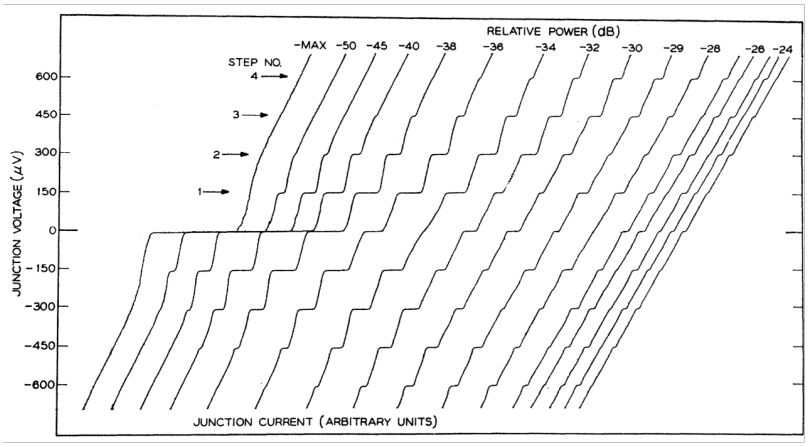


Figure 45. DC Voltage-current curve of a Nb-Nb point contact Josephson junction (credit: ...).

This is quite amazing. You are observing some *quantum* condition satisfied that is manifested into *macroscopic* voltage current characteristic. A pair of electrons cross the junction when

15.3 Two Josephsons on a ring: SQUID

Suppose a superconducting ring geometry made of two Josephson junctions (junction 1 & junction 2) with a threading magnetic field and a current feed from one side, as shown in Fig. 42. Let say the phase difference across one junction $\Delta\theta_1$ and that of the other $\Delta\theta_2$. They may set to be equal when the junctions are balanced and having $I_{c1} = I_{c2} = I_c$. However, one can easily expect that things will change once a magnetic field is applied. For instance, starting from the flux,

$$\Phi = \int \mathbf{B} \cdot d\mathbf{s} \tag{15.4}$$

$$= \oint \mathbf{A} \cdot d\mathbf{r} \tag{15.5}$$

$$= \frac{\Phi_0}{2\pi} \int \nabla\theta \cdot d\mathbf{r} \tag{15.6}$$

$$= \frac{\Phi_0}{2\pi} (\Delta\theta_1 - \Delta\theta_2) \,. \tag{15.7}$$

Thus the magnetic flux through the ring leads to unequal phase differences across the two junctions.

By the way, one may recall from Optics that the above equation has the same form as that in Fraunhofer diffraction,

$$\Delta\theta_1 - \Delta\theta_2 = 2\pi \frac{d \sin\theta}{\lambda} \tag{15.8}$$

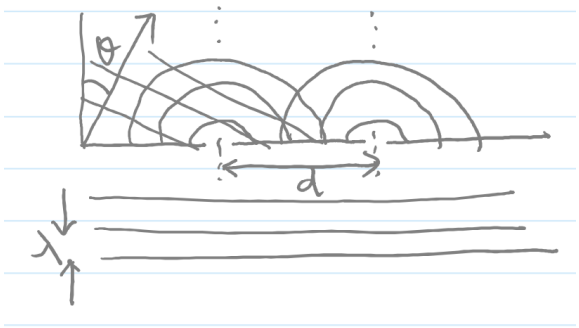


Figure 46. Schematic of Fraunhofer diffraction.

Back to our business, we can also write down the total current across the junctions as

$$I = I_c \sin(\Delta\theta_1) + I_c \sin(\Delta\theta_2), \tag{15.9}$$

if we assume that the two junctions share the same characteristic, e.g., $I_c = I_{c1} = I_{c2}$.

They have common terms of phase difference such that we can combine the two equations. To exploit that, we may first rewrite Eq. (15.7) into the following two equations,

$$\Delta\theta_1 = \Delta\theta + \frac{\pi\Phi}{\Phi_0}, \tag{15.10}$$

$$\Delta\theta_2 = \Delta\theta - \frac{\pi\Phi}{\Phi_0}, \tag{15.11}$$

by introducing a new parameter $\Delta\theta$. This $\Delta\theta$ is the phase difference across each junction in the absence of magnetic flux, and is given by $\Delta\theta = \sin^{-1}(I/2I_c)$. Now we can rewrite Eq. (15.9) as

$$I = I_c \sin(\Delta\theta_1) + I_c \sin(\Delta\theta_2) \tag{15.12}$$

$$= I_c \sin\left(\Delta\theta + \frac{\pi\Phi}{\Phi_0}\right) + I_c \sin\left(\Delta\theta - \frac{\pi\Phi}{\Phi_0}\right) \tag{15.13}$$

$$= 2I_c \sin(\Delta\theta) \cos\left(\frac{\pi\Phi}{\Phi_0}\right) \tag{15.14}$$

What we can see here is that the critical current for the SQUID in a magnetic field has changed modulating with the flux, and the new critical current is given by

$$I_c^{new} = 2I_c \cos\left(\frac{\pi\Phi}{\Phi_0}\right), \tag{15.15}$$

instead of $2I_c$ for the two junctions.

Figure 47 summarise the results in graphical form. On the right upper panel, the new critical current for the SQUID is plotted as a function of magnetic flux. One may consider this represents a quantum interference pattern of the supercurrents passing across each junction.

Suppose that you apply a current larger than the zero-field critical current (see Fig. 47 left, dashed grey horizontal line). As one changes the magnetic flux, the critical current changes (different green curves), thus the resulting voltage changes. One would therefore expect the voltage response will also osscillate as a function of the flux or an applied magnetic field. In the lower panel, the voltage variation with the flux, corresponding to the the critical current, is shown. The SQUID can be used to detect tiny magnetic field or its variation.

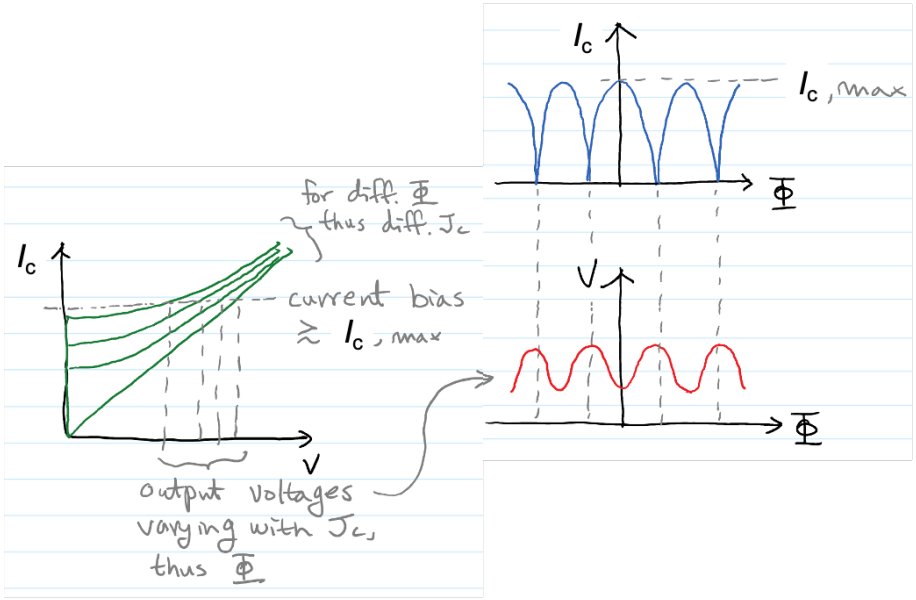


Figure 47. Left: current versus voltage graph. Right upper: critical current as a function of magnetic flux. Right lower: voltage variation as a function of the flux.

Part III
Microscopic Description

Based on several key observations and experiments that cannot be explained by the phenomenological theories, we develop a theory which accounts for microscopic mechanism of superconductivity.



Figure 48. What is this photo doing here? (Photo from thepreppyfoodie.tumblr.com)

16 Lattice, Paring and Gap

The GL theory has enabled us to describe a number of unique observations with superconductors including flux lines, type-I and type-II distinction, fluxoid quantisation, Josephson junction and so on.

However, there also remain several experimental observations that resort to certain microscopic mechanism of superconductivity for which the GL theory has no word to say. Above all, how do we explain some are superconducting elements while other not from the periodic table? Highly conducting metals at room temperature such as noble metals (Cu, Ag, Au) or alkali earth metals (Li, Na, K) tend to remain non-superconductig. On the other hand, less conducting metals such as $4d$ (Nb, Mo, Y) or $5d$ (La, Hf, Ta, W, Ir) tend to show relatively high T_c. Magnetic $3d$ and lanthanide metals are generally not become superconducting. Some semiconductors and semimetals in group III and IV become conducting under pressure which can undergo a superconducting phase transition upon cooling. In a sense, superconductivity looks like a rule than an exception. And the key question is 'what determines T_c?'.

Furthermore, there are experimental results which must resort to microscopic mechanism that is veiled. You should, for instance, be able to recall immediately at least two, that there is an energy gap at low temperatures (e.g., from heat capacity) and that two electrons are somehow involved (e.g., from flux quantisation for a ring). In this lecture, we will go through details of those unexplained experimental observations, along with a few other related but so far unseen experiments such as microwave absorption and tunneling. The latter experiments reaffirm the presence of an energy gap. However, we will also review one experimental observation that provides the key to the microscopic interaction in superconductivity, the Isotope effect.

16.1 Isotope effect

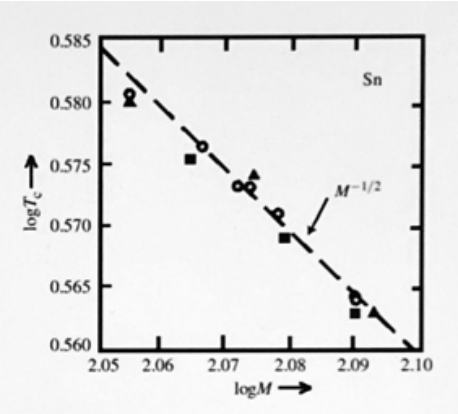


Figure 49. Superconducting transition temperature of Sn with different levels of isotope enrichment. M refers to the average mass of ions.

We are aware of that different isotopes do not make difference in the chemistry in materials. Their main effects could be by far some small quantitative change in phonon

spectrum. It is therefore a surprise that superconducting materials prepared with isotope enrichment, so that they have different average ion masses, show a systematic change in the transition temperature, as shown in Fig. 49

The transition temperature was found to follow

$$T_c \propto M^{-1/2}, \tag{16.1}$$

as shown as a straight line in the log-log plot of Fig. 49. This result clearly indicates that ions play an essential role in superconductivity. It tells that the lighter the ionic mass become, the higher T_c one can get. What does this imply?

16.2 Two electrons

This was shown before when discussing a quantum of magnetic flux is given by $h/2e$ rather than h/e.

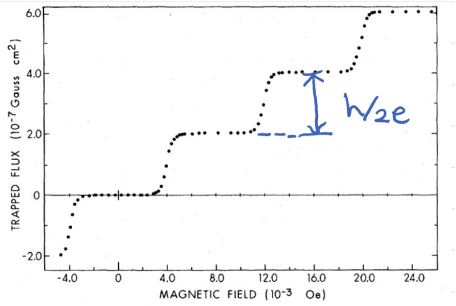


Figure 50. Trapped flux quanta as a function of an applied magnetic field.

To be bold, one may imagine two electrons are somehow bound as a pair to show the observed flux quantisation. How can two electrons, interacting via repulsive Coulomb interaction, bind together? There must be a gluing mechanism. Ions, or more properly their motion may serve as glue. How? Imagine an electron moving through a positively charged ionic lattice. Due to the negative charge of an electron, ions would be attracted to it distorting the underlying lattice. Then, another electron around could find the distorted lattice appealing to gain energy, thereby drawn to it and thus the preceding electron. This process is schematically drawn in Fig. 51.

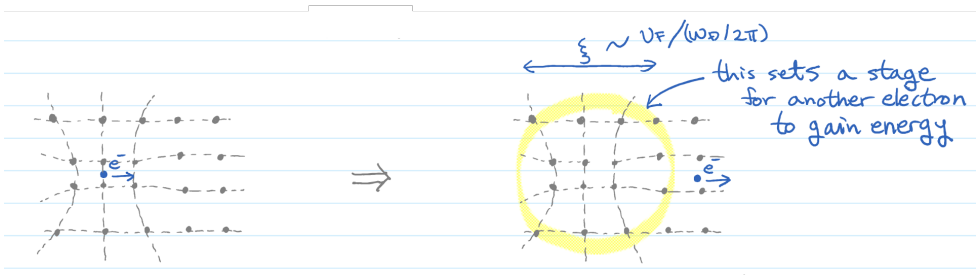


Figure 51. A possible scenario for pairing two electrons via ionic lattice vibration.

OK, this is just a possible, speculative scenario. However, in the subsequent lectures it will be shown quite the right picture. Moreover, it may provide a reasonable explanation of the isotope effect. The energy gain for another electron can be set to $\sim \hbar\omega_D$ where ω_D is the Debye frequency. The Debye frequency is inversely proportional to the square root of the ionic mass, that is, $\omega_D \propto M^{-1/2}$. So, if the phonons are playing a role of glue, and its effects are characterised by its Debye frequency, then one can expect that the materials wih larger Debye frequency would have higher T_c, as observed in the Isotope effect.

16.3 Gap to the excitations

The energy gap as observed in the heat capacity can be shown more clearly in Arrhenius plot, Fig. 52. If you go back to the earlier heat capacity data, you'd recognize that heat capacity falls more rapidly than T^3 at lowest temperature region. The fact that the data sits on a straight line indicates an activated (exponential) behaviour due to the presence of an energy gap to the excitations.

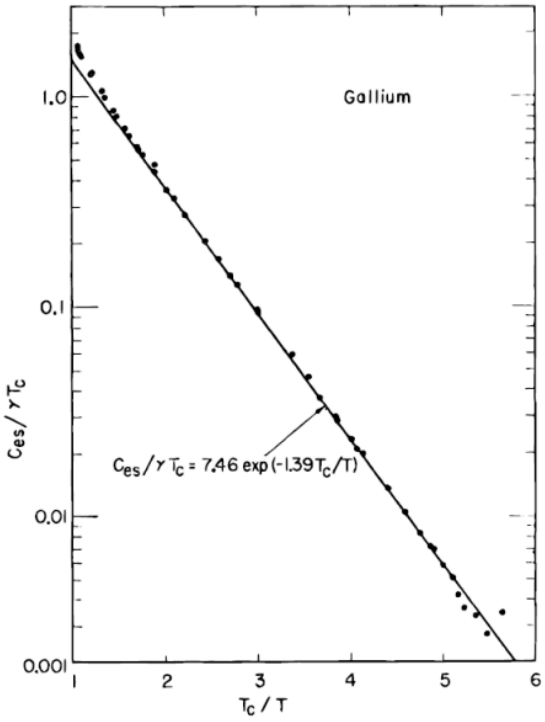


Figure 52. Molar heat capacity of Gallium in an Arrhenius plot.

The energy gap is also observed in other types of experiments. For instance, microwave absorption or tunneling conductance, as shown in upper and lower panel of Fig. 53, respectively. So, why the gap in energy? Wouldn't you be tempted to say that there must be some energy cost needed to break a bounded pair of electrons? It is not a crazy thought. The actuality is much more complicated than that, but that should be a good start.

That idea may also align with the further observation shown in Fig. 54 where it shows temperature evolution of the size of the gap. This almost look like an order parameter

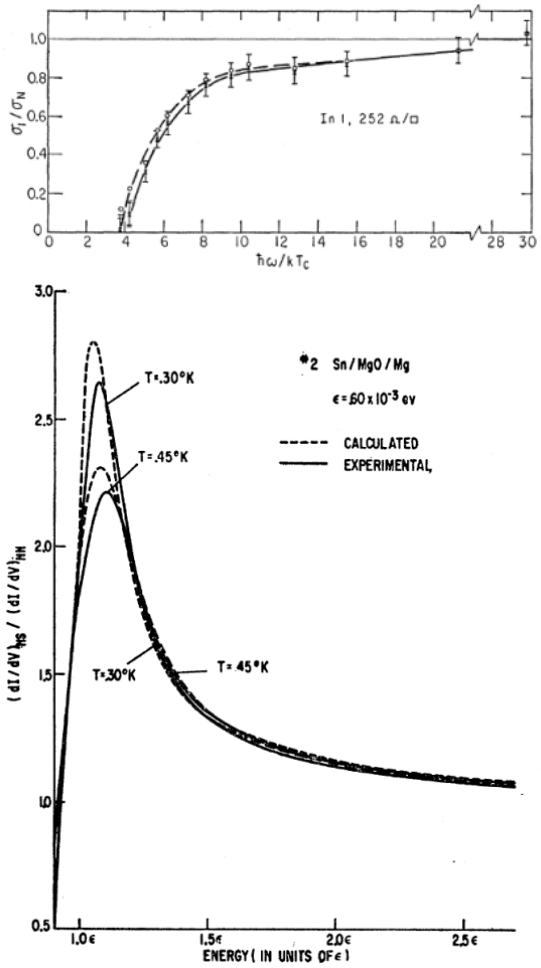


Figure 53. Microwave absorption (upper) and tunneling conductance (lower).

development. Are they related? Absolutely. But, again, our naive expectation here is only a guidance, and the correct description is more subtle and profound as we will see.

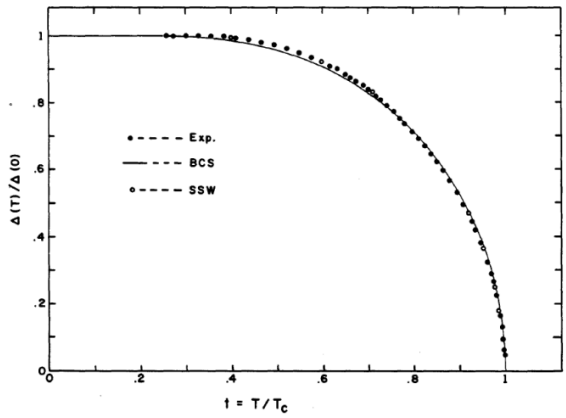


Figure 54. Energy gap in a superconductor as a function of temperature.

17 Attractive

From a series of experiments, we have found a clue on the microscopic mechanism of superconductivity. The question we need to address is how to make two electrons form a bound pair mediated by lattice vibration or phonons. To form a bound state, one needs to show that there are somehow, effectively attractive interactions between them. On top of this, having attractive interactions alone does not always guarantee a bound state.

The starting point should be anyway to understand electronic states in metals because superconductivity at low temperature we learned is given birth from high-temperature metallic phase on cooling. So, what are metals? Some people define a metal as a substance with Fermi surface. You may consider a grossly simplified theory of metal as a gas of independent electrons that are fermions. From Statistical Physics, we learned that fermions form a Fermi sphere. Thus, the electrons in this theory follow the description you learned for a gas of fermions in statistical physics besides possessing an electric charge (See Fig. 55 for a summary). This should be true only if you can find a way to disregard strong Coloumb interaction between electrons, which we will discuss shortly. Then, there arises more hope to find a way to introduce a net attraction.

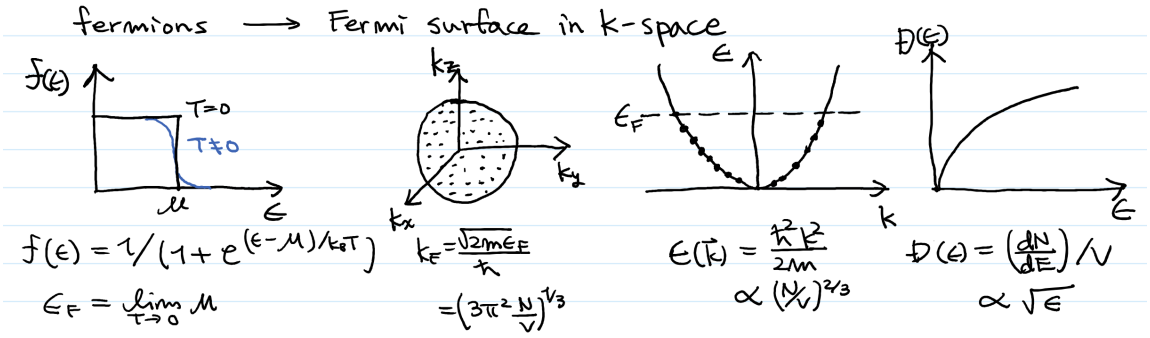


Figure 55. A summary of the free electron model which may approximate metals.

In this lecture, we will see that in the presence of Fermi sphere two electrons added on would form a bound pair if the two electrons are interacting, however small, attractively. To repeat, two attracted electrons would form a bound pair if there exists a Fermi sphere. This was a crucial theoretical insight put forward by L. Cooper, which subsequently lead to the development of the BCS theory.

17.1 Screening and polarization

There are several ways to dismiss Coulomb interaction in metals[22]. To start with, one can wonder the role of positively charged ions forming a lattice. It turned out that if positively charged ions are arranged in a periodic array of lattice, the electrons are not scattered by them due to their wave nature[23]. The lattice would only contribute to slightly modify

[22] There are metals where this is not possible; called strongly correlated electron systems, a major research direction in contemporary physics.

[23] This is what is called the Bloch theorem. If you feel it needed, you may refer to Simon's book on The Oxford Solid State Basic, which you can access via ResourceList.

the dispersion of electrons as shown in Fig. 56. For a metal, they will retain a Fermi surface though the shape is different from a sphere, and much of what you'd expect for non-interacting electrons would still apply.

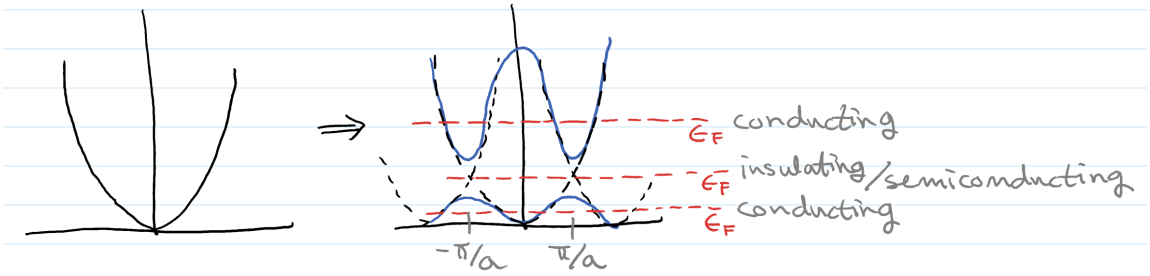


Figure 56. From a free electron model (left) to nearly free electron model with a periodic lattice (right) which allows distinction of transport properties of matter.

Now, what about the interaction between the electrons? They can be effectively screened. Two electrons repel strongly each other via a Coulomb potential,

$$V(\mathbf{r}) = \frac{e^2}{4\pi\epsilon_0|\mathbf{r}|} \tag{17.1}$$

where $\mathbf{r}$ is a vector connecting the two electrons. However, the interactions between the electrons in a metal should be substantially screened for they tend to stay away from one another due to Pauli principle and Coulomb repulsion itself. The screening from one of the simplest model, the Thomas-Fermi model, is given as

$$V_{\mathrm{TF}}(\mathbf{r}) = \frac{e^2}{4\pi\epsilon_0|\mathbf{r}|}e^{-|\mathbf{r}|/r_{\mathrm{TF}}}, \tag{17.2}$$

where the screening length r_{TF}, the Thomas-Fermi length, is order of atomic spacing. It thus replaces a long-range Coulomb repulsion by a much weaker, short-ranged one.

One can also start from a Fourier transformed expression of the Coulomb potential,

$$V(\mathbf{q}) = \frac{1}{\Omega}\int V(\mathbf{r})e^{i\mathbf{q}\cdot\mathbf{r}}d\mathbf{r} \tag{17.3}$$

$$= \frac{e^2}{4\epsilon_0 q^2}. \tag{17.4}$$

In a polarisable medium having dielectric susceptibility, this becomes

$$V(\mathbf{q},\omega) = \frac{e^2}{4\epsilon_r(\mathbf{q},\omega)\epsilon_0 q^2} \tag{17.5}$$

In the Thomas-Fermi approximation,

$$\epsilon_r = 1 + 1/q^2 r_{\mathrm{TF}}^2 \tag{17.6}$$

In any case, we shall accept that electrons in a metal can be considered nearly free, where their wavefunction not scattering off ionic lattice.

17.2 Two interacting electrons

We shall consider a situation of bringing two interacting electrons, 1 and 2, onto a Fermi sphere, following the Hamiltonian,

$$\mathcal{H} = \frac{1}{2m_e}\left(\mathbf{p}_1^2 + \mathbf{p}_2^2\right) + V\left(\mathbf{r}_1 - \mathbf{r}_2\right), \tag{17.7}$$

Their total wavefunction should be antisymmetric upon exchanging the two electrons. We shall assume that the orbital wavefunction is symmetric such that the spin part an odd function of spin singlet[24].

$$\Psi\left(\mathbf{r}_1, \sigma_1, \mathbf{r}_2, \sigma_2\right) = \psi\left(\mathbf{r}_1, \mathbf{r}_2\right)\phi\left(\sigma_1, \sigma_2\right) \tag{17.8}$$

Assuming that the ground state has the total momentum zero (Bloch's argument), the two electrons should have equal and opposite momenta. Then, we can expand the orbital part of the wavefunction in terms of the Bloch waves[25] as

$$\psi(\mathbf{r}) = \sum_{\mathbf{k}} \psi_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}}, \tag{17.9}$$

where $\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$. In doing so, we have dropped off $e^{i\mathbf{k}_{\rm cm}\cdot\mathbf{R}}$, where $\hbar\mathbf{k}_{\rm cm}$ corresponds to the total momentum of the pair and the center of mass $\mathbf{R} = (\mathbf{r}_1 + \mathbf{r}_2)/2$, as we have just assumed that $\mathbf{k}_{\rm cm} = 0$. Likewise, we can expand the potential as

$$V(\mathbf{r}) = \frac{1}{\Omega}\sum_{\mathbf{k}} V_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}} \tag{17.10}$$

We shall now substitute the ψ and V the Schrodinger equation,

$$\left(-\frac{\hbar^2}{m_e}\nabla^2 + V(\mathbf{r})\right)\psi(\mathbf{r}) = E\psi(\mathbf{r}). \tag{17.11}$$

Evaluating the above will lead to

$$(2\epsilon_{\mathbf{k}} - E)\psi_{\mathbf{k}} + \frac{1}{\Omega}\sum_{k'} V_{\mathbf{k}-\mathbf{k}'}\psi_{\mathbf{k}'} = 0, \tag{17.12}$$

where $\epsilon_{\mathbf{k}}$ represent the kinetic energy per electron.

Exercise 34

You can also derive Eq. (17.12) starting from using slightly different notation. For instance, you can start from $|\psi\rangle = \sum \psi_{\mathbf{k}}|\mathbf{k}\rangle$, where $\langle\mathbf{r}|\mathbf{k}\rangle = (1/\Omega)e^{i\mathbf{k}\cdot\mathbf{r}}$.

[24] So here ψ is literally a spatial part of the wavefunction of two electrons, not the complex order parameter or macroscopic wavefunction we used in Part II. Please use the notation bearing in mind the context.

[25] This refers to the electrons in a metal where they do not scatter off ionic lattice due to their wave nature.

Cooper assumed the effective interaction

$$V(\mathbf{k},\omega) = -|V_0| \tag{17.13}$$

dropping the $\mathbf{k}$ and ω dependence. To repeat, Cooper assumed a short ranged $V(\mathbf{r})$, even a delta-function like potential such that V is constant in k-space. This is indeed a bit drastic assumption though. Equation (17.12) then becomes

$$(2\epsilon_{\mathbf{k}} - E)\psi_{\mathbf{k}} - \frac{|V_0|}{\Omega}\sum_{k'}\psi_{\mathbf{k}'} = 0, \tag{17.14}$$

which you can rewrite as

$$\psi_{\mathbf{k}} = \frac{|V_0|}{\Omega}\frac{1}{2\epsilon_{\mathbf{k}} - E}\sum_{\mathbf{k}'}\psi_{\mathbf{k}'}. \tag{17.15}$$

Summing over $\mathbf{k}$ on both sides, we finally obtain

$$\frac{|V_0|}{\Omega}\sum_{\mathbf{k}}\frac{1}{2\epsilon_{\mathbf{k}} - E} = 1. \tag{17.16}$$

So, this is a quite general equation for interacting two Bloch electrons. In the following, we will see the consequence of this equation when applied to a Fermi sphere; we will see that however weakly attracting the two electrons, they will form a bound pair once added onto a Fermi sphere.

17.3 Two electrons added on a Fermi sphere

In metals, all states with $|\mathbf{k}| < \mathbf{k}_F$ are occupied so that $\psi_{\mathbf{k}}$ is non-zero only if $|\mathbf{k}| > \mathbf{k}_F$. Thus,

$$\frac{|V_0|}{\Omega}\sum_{|\mathbf{k}|>\mathbf{k}_F}\frac{1}{2\epsilon_{\mathbf{k}} - E} = 1. \tag{17.17}$$

A bound state would exist for $E < 2E_F$. Let us replace the summation above into an integral,

$$|V_0|\int_{E_F}^{E_F+\hbar\omega_D} d\epsilon \frac{D(\epsilon)}{2\epsilon - E} = 1, \tag{17.18}$$

where D is the density of states. We have assumed based on the isotope effect that the two electrons are interacting via phonons so that the upper limit is bounded by the Debye energy $\hbar\omega_D$. These are represented in Figs. 57 and 58.

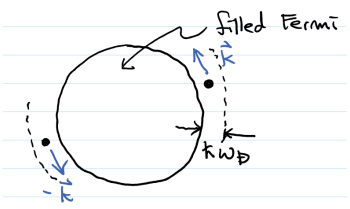


Figure 57. Graphical representation of the Cooper Problem.

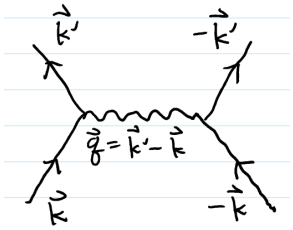


Figure 58. A diagram representing the interaction of two electrons via exchange of phonons.

To solve the above, let us define a binding energy $\delta = 2E_F - E$. We may assume that the density of state does not vary much across the energy scale of $\hbar\omega_D$ such that $D(E) \simeq D(E_F)$ can be brought out of the integral. Then,

$$|V_0|D(E_F)\int_{E_F}^{E_F+\hbar\omega_D} d\epsilon \frac{1}{2\epsilon + (\delta - 2E_F)} = 1, \tag{17.19}$$

It is straightforward to evaluate the integral to obtain

$$\frac{2\hbar\omega_D}{\delta} = \exp\left(\frac{2}{|V_0|D(E_F)} - 1\right). \tag{17.20}$$

If we further assumes $|V_0|D(E_F) \ll 1$, which is called the weak coupling approximation, it becomes

$$\delta \simeq 2\hbar\omega_D \exp\left(-\frac{2}{|V_0|D(E_F)}\right), \tag{17.21}$$

or equivalently,

$$E \simeq 2E_F - 2\hbar\omega_D e^{-2/|V_0|D(E_F)}. \tag{17.22}$$

So, there *is* a binding energy regardless how small $|V_0|$ is. Put otherwise, though the two electrons were brought about with their kinetic energy exceeding the Fermi energy, there is a bound state with negative energy with respect to the Fermi sphere. Note that in most classic superconductors, $|V_0|D(E_F) < 0.3$ so that the weak-coupling approximation is valid.

Equations (17.21) or (17.22) express the tendency to form a pair in terms of the Debye energy, electron-phonon interaction and density of states. This already gives us an idea which kinds of materials would enter a superconducting state and which ones at higher temperature.

Exercise 35

Guess the impication of the Eqs. (17.21) or (17.22). We will come back to this point later.

This is how Cooper showed that two electrons in a metal can be bound together. The bound electrons are called a Cooper pair. Many scientists at the time have guessed that somehow two electrons should be paired via phonon exchange. However, it was Cooper who first showed that it is indeed possible. Now the questions is how to make all electrons to form such Cooper pairs, which Bardeen, Cooper and Schrieffer provided the answer through their formulation of the so-called BCS theory.

18 Ground for Many-Body

How can it be even possible to obtain a wavefunction for the whole of interacting electrons? Solving just for a single electron in a hydrogen atom is far from being simple. For those electrons in their superconducting state, one can guess a form of the wavefunction resorting to the insights we learned from experiments. Then one can further elaborate on it by looking to consistency. This may all sound a bit too generic. So, let's just get it started.

We will first write down a generic form of the ground-state wavefunction. Then, by minimising the ground-state energy, we will quantify the elements in the wavefunction. It was Schrieffer, then a young PhD student, who first wrote down the ground-state wavefunction for a superconductor. It is called the BCS wavefunction following the initials of the three figures who proposed together the full theory.

18.1 From two to many: the $|\texttt{BCS}\rangle$

Let us denote the BCS wavefunction as $|\texttt{BCS}\rangle$. We will construct it from the Fermi sea, the ground state of a metal, which we denote as $|\texttt{FS}\rangle$. How do we construct a Fermi sea? One can do by adding electrons onto a vacuum, $|0\rangle$. There is a useful formalism for such task, called the second quantisation[26], which we will get only a flavour here. It is a kind of generalisation of what you might have encountered when learning ladder operators to describe a harmonic oscillator in Quantum Mechanics, where you have added or subtracted energy quantum via raising or lowering operators, respectively.

Suppose you add an electron with momentum $\mathbf{k}$ and spin σ to a whatever system. We denote an operator that does such a job as $c^\dagger_{\mathbf{k}\sigma}$, and call this a creation operator. You may also suppose to remove an electron with momentum $\mathbf{k}$ and spin σ from a system, which we denote as $c_{\mathbf{k}\sigma}$ and call an annihilation operator. Finally, we define a number operator $c^\dagger_{\mathbf{k}\sigma}c_{\mathbf{k}\sigma}$, which, when applied to an occupied state, returns an eigenvalue of 1, while otherwise 0.

As such electron creation and annihilation operators are fermionic operators, they should obey the anticommutation rule, i.e.,

$$\{c_{\mathbf{k}\sigma}, c^\dagger_{\mathbf{k}'\sigma'}\} \equiv c_{\mathbf{k}\sigma}c^\dagger_{\mathbf{k}'\sigma'} + c^\dagger_{\mathbf{k}'\sigma'}c_{\mathbf{k}\sigma} \tag{18.1}$$

$$= \delta_{\mathbf{k}\mathbf{k}'}\delta_{\sigma\sigma'}, \tag{18.2}$$

and

$$\{c_{\mathbf{k}\sigma}, c_{\mathbf{k}'\sigma'}\} = \{c^\dagger_{\mathbf{k}\sigma}, c^\dagger_{\mathbf{k}'\sigma'}\} \tag{18.3}$$

$$= 0. \tag{18.4}$$

Now you can imagine how to write the Fermi sea ground state of a metal. You prepare a vacuum and add an electron one by one until it fills up to the Fermi level. You can write

$$|\texttt{FS}\rangle = \Pi_{\mathbf{k}\leq\mathbf{k}_F} c^\dagger_{\mathbf{k}\uparrow}c^\dagger_{\mathbf{k}\downarrow}|0\rangle. \tag{18.5}$$

[26] This will be a subject of the Quantum Many-Particle Theory in the next semester.

For a superconductor, one wishes to have a different construction by making use of a pair whose electrons have opposite momenta and spin, i.e.,

$$c^\dagger_{\mathbf{k}\uparrow} c^\dagger_{-\mathbf{k}\downarrow} |0\rangle. \tag{18.6}$$

How can we elaborate on this to find the correct many-body wavefunction? BCS proposed the following trial wavefunction:

$$|\texttt{BCS}\rangle = \Pi_{\mathbf{k}} \left(u_k + v_k c^\dagger_{\mathbf{k}\uparrow} c^\dagger_{-\mathbf{k}\downarrow} \right) |0\rangle., \tag{18.7}$$

where $u_{\mathbf{k}}$ and $v_{\mathbf{k}}$ are allowed to be complex numbers satisfying $|u_{\mathbf{k}}|^2 + |v_{\mathbf{k}}|^2 = 1$ Why don't we take a pause and reflect on the implication of this wavefunction? Or you may guess what has been implemented into this wavefunction by BCS.

One can see from the above that $|v_{\mathbf{k}}|^2$ is the probability for a pair being occupied. But then, what is the role of $u_{\mathbf{k}}$? Try to expand the above for a few $\mathbf{k}$s. You'll recognise that, due to $u_{\mathbf{k}}$, the $|\texttt{BCS}\rangle$ contains a superposition of all possible states with different number of pairs. Put otherwise, one can write $|\texttt{BCS}\rangle$ as $\sum_N \lambda_N |\psi_N\rangle$, where $|\psi_N\rangle$ represents an N-particle state.

It can be seen that among all possible N-particle states, $|\lambda_N|^2$ peaks around

$$\bar{N} = \sum_k 2|v_k|^2 \tag{18.8}$$

with width

$$\langle \left(N - \bar{N}\right)^2 \rangle = \sum_{\mathbf{k}} 4|u^2_{\mathbf{k}}|v_{\mathbf{k}}|^2. \tag{18.9}$$

So why should $|\texttt{BCS}\rangle$ contain all possible states with different number of pairs? This is related to that a superconducting state is characterised by having a single fixed phase. There is a sort of uncertainty relation between the number of particles and the phase. By letting the number of particles fluctuate, the phase can be fixed to a single value. We may have time to elaborate on this later.

Exercise 36

Show Eq. (18.8) by evaluating the expectation value, $\langle \sum_{k,\sigma} n_{k,\sigma} \rangle$, over the BCS ground state wavefunction, Eq. (18.7).

Of course, writing down this trial wavefunction, Eq. (18.7), is not sufficient. One wishes to determine u and v. How shall we proceed to do that?

18.2 Variational method

In order to determine the coefficients u and v in the ground-state BCS wavefunction, we may employ a variational method, where one seeks the energy minimum via

$$\delta E = \delta \langle \psi | \mathcal{H} | \psi \rangle = 0. \tag{18.10}$$

To do so, we should have a relevant Hamiltonian at hands, which we may construct from the interaction form of the two electrons as following.

$$\mathcal{H} = \underbrace{\sum_{\mathbf{k},\sigma} \epsilon_{\mathbf{k}} n_{\mathbf{k}\sigma}}_{K.E.} \underbrace{- |V| \sum_{\mathbf{k},\mathbf{k}'} c^{\dagger}_{\mathbf{k}\uparrow} c^{\dagger}_{-\mathbf{k}\downarrow} c_{-\mathbf{k}'\downarrow} c_{\mathbf{k}'\uparrow}}_{interaction}, \tag{18.11}$$

where the first term on the right-hand side refers to the sum of kinetic energies of all electrons and the second term to the sum of the interaction energies. Apparently, this Hamiltonian should represent the situation significantly reduced from the whole of reality, but may still contain decisive terms responsible for superconductivity. This Hamiltonian is often called the pairing Hamiltonian or reduced Hamiltonian.

One is then able to obtain the energy by evaluating $\langle \mathtt{BCS}|\mathcal{H}|\mathtt{BCS}\rangle$, which essentially reduces to evaluating $\langle n_{\mathbf{k}\sigma}\rangle$ and $\langle c^{\dagger}_{\mathbf{k}\uparrow} c^{\dagger}_{-\mathbf{k}\downarrow} c_{-\mathbf{k}'\downarrow} c_{\mathbf{k}'\uparrow}\rangle$. With some algebra, which is left for exercise, one can express the energies in terms of the coefficients of the BCS wavefunction as

$$\langle n_{\mathbf{k}\uparrow}\rangle = 2|v_{\mathbf{k}}|^2 \tag{18.12}$$

$$\langle c^{\dagger}_{\mathbf{k}\uparrow} c^{\dagger}_{-\mathbf{k}\downarrow} c_{-\mathbf{k}'\downarrow} c_{\mathbf{k}'\uparrow}\rangle = v_{\mathbf{k}} v^{\dagger}_{\mathbf{k}'} u_{\mathbf{k}'} u^{\dagger}_{\mathbf{k}}. \tag{18.13}$$

This leads us to the following expression for the energy,

$$E = 2\sum_{\mathbf{k}} \epsilon_{\mathbf{k}} |v_{\mathbf{k}}|^2 - |V| \sum_{\mathbf{k}\mathbf{k}'} v_{\mathbf{k}} v^{\dagger}_{\mathbf{k}'} u_{\mathbf{k}'} u^{\dagger}_{\mathbf{k}}. \tag{18.14}$$

Exercise 37

Prove Eqs. (18.12) and (18.13) using the fermion anti-commutation relation and the properties of the creation and annihilation operators.

Minimising the energy, δE, can be obtained using a method of Lagrange multipliers, which theoretically minded students may try, but we will skip this procedure. The evaluation leads to the following pair of coupled liner equations,

$$(\epsilon_{\mathbf{k}} - \mu)\, u_{\mathbf{k}} + \Delta v_{\mathbf{k}} = E_{\mathbf{k}} u_{\mathbf{k}} \tag{18.15}$$

$$\Delta^* u_{\mathbf{k}} - (\epsilon_{\mathbf{k}} - \mu)\, v_{\mathbf{k}} = E_{\mathbf{k}} v_{\mathbf{k}}, \tag{18.16}$$

where μ is the chemical potential[27] and the so-called BCS gap parameter Δ is defined as

$$\Delta = |V| \sum_{\mathbf{k}} u_{\mathbf{k}} v^{\dagger}_{\mathbf{k}}. \tag{18.17}$$

[27] The Fermi energy E_F is defined as the chemical potential μ at $T = 0$. Since any reasonable temperature is far below than the Fermi temperature that is tens of thousands of kelvin, the chemical potential is almost always well approximated to the Fermi energy. So, we will not make any formal distinction between them and use interchangeably in this lecture.

The meaning of Δ may appear clearer when one rewrites it back in terms of the expectation value of the operators as

$$\Delta = |V| \sum_{\mathbf{k}} \langle c_{-\mathbf{k}\downarrow} c_{\mathbf{k}\uparrow} \rangle. \tag{18.18}$$

Solving the above pair of linear equations leads us to the eigenvalues $\pm E_{\mathbf{k}}$, where

$$E_{\mathbf{k}} = \sqrt{(\epsilon_{\mathbf{k}} - \mu)^2 + |\Delta|^2}. \tag{18.19}$$

The coefficients can now be written as

$$|u_{\mathbf{k}}|^2 = \frac{1}{2}\left(1 + \frac{\epsilon_{\mathbf{k}} - \mu}{E_{\mathbf{k}}}\right), \tag{18.20}$$

$$|v_{\mathbf{k}}|^2 = \frac{1}{2}\left(1 - \frac{\epsilon_{\mathbf{k}} - \mu}{E_{\mathbf{k}}}\right). \tag{18.21}$$

These are plotted in Fig. 59. Note that this looks alike the Fermi-Dirac distribution for

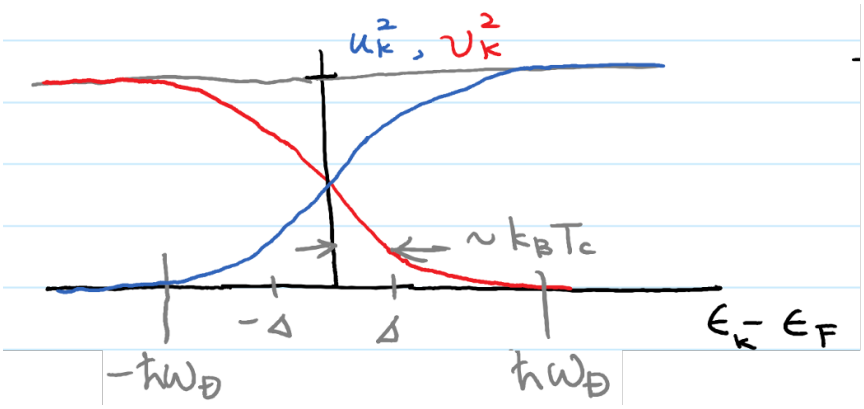


Figure 59. The coefficients of the BCS wavefunction at $T = 0$.

electrons in metals at $T = T_c$. Indeed the the difference should be hardly noticeable. But note that this graph for a superconductor is for $T = 0$!

For self-consistency, one can combine Eqs. (18.15) and (18.17) to obtain

$$\Delta = |V| \sum_{\mathbf{k}} \frac{\Delta}{2E_{\mathbf{k}}}, \tag{18.22}$$

or

$$1 = |V| \sum_{\mathbf{k}} \frac{1}{2E_{\mathbf{k}}}, \tag{18.23}$$

known as the BCS gap equation, which is central to the BCS theory. This allows us to obtain the gap parameter Δ at $T = 0$ as we wil see shortly.

By the way, we rushed a bit to reach the conclusion. As you now have an overall idea of the procedure starting from the assumed Hamiltonian through to finding the minimum energy condition to the BCS gap equation, I hope you reflect on this subsection again by trying to fill some gaps by, e.g., deriving Eqs. (18.20) and (18.21).

Exercise 38

Prove Eqs. (18.20) and (18.21).

18.3 Zero temperature gap

To solve Eq. (18.23), let us replace the summation to an integration, particularly within the thin shell of the Bloch stats around the Fermi energy with $|\epsilon_{\mathbf{k}} - \mu| < \hbar\omega_D$, as

$$\sum_{\mathbf{k}} \to 2 \int_0^{\hbar\omega_D} D(E_F) d\epsilon. \tag{18.24}$$

Then the gap equation can be written as

$$1 = D(E_F)|V| \int_0^{\hbar\omega_D} \frac{d\epsilon}{\sqrt{\epsilon^2 + |\Delta|^2}}, \tag{18.25}$$

Through evaluating the integral on the right, under the assumption of weak-coupling, $D(E_F)|V| \ll 1$, one is led to

$$\Delta = 2\hbar\omega_D e^{-1/D(E_F)V}. \tag{18.26}$$

Exercise 39

Prove Eq. (18.26). When evaluating the integral, you may obtain a form of $\sinh(1/x)$ which can be approximated to $e^{1/x}/2$ for $x \ll 1$.

You may notice that this expression is similar to the binding energy you obtained for a pair of electrons. Δ can be considered an energy scale for superconductivity, and you should notice that it is much less than the Debye energy due to the exponential term. This explains why superconducting T_c is much lower than the typical Debye temperatures.

A net attractive interaction can occur if the phonon-mediated interaction is stronger than the screened Coulomb one. Electronic response is considered instantaneous since the plasma frequency is much higher. One may note that the effective interaction between the electrons is not changing with the mass of ions. What gives rise to the isotope effect is the thickness of the energy shell $\hbar\omega_D$.

19 Quasi-Particle Excitation

Having found the ground state and its energy, the next step is to find ways to describe the excited states, which will allow us to calculate finite temperature properties of superconductors. To do so, we will follow yet another approach than the variational method. We will diagonalise the pairing Hamiltonian using a mean-field approximation. Diagonalising a many-body Hamiltonian is mostly unrealistic except a few cases. So, we will first reduce the many-body Hamiltonian into a further simpler form using a mean-field approximation that is known how to diagonialise.

Recall that the BCS ground state is a peculiar superposition of many electrons filled and emptied. It wouldn't be fair to suppose that excitation from this ground state is simply a set of two normal electrons broken out of a pair, even though they retain some original characters of the normal electrons. What we'd show within the mean-field approximation is that the excitation can be a set of non-interacting, new quasi-particles (that is a superposed state of adding and removing electrons) that exist above the BCS ground state. This idea is quite analogous to deriving phonons from lattice.

19.1 Canonical transformation

We start by adopting the following approximation[28]

$$c^\dagger_{\mathbf{k}\uparrow}c^\dagger_{-\mathbf{k}\downarrow}c_{-\mathbf{k}'\downarrow}c_{\mathbf{k}'\uparrow} \approx \langle c^\dagger_{\mathbf{k}\uparrow}c^\dagger_{-\mathbf{k}\downarrow}\rangle c_{-\mathbf{k}'\downarrow}c_{\mathbf{k}'\uparrow} + c^\dagger_{\mathbf{k}\uparrow}c^\dagger_{-\mathbf{k}\downarrow}\langle c_{-\mathbf{k}'\downarrow}c_{\mathbf{k}'\uparrow}\rangle \tag{19.1}$$

Then the pairing Hamiltonian can be replaced by the following mean-field Hamiltonian,

$$\mathcal{H}_{\mathrm{MF}} = \sum_{\mathbf{k},\sigma}(\epsilon_{\mathbf{k}}-\mu)\,c^\dagger_{\mathbf{k}\sigma}c_{\mathbf{k}\sigma} - |V|\sum_{\mathbf{k}\mathbf{k}'}\left(\langle c^\dagger_{\mathbf{k}\uparrow}c^\dagger_{-\mathbf{k}\downarrow}\rangle c_{-\mathbf{k}'\downarrow}c_{\mathbf{k}'\uparrow} + c^\dagger_{\mathbf{k}\uparrow}c^\dagger_{-\mathbf{k}\downarrow}\langle c_{-\mathbf{k}'\downarrow}c_{\mathbf{k}'\uparrow}\rangle\right). \tag{19.2}$$

Using

$$\Delta = |V|\sum\langle c_{-\mathbf{k}\downarrow}c_{\mathbf{k}\uparrow}\rangle \tag{19.3}$$

the mean-field Hamiltonian can be rewritten as

$$\mathcal{H}_{\mathrm{MF}} = \sum_{\mathbf{k},\sigma}(\epsilon_{\mathbf{k}}-\mu)\,c^\dagger_{\mathbf{k}\sigma}c_{\mathbf{k}\sigma} - \sum_{\mathbf{k}}\left(\Delta^* c_{-\mathbf{k}\downarrow}c_{\mathbf{k}\uparrow} + \Delta c^\dagger_{\mathbf{k}\uparrow}c^\dagger_{-\mathbf{k}\downarrow}\right). \tag{19.4}$$

This quadratic form of Hamiltonian is known to be solved exactly by taking a Bogoliubov-Valatin transformation. To do so, one introduces new operators in terms of the electron creation and annihilation operators as,

$$\gamma^\dagger_{k\uparrow} = u^*_k c^\dagger_{k\uparrow} - v^*_k c^\dagger_{-k\downarrow} \tag{19.5}$$

$$\gamma^\dagger_{k\downarrow} = u^* c^\dagger_{-k} + v^*_k c_{k\uparrow}, \tag{19.6}$$

for which one can also define corresponding annihilation operators[29]. What is the meaning of these operators? If you look into the first one of $\gamma^\dagger_{k\uparrow}$, it is a superposition of two cases

[28] This is the result derived from Wick's theorem that you will find in Quantum Many-Particle Systems module. We use this without proof.

[29] Don't be swamped by complex notations. All books use different notation for γ operators! Little emphasis on mathematical formality in this module, but we take only their physical meanings correctly.

where an electron with k and $\uparrow$ is created with probability of $|u|^2$ while an electron with $-k$ and $\downarrow$ is removed with probability of $|v|^2$ (the latter is equivalent to say 'creating' a hole). Similar for the $\gamma^\dagger_{k\downarrow}$ (and one can also define corresponding annihilation operators). However, all this gets their meaning because the mean field Hamiltonian can be written in terms of the new operators as

$$\mathcal{H}_{\text{MF}} = \sum_{\mathbf{k}} E_{\mathbf{k}} \left(\gamma^\dagger_{\mathbf{k}\uparrow}\gamma_{\mathbf{k}\uparrow} + \gamma^\dagger_{-\mathbf{k}\downarrow}\gamma_{-\mathbf{k}\downarrow} \right) \tag{19.7}$$

where all constant energy terms (for ground state) dropped for simplicity. So this Hamiltonian considers the BCS ground state as a vacuum. From this Hamiltonian, one can say that the excited states above the ground state are creating non-interacting γ particles whose nature is a mixture of electrons and holes. Again, this is analogous to the case of phonons. Without temperature, the lattice is considered a vacuum. By raising temperature, you add phonons to this vacuum.

Exercise 40

Show that the new quasi-particles, Bogoliubons, are fermionic. Also show that they are not present in the BCS ground state. (Skip this exercise if you like; non-examinable)

19.2 Finite temperature

At finite temperatures, the occupancy of those quasiparticles is determined by the Fermi-Dirac distribution, such that

$$\langle \gamma^\dagger_{\mathbf{k}\uparrow}\gamma_{\mathbf{k}\uparrow} \rangle = f(E_{\mathbf{k}}) \tag{19.8}$$

where $f(E) = 1/(e^{E/k_BT} + 1)$. One can generalise the definition of the gap into finite temperature as

$$\Delta = |V| \sum_{\mathbf{k}} \langle c_{-\mathbf{k}\downarrow} c_{\mathbf{k}\uparrow} \rangle \tag{19.9}$$

$$= |V| \sum_{\mathbf{k}} u_{\mathbf{k}} v^*_{\mathbf{k}} \left(1 - 2f(E)\right) \tag{19.10}$$

The BCS gap equation at finite temperature then becomes

$$1 = |V| \int_0^{\hbar\omega_D} \frac{D(E_F)d\epsilon}{\sqrt{\epsilon^2 + |\Delta|^2}} \tanh\left(\frac{\sqrt{\epsilon^2 + |\Delta|^2}}{2k_BT} \right). \tag{19.11}$$

One can immediately check the consequence of this equation. For instance, taking the limit $\Delta \to 0$, which corresponds to approaching T_c from below, one obtains

$$k_BT_c = 1.13\hbar\omega_D e^{-1/D(E_F)|V|} \tag{19.12}$$

Also, taking T=0 leads to

$$2\Delta(0) = 3.52 k_B T_c. \tag{19.13}$$

The zero-temperature gap is found comparable to kTc. By measuring both the Δ and T_c, this prediction with the particular numerical factor of 1.76 was confirmed to be correct.

19.3 Tunneling: Normal-to-superconductor

We are given the expression for the density of states with a gap in a superconductor. It turned out that the tunneling experiment is a useful way of measure the profile of density of states and also the size of the gap. In tunneling experiments, one constructs a contact of a superconductor with a metal, or with other superconductor, and measure currents against voltage through the contact. To understand how this would help us understand the density of states and the gap, have a look into the schematic drawing of the density of states in different conditions as shown in Fig. 60.

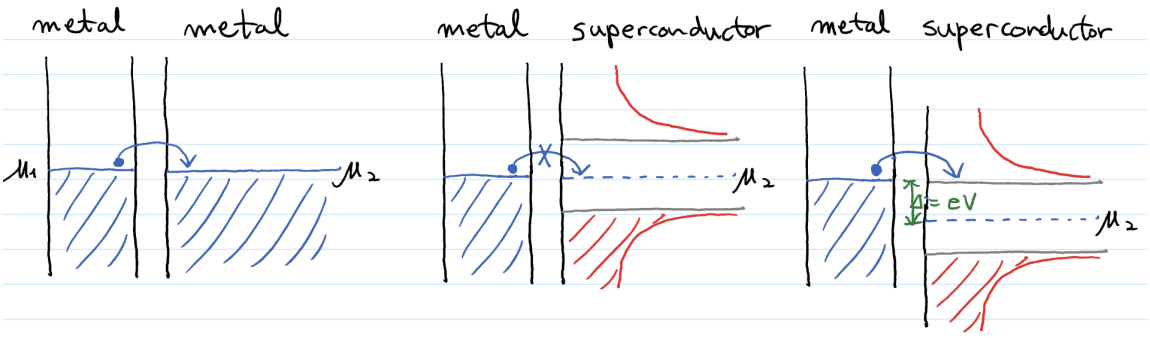


Figure 60. Energy versus density of states for three different contacts of metal-metal, metal-superconductor and superconductor at zero bias, and metal-superconductor at threshold bias.

From the density of states, one can schematically draw I-V curve or dI/dV-V curves[30]. See Fig 61, for example. On the left plots current as a function of voltage at zero temperature (red) and finite temperature (blue). Two features are immediately recognised. First, there is no current up to certain finite value of voltage, $V = \Delta/e$. At sufficiently high voltage, the current response approaches the behaviour expected for ohmic contact. One may further note that just above the threshold voltage, the current grows very rapidly.

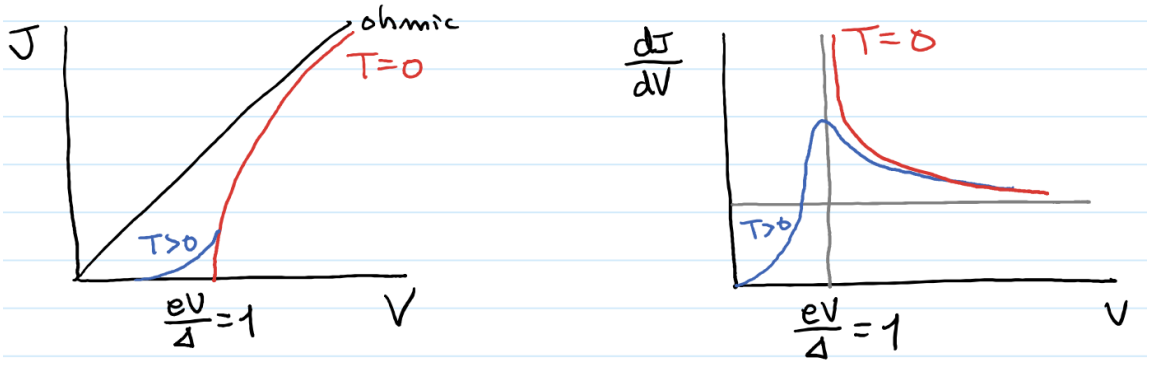


Figure 61. Current-voltage characteristic (left) and differential conductance versus voltage (right).

[30] Again, sorry for mixed notation. I have used both J and I as currents in this note. dI/dV is called the differential conductance.

Given the description of the data themselves, now we need to interpret them. The finite voltage required can be understood from the middle panel in Fig. 60. While in sufficiently low voltage, the electrons at the Fermi level cannot tunnel through to the superconductor because there is no available state. By applying a voltage, one can move vertically the relative position of metal and superconductor, and once the voltage is high enough so that the Fermi level on the metal side matches the level of upper or lower band in the superconductor, electrons can jump to conduct current. Once the voltage is larger than enough, there would be little distinction between the metal-superconductor and the metal-metal tunneling.

Note that the density of states just above or below the gap is very large. So, once the energy level matches, the current would shoot up to fill the states, which you observe as a rapid rise in I just above the threshold V. Conductance dI/dV is a way to measure quantity directly proportional to the density of states. As you see on the right in Fig. 61, the red curve shows a divergence as the voltage is lowered to the threshold value, alike you observe in the middle of Fig. 60. This sharp feature would smear out as temperature is increased, because thermal excitation allows for some electrons to jump over the gap.

19.4 Tunneling: Superconductor-to-superconductor

Now finally superconductor-to superconductor tunnelling. You will find striking difference between the zero-temperature and finite temperature behaviours. For zero temperature, density of states would look like in the left panel of Fig. 62. By applying a voltage, one can move vertical position of each so that above certain value, precisely at $eV = \Delta_1 + \Delta_2$, the tunneling current will show up.

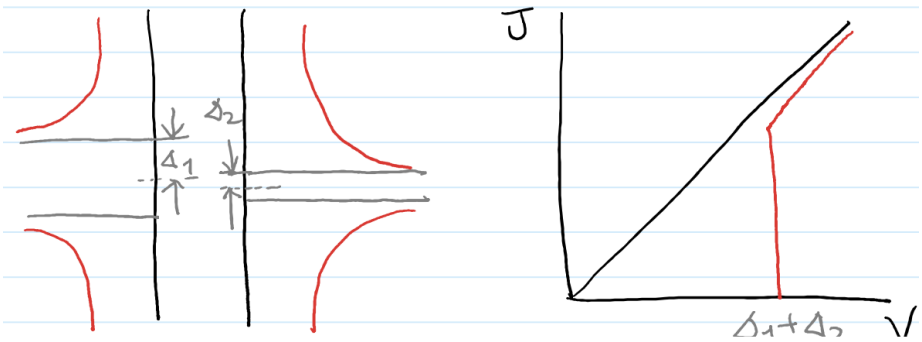


Figure 62. Density of states (left) and I-V characteristic of two different superconductors at zero temperature.

What would you expect if we raise temperature above zero? For their density of states, see the left panel in Fig. 63. Some thermally excited states would overcome the gap to go up, on both superconductors. Then, for tunneling, they don't have to always cross the original gap all again, but they now have some short-cuts. As a result, one would expect the I-V curve as shown on the right in Fig. 63

This will enable us to determine the gap sizes of both superconductors by a single measurement. You can also measure this I-V curves as a function of temperature to obtain

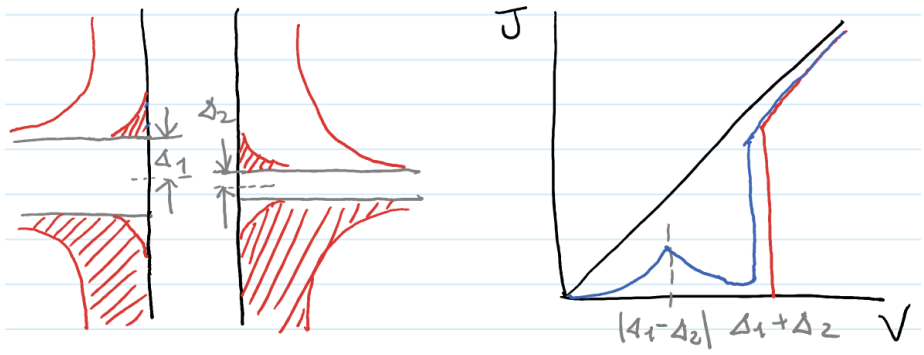


Figure 63. Density of states (left) and I-V characteristic of two different superconductors at finite temperature.

the temperature evolution of the gaps for the two superconductors.

19.5 BCS predictions

Before closing, let us collect a few examples of the predictions from the BCS theory in comparison with the earlier arguments. For instance, one may recall the binding energy in Cooper's treatment of two electrons, which is written by

$$\delta = 2\hbar\omega_{\mathrm{D}} \exp\left(-\frac{2}{V_0 D(E_F)}\right). \tag{19.14}$$

From the BCS theory, the energy gap is given by

$$\Delta = 2\hbar\omega_{\mathrm{D}} \exp\left(-\frac{1}{V_0 D(E_F)}\right). \tag{19.15}$$

You can appreciate their close resemblance.

From thermodynamic consideration by the beginning of the module, we have obtained the condensation energy

$$E = -\mu_0 H_c^2/2. \tag{19.16}$$

From the BCS theory, the condensation energy is given by

$$E = -D(E_F)\Delta^2/2 \tag{19.17}$$

$$= -2\left(\hbar\omega_D\right)^2 D(E_F) \exp\left(-\frac{2}{V_0 D(E_F)}\right). \tag{19.18}$$

Lastly, a length scale that could be obtained form the uncertainty consideration, by associating the binding energy with uncertainty in momentum (via kinetic energy) as

$$\delta \simeq \Delta\left(\frac{\hbar^2 k^2}{2m}\right) = \frac{\hbar^2 k_F \Delta k}{m}, \tag{19.19}$$

where Δ here refers to a difference, not the gap (sorry, but any suggestion for better notation?). Then, the uncertainty in position can be written as

$$\Delta r = \frac{1}{\Delta k} = \frac{\hbar v_{\mathrm{F}}}{\delta} \tag{19.20}$$

From the BCS theory, the coherence length is given by

$$\xi = \frac{\hbar v_{\mathrm{F}}}{\pi \Delta}, \tag{19.21}$$

where Δ here refers to the gap! Again, you may appreciate their resemblance.

20 Explained

You might have noticed that in the previous lecture we have actually answered one big question posed early in the module, that is, why certain elements become superconducting at high or low temperatures while others remain non-superconducting etc. Looking back into the critical temperature,

$$k_B T_c = 1.13\hbar\omega_D e^{-1/D(E_F)|V|}, \tag{20.1}$$

expressed in terms of the Debye frequency (or energy), density of states at the Fermi level and the attractive interaction, one can associate them to the properties of elemental metals.

We have also learned about tunneling experiment in the previous lecture in a qualitative fashion without giving much details behind. Thinking about when the tunneling should happen from material 1 to material 2, one would expect that the state 1 must be occupied while the state in 2 unoccupied, which gives rise to a factor $f_1(1-f_2)$. To calculate the total tunnelling, one should subtract $2 \to 1$ contribution of $f_2(1-f_1)$, which leaves effective $f_1 - f_2$. So, the tunneling from material 1 to 2 can be written as

$$I(V) \propto \int D_1(E) D_2(E+eV) \left[f(e) - f(E+eV)\right] dE, \tag{20.2}$$

where Ds are density of states.

20.1 Heat capacity

We learned from the BCS theory that excitation in a superconductor are understood as non-interacting fermionic quasiparticles of energy $E_k = \sqrt{(\epsilon_k - \mu)^2 + \Delta^2}$. This will allows us to obtain expression for the heat capacity. You may start from

$$C_v = T\frac{\partial S}{\partial T} \tag{20.3}$$

$$= -2k_b T \sum_k \ln\left(\frac{f_k}{1-f_k}\right)\frac{\partial f_k}{\partial T} \tag{20.4}$$

$$= \sum_k 2E_k \frac{\partial f_k}{\partial T}, \tag{20.5}$$

where Fermi-Dirac distribution $f_k = f(E_k)$. To obtain the above, you have needed to use

$$S = -2k_B \sum_k \left[(1-f_k)\ln(1-f_k) + f k_k \ln f_k\right]. \tag{20.6}$$

Equation (20.5) can be evaluated by using $E_k = \sqrt{(\epsilon_k - \mu)^2 + \Delta^2}$, but we skip the procedure to jump to the conclusion:

$$C_v = -\frac{2}{T} D(E_F) \int E^2 \frac{\partial f}{\partial E} d\epsilon + \frac{\partial}{\partial T}\Delta^2 D(E_F) \int \frac{\partial f}{\partial E} d\epsilon. \tag{20.7}$$

The first term equals to the normal state expression when $\Delta \to 0$. The second term gives a discontinuous jump as $T \to T_c$, whose numerical value can be calculated to be

$$\lim_{T\to T_c} \frac{C_s(T) - C_n(T)}{C_n(T)} = 1.43 \tag{20.8}$$

Figure 64 shows an example data of niobium in its normal and superconducting states. Check the numerical agreement between the calculated value and experimental data.

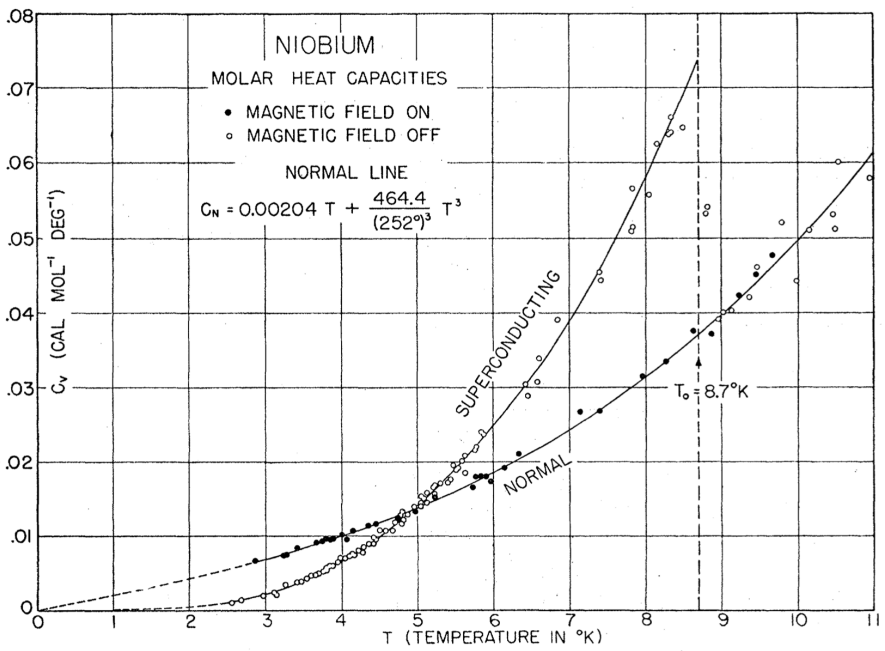


Figure 64. Heat capacity of Niobium in normal and sueprconducting states. (from A. Brown et al., Phys. Rev. 92, 52 (1953)).

20.2 Coherence factor

Many physical properties or our experimental observations for metals involve scattering processes. Electrons may scatter off charged impurities, magnetic ions, nuclear spins or phonons. This could be represented by a perturbing Hamiltonian,

$$\mathcal{H}' = \sum_{k\sigma,k'\sigma'} B_{k\sigma,k'\sigma'} c^\dagger_{k'\sigma'} c_{k\sigma}, \tag{20.9}$$

where $B_{k\sigma,k'\sigma'}$ is the matrix element for the scattering of an electron from state $k\sigma$ to state $k'\sigma'$ ($c^\dagger_{k'\sigma'} c_{k\sigma}$: annihilating $k\sigma$ and creating $k'\sigma'$). In the normal state, each term is independent and the final result is found by adding up transition probabilities $|B_{k\sigma,k'\sigma'}|^2$.

The situation is different for the superconducting state. When we expand $c^\dagger_{k'\sigma'} c_{k\sigma}$ in terms of γ operators, there appear interference effects as different terms involve phase coherent terms. In the end, to calculate the transition probabilities, $|B_{k\sigma,k'\sigma'}|^2$ needs to be multiplied by the so called coherence factors: $(u_{k'} u_k \mp v_{k'} v_k)^2$ for the scattering of quasi-particles and $(v_k u_{k'} \pm u_k v_{k'})^2$ for creation and annihilation of two quasi-particles. The upper sign refers to Case I and the lower to Case II.

For Case I coherence factors, the singularity in the density of states at $E = \Delta$ is cancelled out, and the superconducting response becomes less than the normal. This is the case for thermal conductivity or acoustic attenuation as we will see shortly. On the

other hand, for Case II coherence factors, the superconducting response exceeds the normal response and leads to a peak below T_c. This corresponds to nuclear spin relaxation.

20.3 Thermal conductivity

Thermal conductivity in normal metal is proportional to temperature and is given by

$$\kappa_n = \int_0^\infty \frac{v_F l}{3} \frac{\partial f}{\partial \epsilon} \frac{\epsilon^2}{T} D(E_F) d\epsilon, \tag{20.10}$$

where l is the mean free path. In the superconducting state, physical quantities such as v_F, ϵ and $D(E_F)$ are replaced with those corresponding for a superconductor, and the result becomes

$$\frac{\kappa_s}{\kappa_n} = \frac{\int_\Delta^\infty E^2 \frac{\partial f}{\partial E} dE}{\int_0^\infty \epsilon^2 \frac{\partial f}{\partial \epsilon} d\epsilon}. \tag{20.11}$$

The evaluation of this integral is compared with experimental data for aluminum in Fig. 65. Note that the thermal conductivity drops very rapidly at low temperatures due to the energy gap. The data comprises three different samples whose mean free path differ over two orders of magnitude. However, all the data collapse nicely onto a single, theoretically predicted curve, providing confidence to the theory.

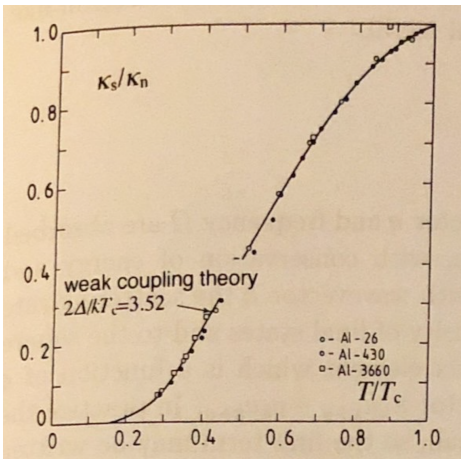


Figure 65. Ultrasonic attenuation in tin (RW Morse and HV Bohm, Phys. Rev. B 108, 1094 (1957)).

20.4 Ultrasound absorption

The ratio of the acoustic attenuation coefficients in the superconducting and normal states can be written in a very simple form as

$$\frac{\alpha_s}{\alpha_n} = \frac{\int_\Delta^\infty \partial f/\partial E) dE}{\int_0^\infty (\partial f/\partial \epsilon) d\epsilon} = \frac{f(\Delta)}{f(0)} = 2f(\Delta). \tag{20.12}$$

Evaluation of this for tin is shown in Figure 66 alongside the experimental data, again in excellent agreement. It is also suppressed strongly at low temperatures due to the energy gap. It may be interesting to think over the meaning of this simple form of Eq. (20.12) by writing down the Fermi-Dirac distribution function with Δ.

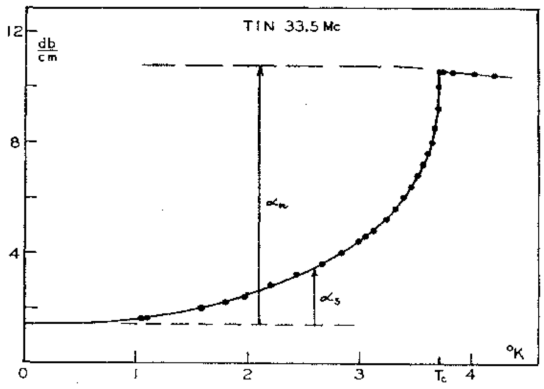


Figure 66. Ultrasonic attenuation in tin (RW Morse and HV Bohm, Phys. Rev. B 108, 1094 (1957)).

20.5 Nuclear relaxation

Electrons can also scatter off nuclei via hyperfine interaction, the effectively magnetic interaction between electron and nuclear spins. This affects the nuclear spin relaxation time in a magnetic field. For instance, when nuclear spins are perturbed by external RF fields, after the removal of the RF fields the nuclear spins will redistribute themselves among the Zeeman split energy levels to meet the thermal equilibrium condition. This occurs as nuclear and electrons spins flip together as a part of their hyperfine interaction. As mentioned before, this belongs to Case II with enhanced superconducting response below T_c compared to metal. Figure 67 shows experimental data alongside the calculation which again match very well.

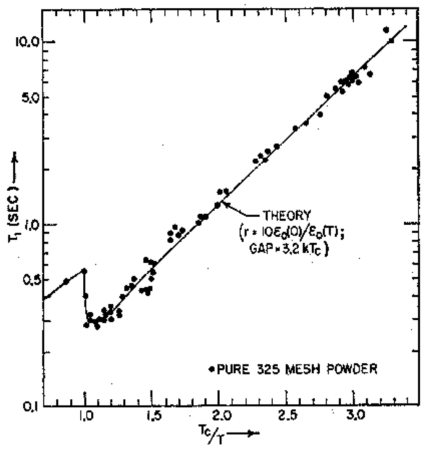


Figure 67. Nuclear relaxation time in aluminum (Y. Masuda and AG Redfield, Phys. Rev. 116, 79 (1959)).

Be careful that this graph should be read with care. Note that x axis is given in $1/T$. The y axis is given in T_1, relaxation time, but the discussion above relates to relaxation rate, $1/T_1$. Then, it makes more sense in the graph. Starting from the left bottom at high

temperature, as temperature is lowered just below T_c, T_1 drops, meaning $1/T_1$ is increased. As temperature is further lowered, $1/T_1$ is strongly suppressed as it appears linear in this Arrhenius plot.

Exercise 41

Read the original paper by Bardeen, Cooper and Schrieffer (via ResourceList). I believe that you will be pleasantly surprised by finding that you could digest so much of this milestone paper in the history of modern physics. Then, you are more or less ready to start enjoying current research papers.

For more details on the discussion here, refer to Chapter 8 of the book by Waldram; you can access a scanned copy from ResourceList via Canvas.

Thank you all for attending this module with curiosity, enthusiasm and patience. I hope you will have a much brighter semester in 2021. Have a warm Christmas and new year celebration.

All The Best,
Mingee

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